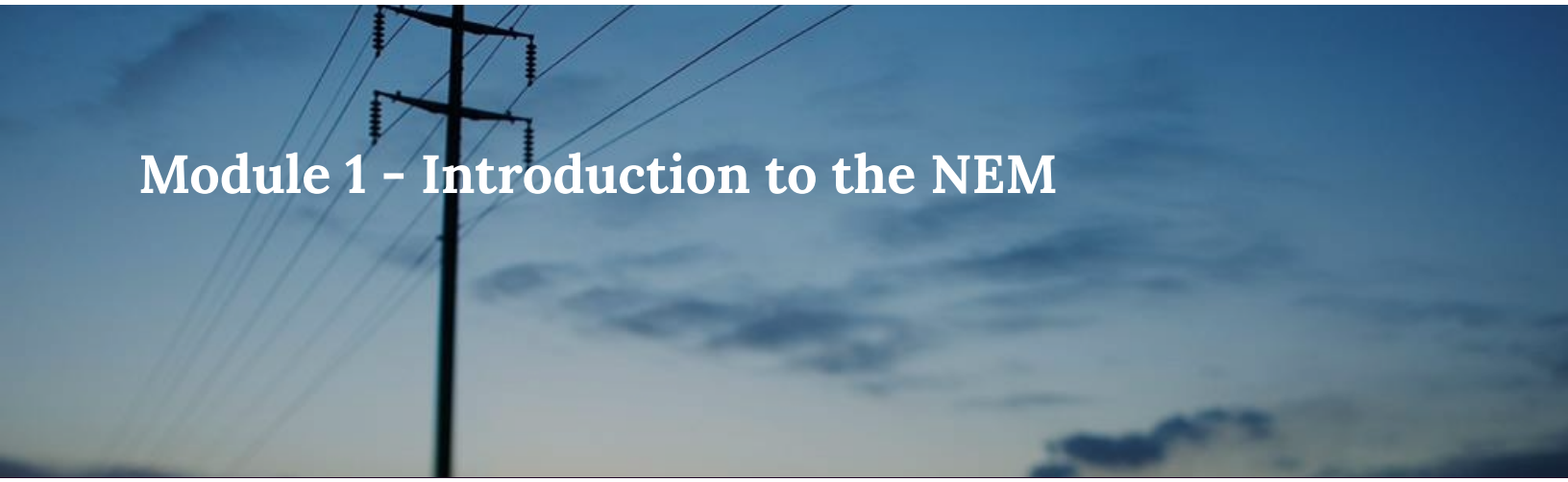

Module 1 - Introduction to the NEM

Welcome to AEMO's National Electricity Market (NEM) Foundations eLearning course. This course is designed to provide a broad understanding of how the NEM operates, both as a physical power system and as a wholesale electricity market.

The course is structured into several topics and can be completed over multiple sittings.

Important information and disclaimer

AEMO provides education courses and materials for general information and education only. They are not to be relied on for business or legal decisions.

We always try to ensure the quality of the information in our course materials, but they may not be accurate, up to date or complete. Energy market rules, procedures and systems are extensive, complex and subject to change, so please obtain detailed, expert advice about their current application to your specific circumstances.

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Module introduction

The electricity system

- Production and transportation
- Units of electrical measurement
- Electricity markets
- The Australian Energy Market Operator (AEMO)
- The NEM - a wholesale exchange
- The retail market
- The financial market
- The history of the electricity industry and the NEM
- Module summary

Lesson 1 of 11

Module introduction

This module addresses the following learning objectives:

- To provide an introduction and overview of the National Electricity Market (NEM)
- To describe the NEM principles and structure
- To describe the role and history of AEMO as it relates to the NEM

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

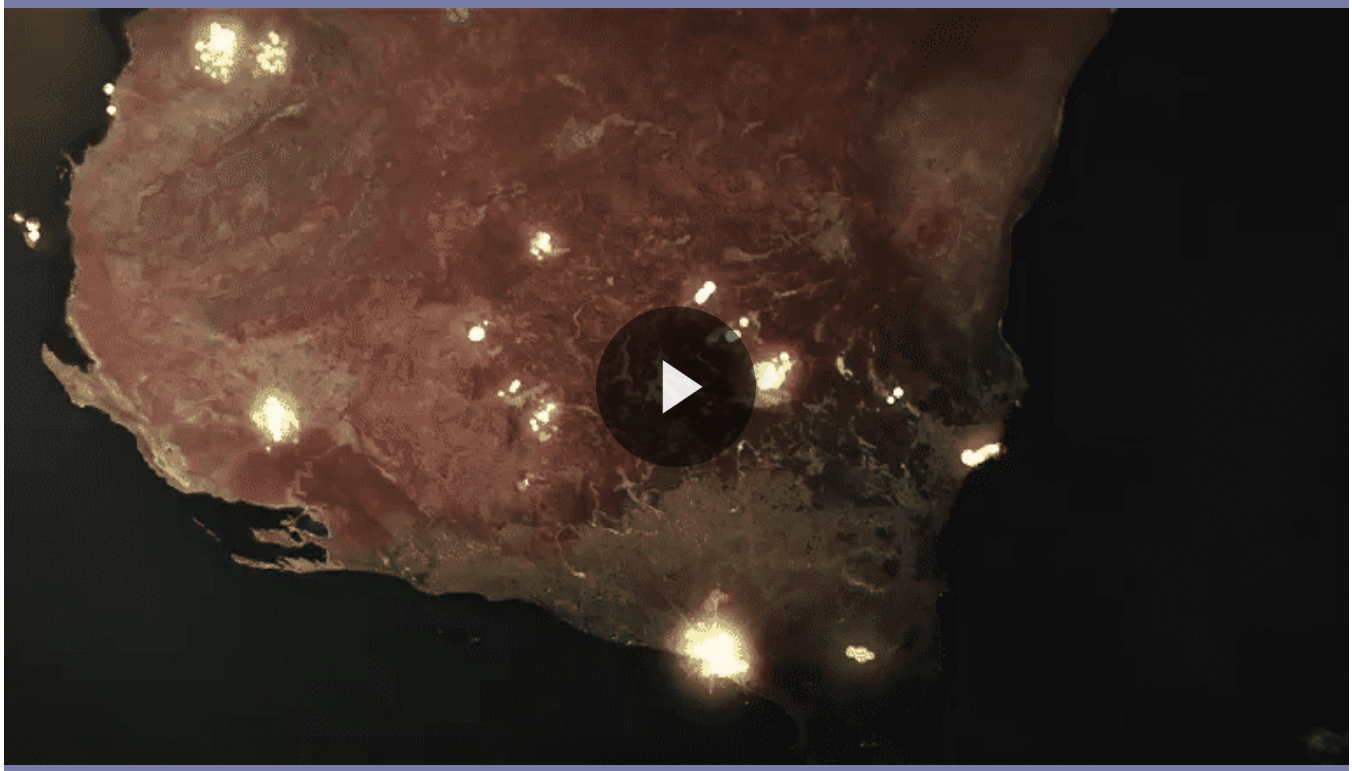
CONTINUE

Lesson 2 of 11

The electricity system

Australia has one of the world's longest interconnected power systems — from Port Douglas in Queensland to Port Lincoln in South Australia and Hobart in Tasmania — a distance of around 5,000 kilometres. This power system is known as the National Electricity Market (NEM).

The following excerpt from the ABC's Great Southern Land documentary series demonstrates the sheer size and complexity of our electricity system.



CONTINUE

Production and transportation

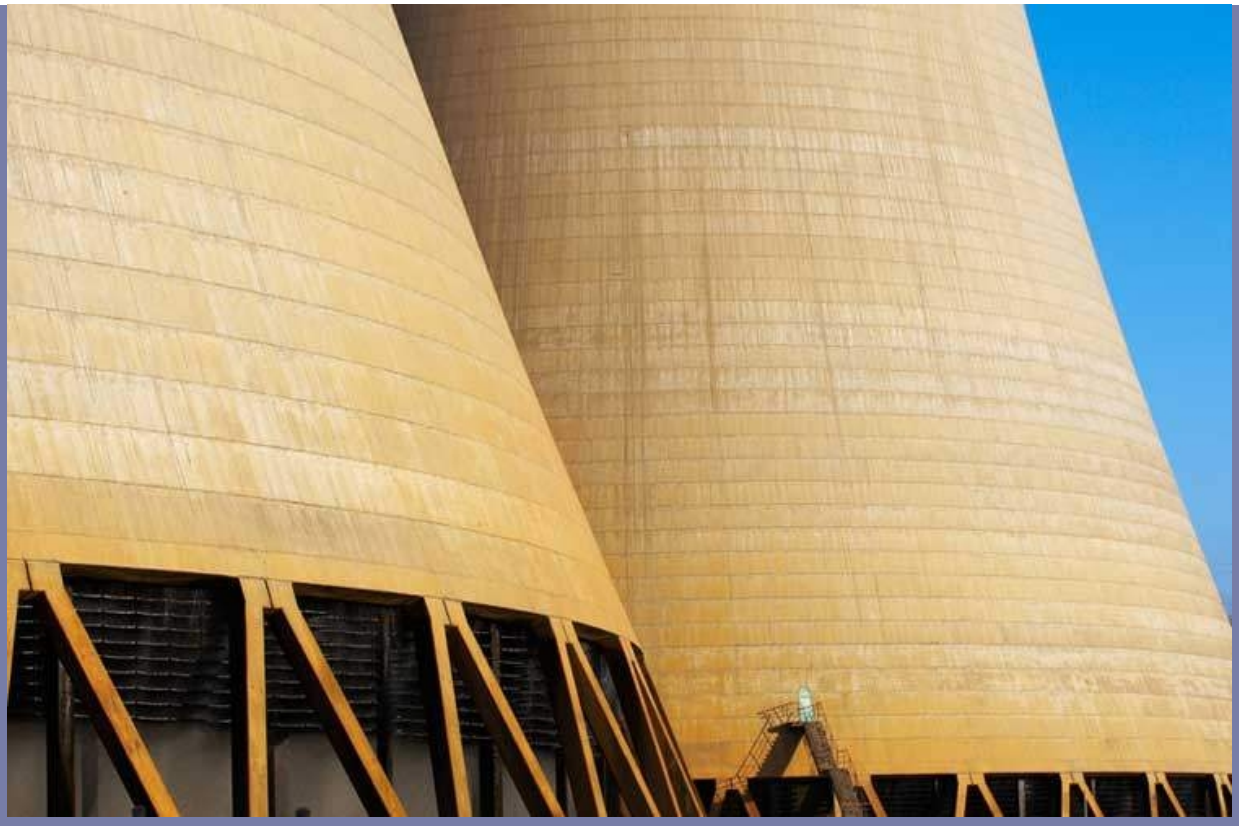
When an appliance is switched on, power is transmitted from a power station to the appliance. Although this occurs instantaneously, a specific sequence of events takes place to ensure delivery of the required electricity.

Let's begin by examining this sequence of events, which starts with electricity generation.

Generators

Generators in modern power stations produce electricity by the mechanical action of large, powerful magnets that spin rapidly inside the huge coils of conducting wire driven by steam, gas, water and wind turbines.

Some of Australia's electricity production relies on burning fossil fuels - mainly gas and coal - which are burned to heat water and produce steam. The steam is then forced under great pressure through a turbine that drives a generator to produce electricity. The complete process involves the conversion of chemical energy to kinetic energy to electrical energy. In a similar way, the kinetic energy of falling water drives turbine blades to produce electrical energy at a hydro-electricity plant, and the kinetic energy of wind drives the blades of a wind-power turbine to produce electricity.



BBC

Transformers

Transformers convert the electricity produced at a generating unit from low to high voltage to enable its efficient transport on the transmission network.

Transmission network

The high voltage current (in Australia, this can be up to 500 kV) leaves the transformer and travels along the transmission network - the high voltage power lines that stretch for vast distances across our interconnected electricity system. Essentially these are bulk transporters of electricity, carrying large supplies of power from power stations to locations closer to consumers.

A tower on the high voltage Transmission network

A tower on a lower voltage Distribution network

Substation transformer

When the electricity has been transported closer to consumers, it enters a substation transformer. Here, the high transmission voltage is reduced, or 'stepped down', to a lower voltage. The electricity has now left the transmission network, and enters what is known as the distribution network.

Distribution network

The electricity travels along the distribution network and into a distribution substation. Here, the high distribution voltage is again reduced to an even lower voltage, suitable for our local distribution lines (the power poles and lines on our streets).

Electricity consumption

The final stop of this journey is consumption, such as powering an appliance in a consumer's home. The voltage at this point of consumption is typically 230-240 volts.

Conclusion

As we've seen, various sectors of the electricity supply industry are involved with the generation, transmission and distribution of electricity. Australia's social, industrial and commercial success depends on a reliable electricity supply.

Energy exists in many forms. A quantity of energy can be changed or converted, but can never be created or destroyed. Electricity is a secondary energy source as it is produced by converting other energy sources like the chemical energy in coal, natural gas and oil. Other primary sources of energy, like the sun and wind, are increasingly being used to produce electricity.

Electricity can be converted readily to heat and light and used to power machines. It can also be transported with relative ease.

These characteristics make electricity a convenient and manageable form of energy, and contribute both to its value as a commodity and its versatility as a source of power.

Next we'll examine how electricity is measured, allowing it to be bought and sold within the National Electricity Market.

CONTINUE

Lesson 4 of 11

Units of electrical measurement

AEMO operates the National Electricity Market (NEM) - a market where electricity is bought and sold. In order to trade in electricity we need a way of measuring it.

The common unit of electrical energy measurement is the kilowatt hour, or kWh for short. Other fundamental electrical units are volts (V) and amps (A).

Electrical units of measurement

Let's take a closer look at these units of measurement.

Volts (V)

Volts is the unit measuring the amount of energy delivered per unit of charge. One volt means that one Joule of energy is delivered by one Coulomb of charge. Voltage can be likened to the pressure in a pipe containing water - the higher the voltage, the higher the pressure on electrons to move out of a source. Voltage is measured in volts, and its symbol is V,

Currents and Amps (A)

Amps (ampere) is the unit that measures the flow rate of electrical charge, known as current. At the atomic level an electron is a single electrical charge. At the macro level, charge is measured in Coulombs, and a current flow rate of one Coulomb passing a point along a wire in one second is one amp. If we expand our water pipe analogy, current can be thought of as the rate of flow of water as is passes through-the pipe.

Both voltage and current are required to power anything.

Watts (W)

The fundamental unit of energy is the Joule, and its symbol is J. The time rate at which energy is delivered (the number of Joules per second) is measured in Watts.

This 'time rate' of energy delivery is also known as 'power', therefore a Watt is a unit of power. An energy delivery rate of one Joule in one second, or one Joule/second, is known as a power of one Watt. Its symbol is W.

The number of watts, or wattage, of an electrical appliance indicates the rate at which the appliance converts electrical energy to another form of energy such as heat or light. One watt is equivalent to one joule of work per second. Both the electrical pressure (voltage) and the number of electrons flowing (current) determine the electrical power or rate of energy conversion. A 60 watt light globe uses 60 watts of electricity to produce light, and a typical electric kettle uses 2,400 watts to produce heat.

Kilowatts (kW) and Megawatts (MW)

Energy consumption is shown on Australian electricity bills in kWh units. A kilowatt (kW) is 1,000 watts, and a kilowatt hour is a power of one kilowatt used for one hour or 1,000 W used an hour.

Another important unit is the megawatt, which is equal to 1 million watts (1 MW). One megawatt hour is the power required to light ten thousand 100 W light globes for one hour.

Ten thousand 100 W light globes for one hour = 10,000 x 100 = 1,000,000 watts for an hour = 1 MWh.

The capacity of a power station will typically be defined in MW e.g 500 MW.

Gigawatts (GW)

A gigawatt is one thousand times bigger than a megawatt.

Hence 1 GW = 1,000 x 1,000,000 Watts = 1,000,000,000 Watts

Joules (J) and Megajoules (MJ)

The fundamental unit of energy is the joule, and its symbol is J. Electricity consumption is measured in watt seconds, or more commonly kilowatt hours (kWh), whereas gas usage is measured in joules per second, or more commonly in megajoules per hour.

A 1000 W electrical appliance running for 3 hours will consume 3000 Wh or 3 kWh.

A 40 MJ/hr gas appliance running for 3 hours will consume 120 MJ.

Thumbnail

Complete the content above before moving on.

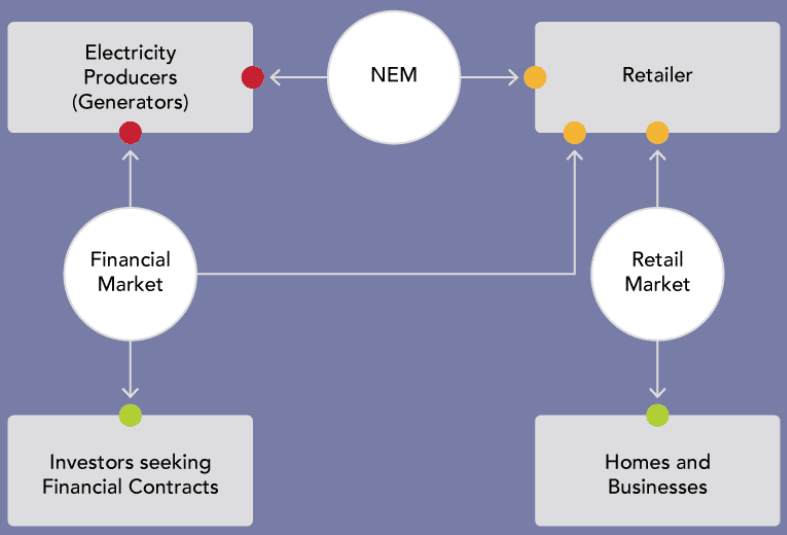
Electricity markets

The NEM supplies around 200 TWh (terawatt hours) of energy to businesses and households each year. There are three key markets that underpin this system: the National Electricity Market, the Retail Market, and the Financial Market.

The **National Electricity Market (NEM)** is a wholesale market involving exchanges between electricity producers (generators) and retailers (companies that purchase electricity from generators and then sell it to homes and businesses).

The **Retail Market** involves electricity retailers selling the energy they have purchased wholesale (via the NEM) to homes and businesses.

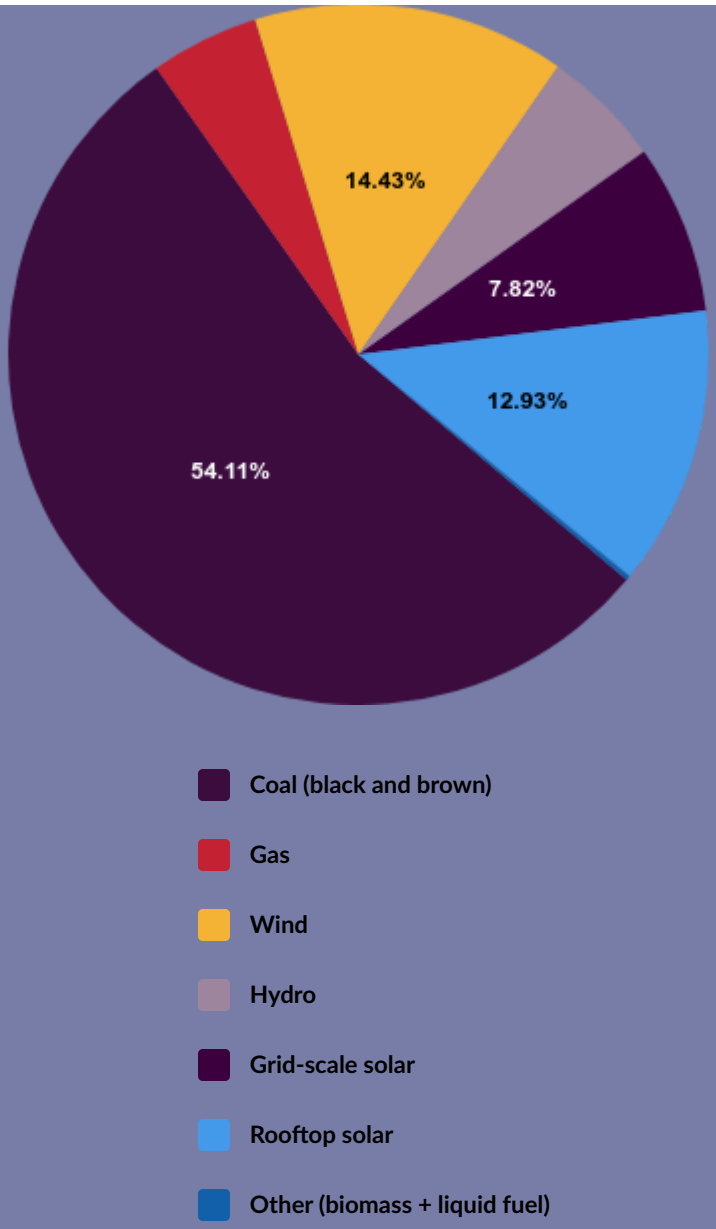
The **Financial Market** describes various contracts set up between electricity producers, retailers and investors, which act as insurance policies by reducing the significant risk of financial exposure faced by market participants due to electricity price volatility that can occur. These financial contracts may lock in a firm price for electricity that will be produced or consumed at a given time in the future.



The National Electricity Market (NEM)

The National Electricity Market (NEM) is a wholesale market that supplies electricity to retailers and end-users in Queensland, New South Wales, the Australian Capital Territory, Victoria, South Australia and Tasmania. Electricity bought and sold in the NEM is transported along the interconnected power system.

Annual generation by technology 2024-25



NEM operations today are based in five interconnected regions that largely follow state boundaries. The assets that comprise the NEM's infrastructure are owned and operated by either state governments or under private business arrangements.

CONTINUE

The Australian Energy Market Operator (AEMO)

The Australian Energy Market Operator (AEMO) was established to manage the NEM and gas markets from 1 July 2009. AEMO's core functions can be grouped into the following areas:

- Electricity Market - Power System and Market Operator
- Gas Markets Operator
- National Transmission Planner
- Energy Market Developer

AEMO held the role of Transmission Network Service Provider (TNSP) for planning in Victoria for many years. This responsibility transitioned to VicGrid on 1 November, 2025.

AEMO's primary responsibility is to balance the demand and supply of electricity in real time by dispatching the generation necessary to meet demand.

AEMO manages the NEM and the power system from two control centres in different states. Both centres operate around the clock and are equipped with identical communication and information technology systems. The entire NEM, or individual regions within it, can be operated from either or both centres. This arrangement ensures continuous supply despite the risks posed by natural disasters or other

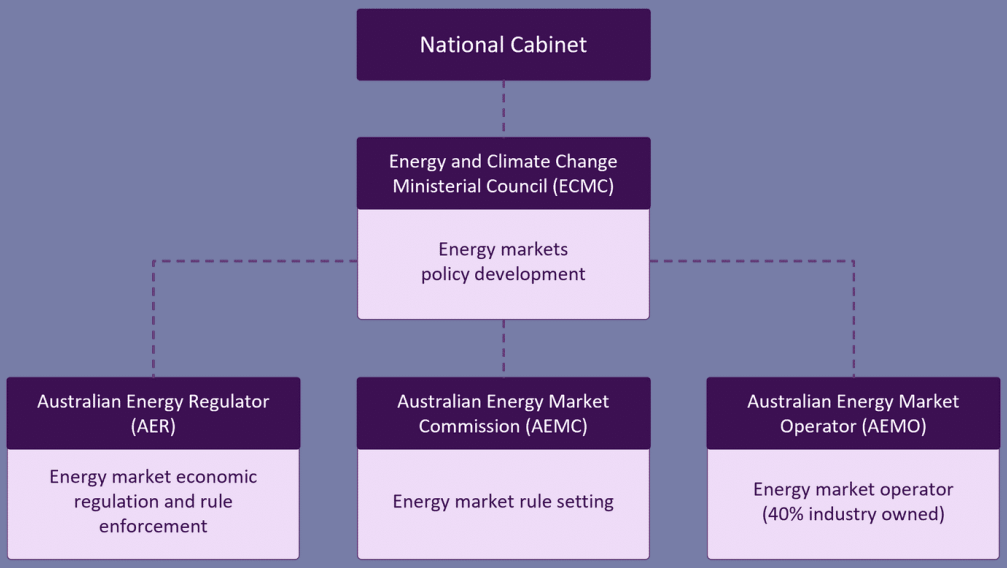
critical events, and provides AEMO with the flexibility to respond quickly to dramatic changes in the market or the power system.

AEMO's place in the energy industry

AEMO operates alongside the Australian Energy Regulator (AER) and the Australian Energy Market Commission (AEMC). The AER oversees economic regulation and compliance with the National Electricity Laws and National Electricity Rules, reports on generator bidding behaviour in the NEM, and regulates electricity transmission and distribution networks in the NEM. The AEMC makes the rules governing the regulation of the energy markets.

In September 2022, the Energy and Climate Change Ministerial Council (ECMC) was established within the streamlined model of Australia's federal relations architecture. This council replaces the former Energy National Cabinet Reform Committee, which was preceded by the COAG Energy Council.

These institutions have assumed many of the electricity regulatory arrangements previously managed by State government authorities.



AEMO's role within the NEM

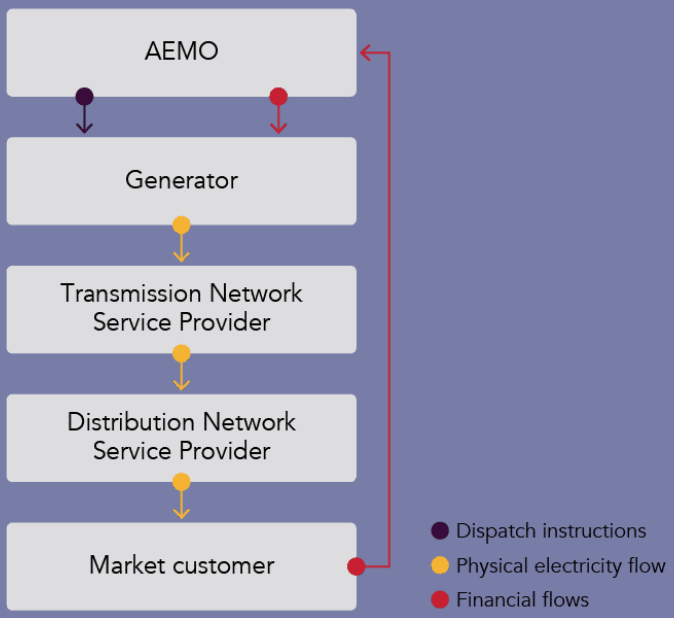
Operating the NEM involves conducting a sequence of activities to facilitate trade between the producers and wholesale consumers of electricity. These activities include establishing demand levels, receiving offers to supply from generators, scheduling generators, dispatching generators into production, calculating the spot price, measuring electricity use and financially settling the market.

AEMO's electricity market and power system operation responsibilities include:

- Managing the NEM
- Overseeing reliability and security of the NEM
- Ensuring supply reserve to meet reliability standards
- Directing generators to increase production during periods of supply shortfall

- Instructing load shedding if required to rebalance supply and demand and protect power system operations
- Implementing reserve trading to maintain supply and reliability levels through demand-side response
- National transmission planning
- Electricity emergency management
- Facilitating retail competition

Energy and Financial Flows



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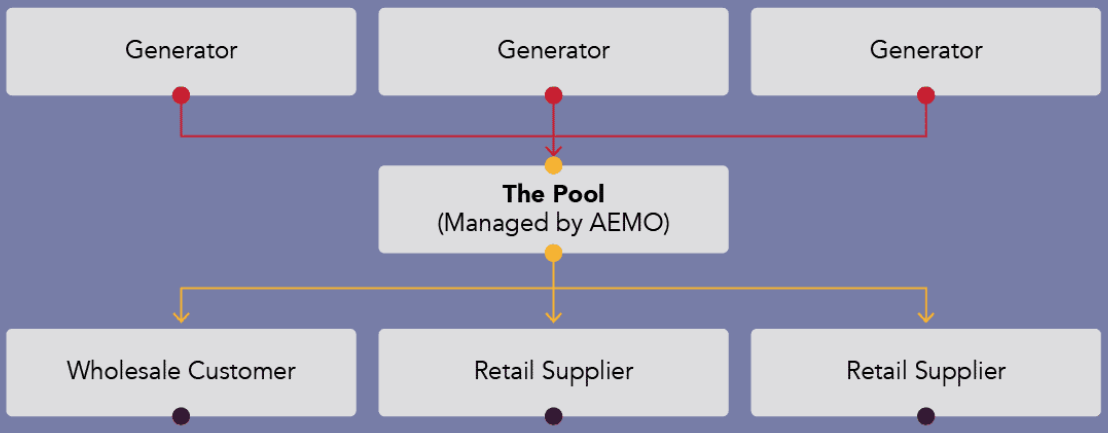
The NEM - a wholesale exchange

The National Electricity Market differs from other markets because electricity cannot be stored within the power system, and it is not possible to distinguish which generator produced the electricity consumed by a particular customer. Because of this, the NEM uses the concept of a pool where all the electricity output from generators is centrally pooled and scheduled to meet customer demand.



The two basic components of the pool are the centrally co-ordinated dispatch process and the spot market. Let's look at each of these in turn.

The centrally coordinated dispatch process

To enable AEMO's systems to facilitate supply, generators must submit, to AEMO, offers indicating the volume of electricity they are prepared to produce for a specified price.

There are three types of bids or offers to supply — daily bids, re-bids and default bids. Daily bids are submitted before 12:30pm on the day before supply is required. Generators may submit re-bids until approximately five minutes prior to dispatch. Re-bids allow generators to change the volume of electricity stated in the original offer, but they cannot change the offer price.

From the bids submitted, AEMO's market systems determine which generators are required to satisfy demand, at what time, and their production levels in a process called scheduling. AEMO stacks the bids in order of rising price, and then schedules them for dispatch into production. Using the rising price stack means that more expensive generators are only scheduled into production as total demand for electricity increases.

At times, the technical capacity of the transmission network may determine which generators are scheduled to meet demand. In these situations, generators may be scheduled out of price order so that demand in a particular area can be satisfied.

The spot market

Wholesale trading in electricity is conducted as a spot market where supply and demand are instantaneously matched in real-time through a centrally-coordinated dispatch process.

Generators offer to supply the market with specific amounts of electricity at particular prices. Offers to produce electricity are received by AEMO and stacked in ascending price order for each dispatch period, which is every 5 minutes. AEMO then

progressively schedules generators into production to meet the forecast demand, starting with the least-cost option. Once the demand forecast has been matched with sufficient generation, AEMO calculates the spot price for that 5-minute dispatch interval.

AEMO uses this spot price as the basis for the settlement of financial transactions for all electricity traded in the NEM.

The National Electricity Rules set a maximum spot price, also known as a Market Price Cap. This is the maximum price at which generators can bid, and is also the price automatically triggered when AEMO directs network service providers to interrupt customer supply in order to keep supply and demand in balance.

AEMO is responsible for calculating the financial liabilities of market participants on a daily basis and for settling transactions weekly for all trade in the NEM. This means that AEMO collects from Market Customers all money due for electricity purchased from the pool, and pays generators for the electricity they produced. The spot price is the basis for all of these transactions.

Prior to 1 October 2021, the NEM was settled in 30-minute intervals. A dispatch price was determined every five minutes, and the six dispatch prices within a half-hour period were averaged to determine the spot price for each NEM region.

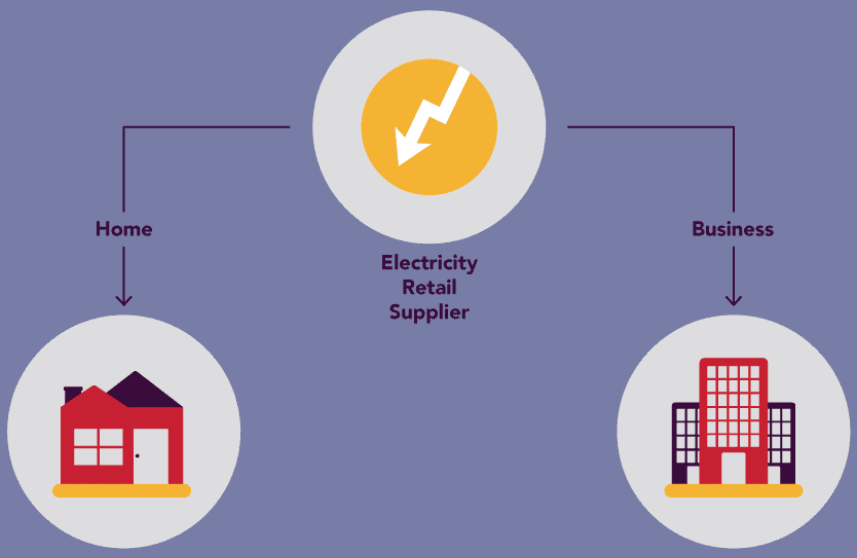
The implementation of Five-Minute Settlements on 1 October 2021 saw the alignment of the dispatch price and the spot price at 5-minute intervals.

CONTINUE

The retail market

The Retail Market involves electricity retailers and end-users (homes and businesses). End-users purchase power from their electricity retail supplier, who in turn purchases power from the NEM pool.

The amount the end-user pays for their electricity is made up of several charges including the cost of the electricity transmission and distribution networks. The cost of the electricity itself is based on the market spot price.



Electricity is purchased by the retailer and transported to home and businesses.

Electricity is purchased by the retailer from the NEM (the pool). It is transported via the power system to our homes and businesses.

The cost of this electricity is determined by the spot price. This is constantly changing so the retail supplier takes on the risk of this volatile market.

The retail supplier signs up end-users to their customer base, usually on a contract. These contracts need to cover potential fluctuations in the spot price, otherwise the retail supplier loses money.

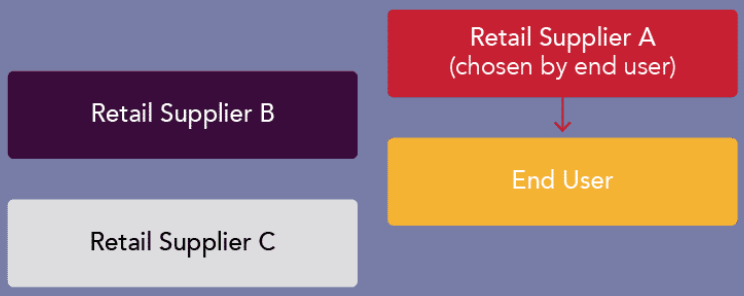
The final bill received by end-users takes into account the cost of the transmission and distribution network.

Full retail competition

Since the NEM commenced, electricity consumers have progressively gained the right to choose their own supplier. This right to choose is known as Full Retail Competition. Delivering full retail competition has required new information

technology systems to process customer transfers between registered retailers in the NEM. These systems are operated by AEMO, and contain one of the largest metering databases in the world. They accept data from a variety of electricity meter types and can process information from up to 11 million meters.

Retailers are billed on the total electricity consumed at a particular network connection point; if a customer changes their electricity retailer, this is recorded in AEMO's database so that all subsequent consumption is allocated to the new retailer.



Meter readings for this end-user are now charged to Electricity Retailer A, who has to settle with AEMO each week. AEMO then pays the generators who created the electricity.

The retailer makes money by then billing the end-user.

CONTINUE

The financial market

Fluctuating spot prices for electricity create substantial risk for parties who trade in the NEM's spot market. When the price is low, generators may not be fully compensated for the power they supply. On the other hand, retail customers will not want to pay for energy which is priced too high. The Financial Market uses hedge contracts to mitigate these risks by benefiting both parties in times of price volatility.

Hedge contracts are typically agreements between generators and market customers. These contracts operate independently of both the NEM and AEMO's administration. The details of hedge contracts are not factored into the balancing of supply and demand, and are not regulated under the National Electricity Rules.

Hedge contracts are financial instruments that market participants can use to manage the financial risk that results from potential volatility of the spot price.

For example, the most common type of contract, a *Swap* contract, exists where two parties agree to exchange cash against a spot price outcome in the NEM so that a defined quantity of electricity over a nominated period is effectively valued at an agreed strike price. Under such an agreement, generators pay market customers the difference when the spot price is above the strike price. When the spot price is below the strike price, market customers pay generators the difference.

Spot Market + Price Volatility

= Financial Markets (to manage risk)

Complete the content above before moving on.

The history of the electricity industry and the NEM

Pre 1990

Prior to 1990, each state had its own vertically integrated state-owned authority responsible for the generation, transmission and distribution of electricity.

Investment in new generation was largely driven by each state government and the state electricity authorities. Electricity prices were set by state government regulation to cover industry costs plus any returns required by the government.

1990

In 1990 an Industry Commission Inquiry initiated a national electricity industry reform process. The Inquiry revealed that major improvements in national productive output could be achieved by:

- Separating generation, transmission, distribution and retail supply
- Introducing competition into the electricity market
- Extending the States' interconnected power systems

The goals of the reform were to:

- Increase competition in the industry
- Promote greater choice for electricity consumers
- Provide timely information to market participants regarding appropriate timing for investing in more generation and network developments

1996

In 1996 the National Electricity Law was adopted by five jurisdictions when they signed the National Electricity Market Legislation. Midnight on 13 December 1998 marks the commencement of the NEM. Electricity industry reform continued through the 1990s and into the 21st century.

2005

In 2005, a new regulatory regime was formed comprising an amended National Electricity Law, the National Electricity Regulations, the National Electricity Rules, the Australian Energy Market Commission (or AEMC) and the Australian Energy Regulator (or AER).

2009

AEMO - the Australian Energy Market Operator — commenced operation in July 2009. AEMO operates the wholesale electricity market (the NEM), develops the NEM for improved efficiency and coordinates planning of the interconnected power system. AEMO's primary responsibility is to maintain system security by balancing electricity demand and supply.

With respect to the NEM, AEMO has two core roles:

- Power System Operator
- Market Operator

The market requirements determine how the power system is operated.

The Present, and into the future

Electricity industry reforms continue to this day.

There is now one centralised point of sale for participating regions (NSW, QLD, VIC, SA, TAS), one centralised payment system and one centralised dispatch process.

There is also competition in the marketplace, with sellers competing against each other to provide the cheapest electricity to consumers.

The power system is rapidly transitioning away from carbon-emitting generation assets towards renewable energy sources.

CONTINUE

Module summary

Congratulations! You've now completed **Module 1 - Introduction to the National Electricity Market**.

This module has introduced you to Australia's electricity system, and has defined the basic units of electrical measurement. You should also now be familiar with the definition and role of AEMO, and have an understanding of the National Electricity Market. Finally, this module introduced the Retail and Financial Markets before reviewing the history of the National Electricity Market and the electricity industry in general.

You may now review any section of this module via the main menu on your left, or exit this module and proceed to **Module 2 - NEM Rules and Regulations.**

Module 2 - NEM Rules and Regulations

- Module introduction
- Regulatory overview
- Governing bodies
- Registered participant roles and responsibilities
- Rules enforcement
- Power of Choice reforms
- Module summary

Module introduction

Welcome to **Module 2 - National Electricity Market (NEM) Rules and Regulations**.

This module addresses the following learning objectives:

- To explain the regulatory instruments applicable to the NEM (Law, Regulations and Rules)
- To introduce the NEM's governing bodies and their roles
- To provide an overview of the Rules and the Rules change process
- To detail the Registered Participant categories, and their associated obligations under the Rules and requirements for compliance

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

Regulatory overview

The primary pieces of legislation making up the NEM regulatory framework are the National Electricity Law (NEL) and the National Electricity Rules (NER).

Let's examine each of these more closely.

The National Electricity Law

The NEL is the law set out in the Schedule to the National Electricity (South Australia) Act 1996, (an Act passed by the South Australian Parliament), which has been incorporated by statute in each of the other participating jurisdictions. Occasionally there are variations to cater for specific arrangements or conditions in each participating jurisdiction.

Participating jurisdictions in the NEM are South Australia, Queensland, New South Wales, the ACT, Victoria, Tasmania and the Commonwealth.

Briefly, the NEL:

- Defines the NEM
- Sets out a single National Electricity Objective (NEO)
- Confirms that the Energy and Climate Change Ministerial Council is responsible for policy leadership for the NEM
- Outlines the roles and powers of the governing bodies (AEMC, AER and AEMO)
- Describes the process for making and amending the National Electricity Rules

The National Electricity Law is the highest legislative authority in the NEM.

The National Electricity Rules

The National Electricity Rules (known simply as 'the Rules') govern the day-to-day operation of the NEM, and are made and amended under the NEL. The AEMC is responsible for initiating proposed rule changes to the Rules.

The NEL states that the Rules have the force of law, so any person to whom they are expressed to apply must comply.

The Rules cover a variety of subject areas, including:

- NEM participation/ registration
- Regional structure
- Market rules operation, such as the operation of the central dispatch process and spot market
- Power system security
- Network connection, planning and expansion
- Electricity connection for retail customers
- Economic regulation of distribution and transmission services
- Retail Markets
- Metering
- Administrative functions, such as dispute resolution
- Jurisdictional derogations (ie rules that apply only in a particular jurisdiction, or to particular participants)

Other

Other relevant legislation includes the National Electricity Regulations (South Australia), National Energy Retail Law (NERL), National Energy Retail Regulations and National Energy

Retail Rules. The last three, which constitute the National Energy Customer Framework (NECF), regulate the supply of both electricity and gas to end-use customers, including the authorisation or exemption of energy retailers, relationships between distributors and retailers and arrangements for retailer failure.

As of January 2019, the NECF applies in all NEM participating jurisdictions except Victoria, which has retained its own electricity legislation dealing with those matters.

CONTINUE

Governing bodies

The NEM has five key governing bodies:

- Energy and Climate Change Ministerial Council (ECMC)
- Australian Energy Market Commission (AEMC)
- Australian Energy Regulator (AER)
- Australian Energy Market Operator (AEMO)
- Energy Advisory Panel (EAP)

The governance arrangements separate:

- High level policy direction — the ECMC's responsibility
- Rule-making and energy market development — the AEMC's responsibility
- Economic regulation and Rules enforcement —the AER's responsibility
- Operation of the power system and the NEM — AEMO's responsibility
- Oversight and advice to the ECMC on energy security, reliability and long-term planning - EAP's responsibility

Regardless of the function being performed, each of these bodies must adhere to the National Electricity Objective, which is defined as:

> *..to promote efficient investment in, and efficient operation and use of, electricity services for the long-term interest of consumers of electricity with respect to:*
>
> *(a) price, quality, safety, reliability and security of supply of electricity; and*
>
> *(b) the reliability, safety and security of the national electricity system.*

National Electricity Objective

Let's examine each of these governing bodies more closely.

Energy and Climate Change Ministerial Council (ECMC)

The ECMC provides high level policy direction for the energy market, which includes:

- Directing the AEMC to carry out reviews and report back on market development issues
- Initiating proposals to change the Rules
- Publishing statements of policy principles on matters relevant to the AEMC as it performs its functions

The ECMC is not involved in the day-to-day operation of the NEM or the conduct of the NEM regulators.

Australian Energy Market Commission (AEMC)

The AEMC is established under the Australian Energy Market Commission Establishment Act 2004 of South Australia.

Its principal functions include:

- Rule-making, market development and other functions conferred on it, such as under the Rules
- Provision of advice, as requested, to the ECMC
- Conducting reviews into the operation and effectiveness of the Rules

Its role is not limited to the NEM, but also includes gas markets.

The AEMC must note any policy principles when making a Rule change or conducting a Rule review.

In addition, the Reliability Panel has been established by the AEMC under the NEL and in accordance with the Rules. Among other matters, the Panel monitors, reviews and reports on, in accordance with the Rules, the safety, security and reliability of the national electricity system.

Australian Energy Regulator (AER)

The AER regulates electricity markets and networks. Its functions include:

- setting the prices charged by distribution and transmission networks via a periodic revenue determination process
- monitoring wholesale markets to ensure compliance by participants — with the legislation and rules, and to take enforcement action where necessary
- regulating retail markets (other than in Victoria)
- assisting the ACCC with energy-related issues arising under the Competition and Consumer Act, including enforcement, mergers and authorisations

Only the AER can bring proceedings against another person for a breach of the NEL or Rules.

Australian Energy Market Operator (AEMO)

AEMO's key functions, set out in the NEL, include:

- Operating and administering the wholesale electricity markets and the power system in the NEM
- Promoting the development and improving the effectiveness of NEM operation and administration
- Registering participants
- Maintaining and improving power system security
- Facilitating retail customer transfer, metering and retail competition
- Additional advisory functions or declared network functions (as requested by a participating jurisdiction)
- Any other functions stated in the Rules

The Wholesale Electricity Market (WEM) is operated under Western Australian energy legislation.

AEMO is the National Transmission planner, and publishes the Integrated System Plan (ISP). AEMO is required to perform its functions efficiently and on a full cost recovery (i.e., not-for-profit) basis.

AEMO was established to manage the NEM and gas markets from 1 July 2009. It carries out functions previously undertaken by NEMMCO, VENCorp, ESIPC South Australia, REMCo, Gas Market Company and Gas Retail Market Operator.

Energy Advisory Panel (EAP)

The Energy Advisory Panel commenced in July 2023, replacing the Energy Security Board (ESB) which had been operating since 2017.

The EAP's role is to coordinate market bodies' advice to government on issues relating to the security, reliability, and affordability of Australia's east coast energy system.

The EAP comprises the heads of the three energy market bodies, and the Energy Commissioner of the Australian Competition and Consumer Commission (ACCC) as an observer. Expanding the membership to include ACCC is designed to enable better consideration of issues relating to gas supply, retail markets and consumer protections.

Activity: Governing bodies and legislation

Test your knowledge of the NEM's governing bodies and key legislation by matching the institution to its description.

Australian Energy Market Operator (AEMO)	The highest authority in the NEM
Australian Energy Market Commission (AEMC)	Principal functions include amending and making Rules, market development and review
Energy and Climate Change Ministerial Council (ECMC)	High-level policy direction to the energy market
Australian Energy Regulator (AER)	Responsible for Rules enforcement, market monitoring, and economic regulatory functions

National Electricity Rules (the Rules)	Operates and administers the market and system on a cost recovery basis
National Electricity Law (NEL)	Govern the operation of the NEM, and must be followed by all participants

SUBMIT

Other sources of regulation

The NEL and the Rules provide for a uniform framework for the operation and management of the NEM. However some matters fall outside this framework, and are subject to separate regulation.

Australian Securities and Investment Commission (ASIC)

ASIC regulates the ability of NEM participants to trade electricity derivatives and hedge contracts under the Corporations Act 2001 (Cth), including through licensing, market conduct and financial market supervision. These activities are not regulated under the National Electricity Rules, as they fall outside the physical electricity market.

State and Territory laws

Each participating State and Territory:

- Determines the extent of retail competition in its jurisdiction, for instance, whether a default regulated retail tariff applies
- Can develop its own regulatory provisions via licence conditions, codes etc...
- Enforces safety and environmental standards
- Can set generation licence or approval conditions

On the map below, click each state to view the details of the bodies that are relevant to the performance of these residual functions.

New South Wales

ACT

- Independent Competition and Regulatory Commission (ICRC)
- Utilities Technical Regulation (UTR)

NSW

- Independent Pricing and Regulatory Tribunal of NSW (IPART)
- NSW planning, environmental and energy policy functions are administered by relevant NSW Government departments, including the Department of Planning, Housing and Infrastructure and the Department of Climate Change, Energy, the Environment and Water

Tasmania

- Office of the Tasmanian Economic Regulator (OTTER)

South Australia

- Essential Services Commission of South Australia (ESCOSA)
- Office of the Technical Regulator

Victoria

- Essential Services Commission (ESC)
- Energy Safe Victoria (ESV)

Queensland

- Department of Energy and Public Works
- Queensland Competition Authority (QCA)

CONTINUE

Registered participant roles and responsibilities

Chapter 2 of the NER contains the detail of the registration categories, process eligibility and conditions, including the payment of participant fees.

Registered participants include: generators, network service providers, customers, small resource aggregators, metering coordinators, market ancillary service providers, special participants, traders, reallocators and intending participants. The ways energy and services are supplied to and from the grid are changing rapidly and this may mean more changes to registered participant categories in the near future.

A participant can be registered in more than one registration category. Many registration categories and associated services also have multiple sub-classifications, depending on the type of activity or facility and how it interacts with the market. For example:

- Generators will be classified separately for each generating system they operate, depending on whether and how that particular facility is scheduled and dispatched by AEMO in the NEM, and whether any electricity it produces is sold on the wholesale market or bought and consumed locally.
- Network service providers will be classified as transmission or distribution, based on the voltages at which their network operates. Some transmission assets can also be classified as providing market (scheduled) network services, or dedicated connection services.

- A provider of market ancillary services (for frequency control) can be the registered market customer (typically a retailer) for a specific load, the market generator for a specific generation facility, or a market ancillary service provider who can aggregate loads and/or other small resources. In each case, the identified facilities must be capable of providing a specified level of frequency response.

The term 'market participants' includes participants registered in the specified categories of generators, customers, network service providers and aggregators who buy or sell wholesale energy, or provide system support services (called ancillary services) within the NEM.

The NER specifies rights and obligations that may apply generally to all registered participants, or to particular categories and sub-classifications of participants.

AEMO's registration guides contain further information for prospective NEM participants on registration categories, classifications and conditions (see AEMO's website for further details).

Registration categories

The Rules define specific rights and obligations for each type of registered participant.

Let's examine these in more detail.

Generators —

Any person who owns, controls or operates a generating system connected to a transmission or distribution network must register as a generator or an Integrated Resource Provider, except where they meet the exemption criteria.

Each of the registered generating units must be classified as market or non-market depending if the sent-out electricity is sold through the spot market and classified as scheduled, non-scheduled or semi-scheduled depending on how the generating unit participates in central dispatch.

Integrated Resource Provider, Small Resource Aggregator (IRP, SRA)

An IRP, SRA can supply electricity aggregated from one or more small production units, which are connected to a distribution or transmission network at a small resource connection point. A small production unit is owned, controlled and/or operated by a person who AEMO has exempted from the requirement to register as a generator or an integrated resource provider in respect of the production unit. An IRP SRA is a market participant.

Customers

A customer is a registered participant that purchases electricity supplied through a transmission or distribution system and consumed at a market connection point. Electricity may also be generated from small production units at these market connection points.

Retailers and end users who buy (or sell) electricity in the spot market must be registered as Market Customers or Integrated Resource Providers.

Network Service Providers (NSP) - including Special Participant

A person who owns, operates or controls a transmission or distribution system must register with AEMO as an NSP. The Australian Energy Regulator may exempt a person from registering as an NSP if the person meets specific criteria or if they are appointing an intermediary.

When registering, NSPs must categorise each of their network systems as either a transmission system or distribution system. NSPs either earn revenue from a regulated income or from the spot market. The latter must register as market NSPs and classify their services as scheduled network services. Market NSPs must participate in AEMO's central dispatch process.

Traders

Anyone who wants to take part in a Settlements Residue Auction and is not already registered as a customer or generator, must register as a trader.

Reallocators

Anyone that wishes to participate in a reallocation transaction must register as a reallocator. A reallocator must be a “wholesale client” as defined in the Corporations Act 2001 (Cth).

A reallocation transaction is undertaken with the consent of two market participants and AEMO. AEMO credits one market participant with a positive trading amount for a trading interval, and debits a matching negative trading amount to the other market participant for the same trading interval. These may be specified dollar amounts, or calculated by reference to a specified quantity of energy and price. These amounts are typically underpinned by a contract between the credit and debit parties, but AEMO is not a party to that contract.

Demand Response Service Provider

A Demand Response Service Provider is a registered participant that offers demand response into the National Electricity Market (NEM). DRSPs aggregate and dispatch reductions in electricity consumption from customers, which can be used to respond to changes in supply and demand. DRSPs participate in the market by providing demand-side capacity that can be scheduled in a similar way to generation.

Intending Participant

A person may apply for registration as an intending participant if they can reasonably satisfy AEMO that they intend to commence activities that require them to be a registered participant.

Integrated Resource Provider (IRP)

Any person who owns, controls or operates an integrated resource system connected to a transmission or distribution network must register as an integrated resource provider, except where they meet the exemption criteria.
Each of the integrated resource provider's registered production units (bidirectional units or generating units) must be classified as market or non-market depending if the sent-out electricity is sold through the spot market and classified as scheduled, non-scheduled or semi-scheduled depending on how the production units participates in central dispatch.
A person may also register in the category of Integrated Resource Provider to undertake the activities of a Generator, Small Resource Aggregator, and Market Customer.

CONTINUE

Rules enforcement

The Australian Energy Regulator monitors Rules compliance. The AER can undertake:

Administrative action

If the breach is regarded by the AER as relatively minor, the AER may seek commitment from a participant to take administrative action to remedy the breach such as implementing a compliance program, improving internal procedures, or providing staff training.

Statutory action

This could take the form of the AER issuing an infringement notice for breach of a civil penalty provision (requiring payment of a financial penalty).

Alternatively, the AER could institute court proceedings where it has reason to believe the participant has breached a civil penalty provision. The AER could seek a number of remedies, such as an injunction, restraining order or other court direction to remedy the breach.

Court proceedings cannot be issued if infringement notice proceedings are underway. Payment of an infringement notice means no proceedings may be taken for the alleged contravention. Paying an infringement notice is not considered to be an admission of liability.

For further information, see also the AER's 'Compliance and Enforcement Statement of Approach' at www.aer.gov.au.

Finally, the NEL deems officers (i.e. directors and managers) of a participant who knowingly authorise or permit a breach of the NEC Rules or Regulations, to be in breach as well as the body corporate, and may issue proceedings against that officer (regardless of whether or not proceedings are brought against the body corporate). Similarly, if an officer of a body corporate is found to be in breach, the body corporate will also be deemed to be in breach.

Example | Rules enforcement

The AER takes federal court action against companies when they breach the rules. One company that breached clause 4.9.8(a) nine different times in 2012-13 incurred a $400K penalty and was required to submit to independent compliance assessment and enforceable undertakings about automatic generation control, as well as contributing to the AER's court costs.

The rule change process

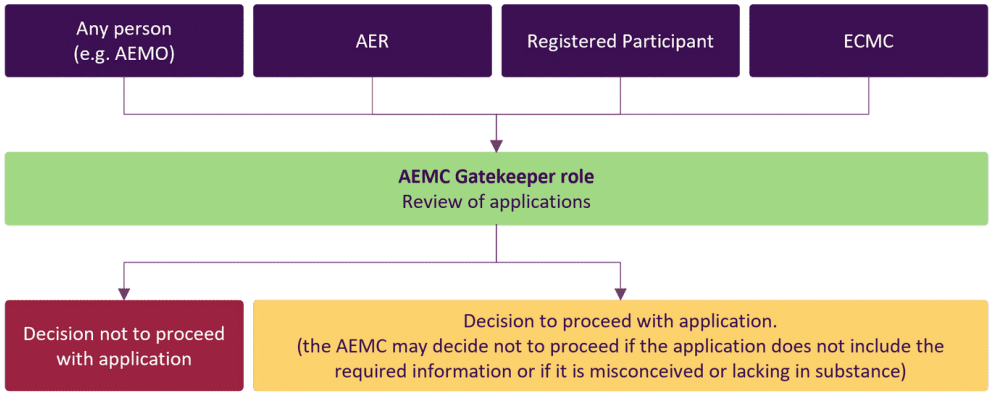
The NEL allows any interested person (including the Reliability Panel and the ECMC) to initiate a Rule change. The Regulations clarify how this process works. The AEMC cannot initiate a Rule change, except where the change is to correct a minor error, or is a non-material change.

There are four stages to the Rule change process:

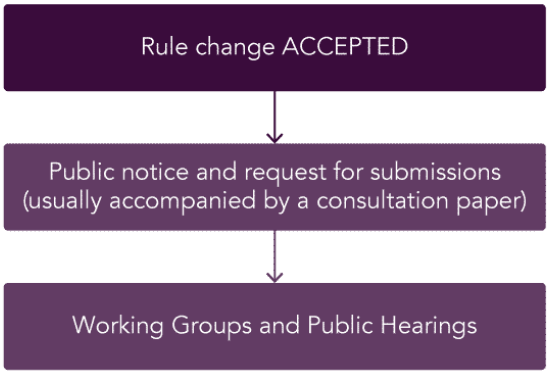
1. Initiation and preliminary assessment of a Rule change proposal
2. Publication and first round of consultation on a proposed Rule change
3. Draft Rule determination and second round of consultation
4. Final determination and amendment of Rule

Explore each of these stages below:

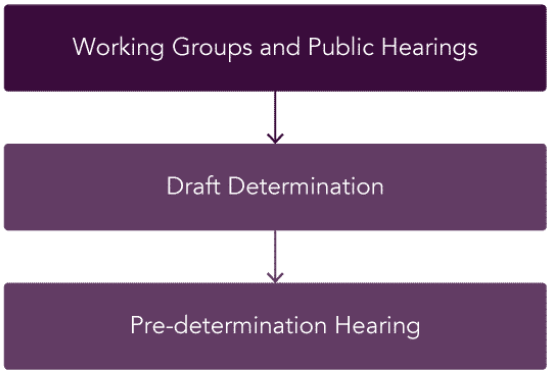
Stage one



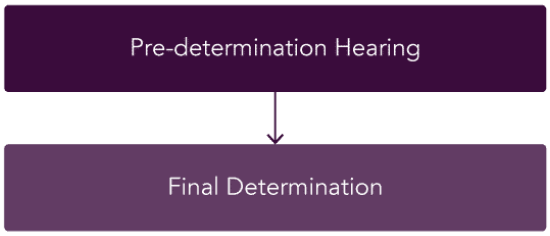
Stage two


Rule change ACCEPTED
Public notice and request for submissions (usually accompanied by a consultation paper)
Working Groups and Public Hearings

Stage three


Working Groups and Public Hearings
Draft Determination
Pre-determination Hearing

Stage four



Rule consultation procedures

The only way for stakeholders to influence Rule change decisions is by participating in the various consultations that must be undertaken by the "consulting party" as outlined in the Rules. (While the "consulting party" is generally AEMO or the AEMC, it can also be a Jurisdictional Regulator, Network Service Provider or the AER.).

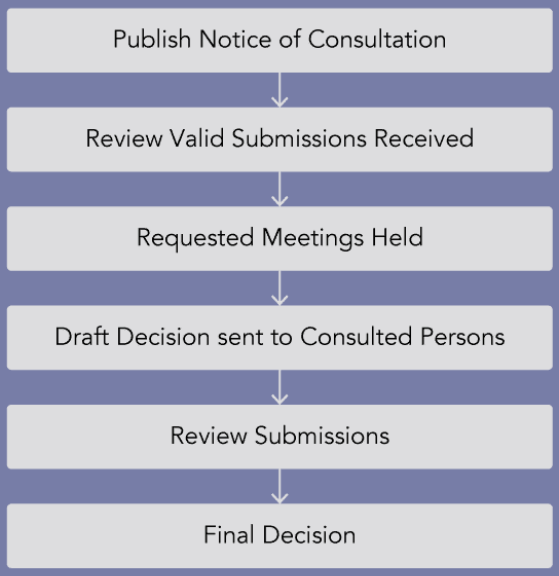
A significant failure to comply with the Rules consultation procedure could cause:

- Non-compliance with the Rules
- Grounds to challenge a decision (irrespective of its merits)
- Decisions being deemed invalidly made

Decisions made by AEMO under the Rules consultation procedure are subject to review through:

- The dispute resolution procedure (in rule 8.2 of the Rules)
- Judicial review in the court system

The Rules consultation process comprises six stages, as illustrated below.



Disputes

The NER sets out a dispute resolution regime which provides a staged negotiation, mediation and expert determination process for registered participants or AEMO to resolve disputes relating to certain matters arising from Rules processes and obligations. The overall NEM dispute resolution process is administered by a dispute resolution adviser appointed by the AER (Adviser). The Adviser maintains a web page on the AER's website.

All registered participants, and AEMO, must have a dispute management system (DMS) that satisfies the criteria outlined in the Rules. This includes nominating a DMS contact as the first point of contact for dispute notification. The Adviser maintains a list of notified DMS contacts, and is responsible for appointing a dispute resolution panel to resolve a dispute if it progresses to that stage.

CONTINUE

Power of Choice reforms

Technological developments and changing consumer behaviour means that the electricity market is facing a period of unprecedented change.

Substantial reforms were implemented following the AEMC's Power of Choice review. The objective of the reforms was to make it easier for consumers to engage in the energy market including:

- Providing cost reflective pricing enabling consumers to choose when to use energy based on price
- Greater competition in metering and related services:
    - By allowing metering services to be provided by parties other than network service providers

The AEMC's website contains more information on the current status of various reforms (https://www.aemc.gov.au/our-work/market-reviews-and-advice).

CONTINUE

Module summary

Congratulations! You've now completed **Module 2 - National Electricity Market (NEM) Rules and Regulations.**

This module has introduced you to NEM governing bodies, and the legislation and rules that empower them. You should also now be familiar with the terms 'National Electricity Law' (NEL) and 'National Electricity Rules' (the Rules), and understand the roles and obligations of various registered participants. Finally, this module introduced the Rules change and dispute resolution processes.

You may now review any section of this module via the main menu, or exit this module by pressing the button below and proceed to **Module 3 - The Network**.

Module 3 - The Network

- Module introduction
- The network
- Economic regulation and pricing
- Transmission loss factors
- Power system security
- Network planning
- Module summary

Module introduction

This module addresses the following learning objectives:

- To describe how the transmission of electricity takes place
- To provide information on network ownership
- To introduce network security concepts
- To discuss network planning
- To provide an explanation of network losses and their impact on the market
- To explain how the network is regulated in the NEM
- To discuss network revenue and pricing arrangements

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

The network

The network - a definition

The network refers to the infrastructure that transmits electricity from generation sources to consumers. It comprises the poles and wires, as well as the apparatus, equipment, plant and buildings used to transmit electricity to customers, excluding any connection assets.

The overall system is made up of two main sections: the transmission system and the distribution system. Transmission is the high voltage, large power lines that can be considered similar to a freeway. Distribution is lower voltage and delivers power into our suburbs, like smaller roads.

While the product itself (electricity) is a free market commodity, its transportation and delivery is still considered a "natural monopoly" so it remains regulated. This prevents companies from building their own networks, clogging the landscape with private power poles and distribution lines.

Why have a network?

While domestic solar panels are widespread, not everyone can afford them, or the backup required to keep the lights on 24 hours a day. A shared network that allows the distribution of bulk generation is the most economically efficient and reliable way to serve customers.

The network fulfils three key roles:

1. It transmits electricity from generators to consumers
2. It facilitates competition by separating electricity generation and transmission
3. It ensures a secure reliable supply of electricity to consumers

The interconnected network

The NEM comprises five interconnected regions (Queensland, New South Wales, Victoria, Tasmania and South Australia). Western Australia and the Northern Territory have electrically separated grids. The networks in each NEM region were originally built as standalone systems by State governments. From the 1990s, the regions began to connect with each other. This means that electricity can now be moved from a region where it is cheaper, into a region where it is more expensive, allowing electricity to flow between regions and facilitating a more competitive market. Each NEM region has a designated 'regional reference node' (RRN) where the regional electricity spot price is set, and all regions contain major generation and demand centres.

In general the RRN is a nominated major transmission substation located at, or close to, the largest load centre within the region. The regional price is calculated at this node and inter-regional energy transfers are also modelled between these regional reference nodes.

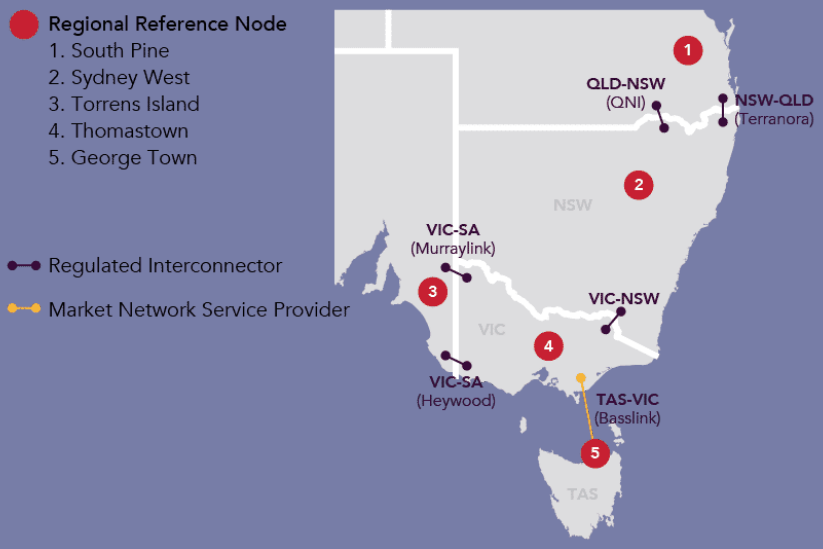
Interconnectors

The high-voltage transmission lines that transport electricity between adjacent NEM regions are called interconnectors. These are used to import electricity into a region when:

- Local generators cannot meet that region's demand, or
- The price of electricity in an adjoining region is low enough to displace local supply

Benefits of connecting the regional networks are:

- Increased supply reliability
- Inter-regional trade
- More efficient sharing of generation resources



Network Service Providers

Network service providers (NSPs) own, operate and control the network. There are two types:

- Transmission network service providers (TNSPs), who own transmission assets. There is only one of these in each NEM region
- Distribution network service providers (DNSPs), who own distribution assets. In some states there are several of these and in other states just one

Under Australian electricity laws, network owners are carefully regulated by the Australian Energy Regulator (AER), which is an arm of the Australian Competition and Consumer Commission (ACCC). Network owners are unable to own generation assets, or sell energy like a retailer. This 'ring fencing' was set up to ensure that network owners can't exploit their position as a natural monopoly.

Click through the below map to view the TNSPs and DNSPs for each State.

Queensland

TNSPs

- Powerlink Queensland

DNSPs

- Ergon Energy
- Energex

NSW and ACT

TNSPs

- TransGrid

DNSPs

- Ausgrid
- Essential Energy
- Endeavour Energy
- Evoenergy (ACT)

Victoria

TNSPs

- AusNet Services (asset owner / operator)
- VicGrid (planning function)

DNSPs

- Jemena
- CitiPower
- Powercor
- AusNet Services
- United Energy

South Australia

TNSPs

- ElectraNet

DNSPs

- SA Power Networks

Tasmania

TNSP and DNSP

- TasNetworks
- From 1 July 2026 Basslink has transitioned to a regulated TNSP, operating as an interconnector between Tasmania and the mainland within the regulated transmission framework.

Basslink differs from typical TNSPs in that it functions as an interconnector between regions rather than a transmission network within a single jurisdiction, and its transition to a regulated model involves unique cross-regional coordination and cost allocation arrangements. Further information is available in guidance published by AEMO and the AER on the Basslink conversion and regulatory framework.

Network access

The National Electricity Rules (the Rules) outline access arrangements for generators and market network service providers who request access to the network. Any party

that satisfies eligibility criteria for participation in the NEM can connect to power system infrastructure.

NSPs and registered participants enter commercial agreements about the terms and conditions of their connection to a network and the provision of network services. The technical terms and conditions of connection to the network are prescribed in the National Electricity Rules. Terms and conditions in the Rules have been set to enable AEMO to meet its obligation to operate the power system securely and reliably.

CONTINUE

Economic regulation and pricing

Regulators

Because providing electricity transmission and distribution services is considered a natural monopoly, pricing for these services is regulated by the Australian Energy Regulator (AER).

In addition to this, TNSP and DNSP licences are also regulated. This means the AER may impose explicit requirements and service standards in addition to those in the Rules.

What can network service providers charge?

NSPs are entitled to recover efficient operating costs and earn a risk-adjusted commercial rate of return on capital required to provide services.

The AER reviews NSP revenue requirements periodically (usually every five years) and sets the maximum allowable revenue (MAR or "revenue cap") an NSP can collect from customers for each year of that regulatory period. This process is known as a "revenue reset" (or sometimes a "price review" for DNSPs).

Paying for the network

Think of the power network as a public road enabling electricity to travel from a generator to your home. Maintaining this 'road' attracts certain costs, and these are

factored into everyone's electricity bill. The proportion of network costs included in a bill depends largely on the customer — e.g. a domestic consumer will have different needs to a wholesale customer (such as an industrial plant or a mine). Use of system charges are broken into transmission use of system charges (or TUOS) and distribution use of system charges (or DUOS). Note, that breakdowns are different in each NEM region.

TRANSMISSION USE OF SYSTEM | DISTRIBUTION USE OF SYSTEM

Market participants at a connection point pay TUOS charges to the TNSP for the network usage at that point. TUOS charges recover the asset, operating and maintenance costs of the transmission network.

TRANSMISSION USE OF SYSTEM | DISTRIBUTION USE OF SYSTEM

Market participants pay DUOS charges to distribution network service providers.

CONTINUE

Transmission loss factors

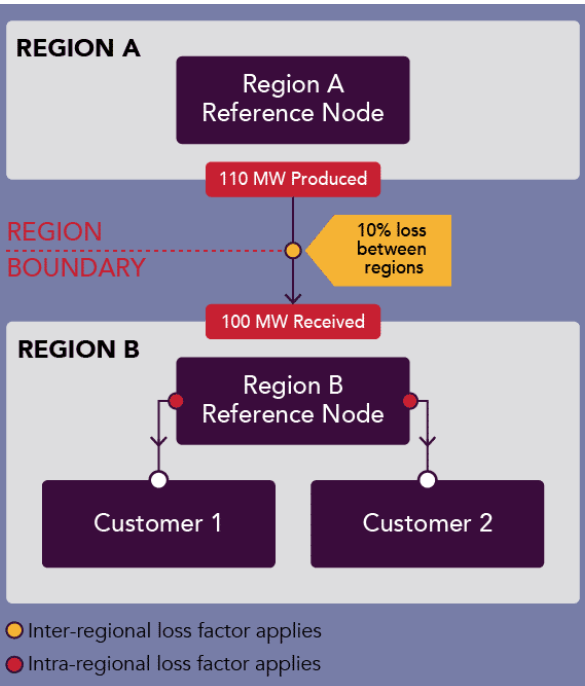
Network losses

As electricity flows through the transmission and distribution networks, a portion the energy is lost as heat due to electrical resistance of conductors.

Network energy losses are factored in at all stages of electricity production and transport to ensure supply meets demand and the power system remains secure. To allow for this loss more electricity is generated than is consumed by end-users.

Electricity losses occur between regions and within regions. These losses are approximately 10 per cent of electricity transported. As an example, to supply 100 MW of electricity to Region B (in the diagram) from Region A, 110 MW of electricity must be exported.

Intra-regional losses occur between the regional reference node (RRN) (where the region spot price is set) and customer connection points. In the diagram, Customer 2 is closer to the regional reference node so Customer 1 would require more energy to be imported to receive the same amount as Customer 2.



Thumbnail

Loss factors and their impact on the market

Losses occur when transmitting electricity from a generator to the consumer due to the physical characteristics of the network. These losses need to be included when calculating electricity prices.

To incorporate the impact of network losses on dispatch and prices, losses are mathematically represented as 'loss factors'. There are several different types of loss factors in the NEM, calculated in different ways for different purposes. These loss factors are grouped into two main categories:

Transmission Loss Factors (TLFs)

In the NEM, all loss factors affecting the transmission network are determined in April annually and applied each year on 1 July. A loss factor for a connection point is determined by calculating the increase in losses for an increase of 1 MW of load or generation at that connection point as supplied from the RRN. These are referred to as Marginal Loss Factors (MLFs) and the reason these are calculated in this way is to match the method of calculating prices required under the NER.
Loss factors are used by AEMO to work out how much electricity retailers should pay for their electricity consumption at a certain connection point, and how much generators should be paid.

Generators also use TLFs to adjust their bid prices to remain competitive even after losses have been considered.

Distribution Loss Factors (DLFs)

DLFs represent the average electrical energy loss between the distribution network connection point and the transmission network connection point to which it is assigned. Unlike TLFs (which are calculated marginally), DLFs are calculated as an average of the losses between the transmission connection point to the load in the distribution network over 12 months. They are calculated annually by the DNSPs and approved by each jurisdiction before being communicated to AEMO.

AEMO uses the DLFs to assist in settlement calculations at each connection point. Retailers use DLFs to verify the amount of energy being consumed at each connection point.

CONTINUE

Power system security

AEMO's highest priority as power system and market operator is managing power system security and reliability.

'Security' is the power system's capacity to continue operating within defined technical limits even if a single power system element (e.g. a generator or interconnector) is disconnected. Such an event is known as a credible contingency.

'Reliability' is the power system's capacity to supply enough energy, even in times of peak demand like hot evenings when air conditioning load is high.

Maintaining power system security means operating it in a way that does not overload or damage any part of it, or risk overload or damage after a contingency event.

AEMO maintains sufficient power reserves to respond to major unexpected credible contingency events such as generation losses, interconnector trips, and other major contingencies.

Review of AEMO's responsibilities

A summary of AEMO's responsibilities in maintaining power system security is as follows:

- **Maintaining power system security** in a secure operating state and within the technical envelope
- **Maintaining reserves** by having in place a means of determining and maintaining sufficient reserve available to restore power system to a satisfactory operating state after a credible contingency
- **Load shedding** by having processes in place to arrest the impact of multiple losses of power system generators.
- **Monitoring the power system** to assess that it is within the technical envelope, that scheduled generators and scheduled loads are dispatched in accordance with the National Electricity Rules, and that power system security is not jeopardised by operations on the interconnected transmission network and distribution networks
- **Reporting on and reviewing** events that pose a significant risk to the power system and major power system incidents
- **Procuring and utilising Ancillary Services** to maintain or restore the satisfactory operating state of the power system

CONTINUE

Network planning

Network planning - upgrading the network

Network planning is the process of investigating the present and future network capability and identifying network upgrade requirements. Long-term network planning is crucial to maintaining a secure power system.

Network upgrades are driven by the need to:

- Maintain system security and reliability
- Increase market efficiency (by reducing network congestion and network losses, etc)

Network service providers (NSPs) wishing to upgrade the network must follow the approval process set out in the Rules. This process requires them to consult publicly on any proposed upgrade requirements and the options to address those requirements.

Throughout this process, NSPs must provide specific information to the public. The final part of the process is ranking and selecting the best upgrade option by applying an economic cost-benefit test known as the Regulatory Investment Test for Transmission (RIT-T).

RIT-Ts specify the types of costs and benefits that must be considered. They also specify the method for selecting the preferred option from the feasible alternatives,

which must include non-network options such as generation or demand-side response.

The Rules also require NSPs to conduct an annual planning review. This review must incorporate load forecasts, existing connection point adequacy, relevant transmission system adequacy, and planning proposals for future connection points.

Network planning - AEMO's role

As AEMO is the National Transmission Planner for the National Electricity Market. It's role is to plan the future development of the power system to ensure electricity remains secure, reliable and affordable as technology and consumer behavior change.

AEMO does this through two key planning plublicstion:

- Electricity Statement of Opportunities (ESOO)

The ESOO looks ahead 10 years and assesses whether there will be enough electricity to supply to meet forecast demand. it includes information on electricity demand, existing and committed generation, and potential reliability risks.

- Integrated System Plan (ISP)

The ISP is AEMO's long-term, whole-of-the-system plan, with a 20-year outlook and beyond. It sets out the most efficient development path for generation, transmission, storage and distributed energy resources, consistent with government emissions policies and reliability standards.

The first ISP was published in 2018, replacing earlier planning documents such as the National Transmission Network Development Plan (NTNDP). The ISP is updated every two years and forms the foundation for major transmission investment decisions.

AEMO's planning work relies on extensive modeling and consultation with industry and other stakeholders to test assumptions and future scenarios.

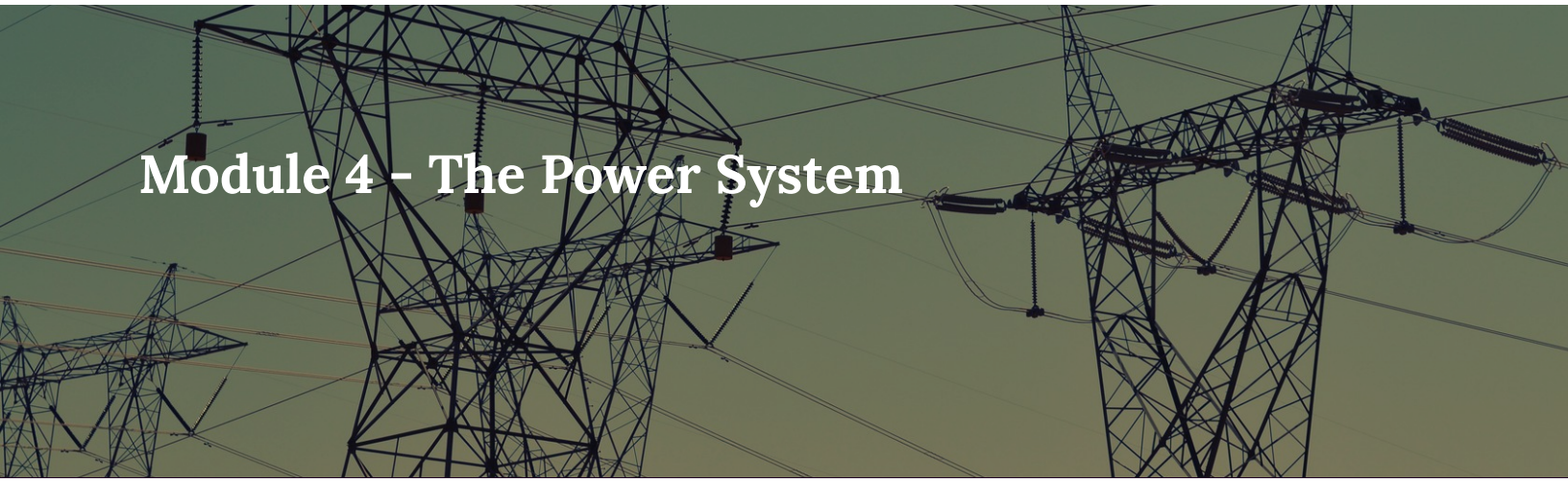
CONTINUE

Module summary

Congratulations! You've now completed **Module 3 - The Network**.

This module has examined the physical aspects of the network, explaining how the network facilitates market competition while maintaining a secure and reliable electricity supply. You should now have a good understanding of network loss factors, and an appreciation of pricing, governance and planning arrangements.

You may now review any section of this module via the main menu, or exit this module and proceed to **Module 4 - The Power System**.



Module 4 - The Power System

- Module introduction
- Managing the power system
- Maintaining a reliable and secure power system
- Credible contingencies
- Maintaining adequate reserves
- Ancillary services
- Demand forecasting
- Module summary

Module introduction

This module addresses the following learning objectives:

- To provide an overview of the physical power system and its components
- To explain the physical characteristics and limits of the power supply system
- To explain the importance of the supply/demand balance in the NEM
- To explain AEMO's role in controlling the power system
- To explain reliable operating state, satisfactory operating state and secure operating state

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

Managing the power system

The term *power system* refers to the control centres, power stations, transmission, and distribution networks that deliver electricity. As the Australian Energy Market Operator, AEMO is responsible for the day-to-day management of this power system.

This module highlights the importance of power system security and reliability, and outlines how AEMO manages the system from its two national dispatch and security centres.

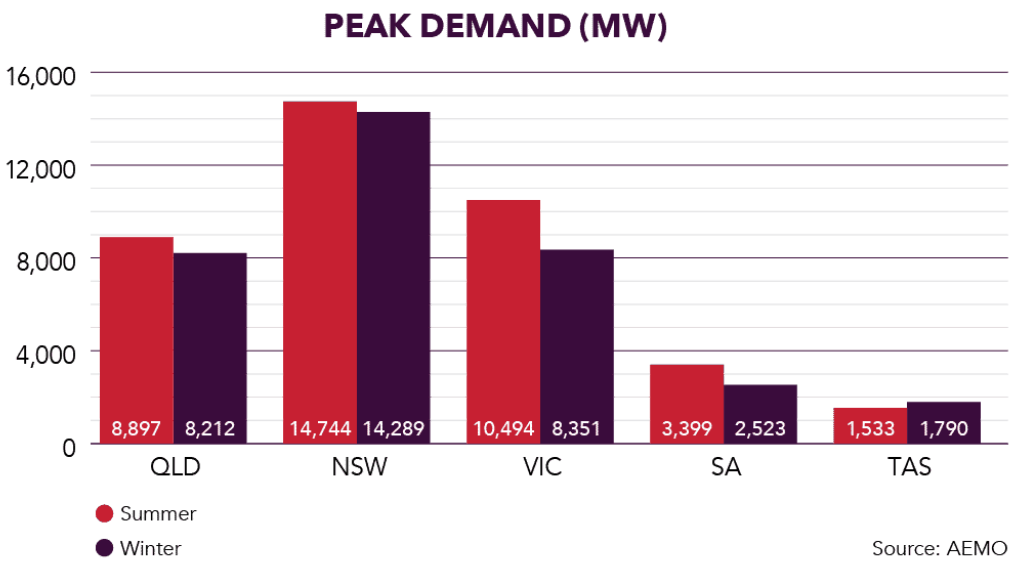
Let's begin by examining two fundamental aspects of the power system - supply and demand.

DEMAND | **SUPPLY**

Demand refers to the total amount of electricity required by all Customers.

Demand varies from region to region depending on population, temperature, and industrial and commercial requirements. It also varies throughout the day, with daily demand peaks (driven by domestic activity) typically occurring between 7:00am and 9:00am and between 4:00pm and 7:00pm.

A typical level of demand for electricity across the NEM is approximately 25,000 megawatts on a business day of average temperatures.



DEMAND

SUPPLY

Supply refers to generating electricity to meet demand. Supply and demand must be equal every second of every day, and AEMO's control centres are set up to ensure that it happens.

To enable AEMO to select the most economically efficient mix of generation at any particular moment, generators are required to submit bids to AEMO indicating the volume of electricity they are prepared to produce for a specified price.

From the bids submitted, AEMO's systems determine which generators are required to satisfy demand, at what time, and their production levels. This process is called scheduling.

Characteristics of Generators

Characteristic	Type			
	Gas and Coal-fired Boilers	Gas Turbine	Water (Hydro)	Renewable (Wind/Solar)
Time to fire-up generator from cold	8-48 hours	20 minutes	1 minute	dependent on prevailing weather
Degree of operator control over energy source	high	high	medium	low
Use of non-renewable resources	high	high	nil	nil
Production of greenhouse gases	high	medium-high	nil	nil
Other characteristcs	medium-low operating cost	medium-high operating cost	low fuel cost with plentiful water supply; production severely affected by drought	suitable for remote and stand-alone applications; batteries may be used to store power

CHARACTERISTICS OF GENERATORS

AEMO's control centres

To supply electricity to users, generators must respond rapidly to changes in demand. AEMO's control systems provide the signals required to maintain this delicate balance.

Two control centres, located in Sydney and Brisbane, operate 24 hours a day, seven days a week. These centres continuously manage the real-time dispatch of scheduled generators and loads to meet electricity demand. They are equipped with identical communication and information systems, enabling either centre to independently manage the NEM and operate the power system if required.

CONTINUE

Maintaining a reliable and secure power system

Security of supply

AEMO's highest priority as system operator is managing power system security. Security of electricity supply is measured by the power system's capacity to continue operating within defined technical limits even if a major power system element, such as a generator or interconnector, fails. Such failures are known as 'credible contingency events'. A credible contingency event is one that AEMO considers reasonably possible under the prevailing conditions, including within the technical envelope. Examples include:

- the unexpected automatic or manual disconnection of, or unplanned reduction in capacity of, a single operating generating unit; or
- the unexpected disconnection of one major transmission asset (e.g. a transmission line, transformer, or reactive plant), except where caused by a three-phase fault anywhere in the power system.

Maintaining power system security ensures the power system (or any part of it) does not become overloaded or damaged after a credible contingency event.

Power system reliability

'Reliability' means the power system's capacity to continue supplying sufficient power to satisfy customer demand, even during a very high demand event like a heat wave or in the event of a credible contingency.

Managing security and reliability

In all but extraordinary circumstances, market forces keep supply and demand in the NEM in balance. However, during periods of supply shortfall when system security or reliability of supply is threatened, the Rules give AEMO the authority to use a variety of methods to restore the supply/demand balance. Methods include demand-side management, the power of direction, load shedding and reserve trading.

Security and reliability directions

When a supply shortfall is expected, AEMO can direct registered generators into production where those generators are known to have withheld some of their total capacity from the market. AEMO only uses this power of direction to protect power system security or supply reliability.

The technical envelope

The following factors must be considered when maintaining the safe and reliable delivery of electricity:

- Transmission limits
- Power System standards
- Electrical losses
- Necessary redundancy of lines

All of these factors create the technical boundaries within which the National Electricity Market is constrained to operate (sometimes referred to as a Technical Envelope). Let's examine each of these more closely.

Transmission Limits

Elements of the transmission System include power lines, transformers, switches and connectors between switches and lines (bus bars). Each element is designed to carry a certain amount of electrical current on a continuous basis. Just as house fuse wire is 'rated' to 8 or 10 amperes of current, transmission elements also have a rating which, if exceeded by too much for too long, like fuse wire, will cause them to melt or sustain serious damage.

Transmission element ratings largely define the physical limits within which equipment must operate.

Power system standards

Alongside the physical limits of the system, electricity must be supplied to customers in a way that meets defined standards of frequency and voltage. Australia's domestic standard is 230V AC (alternating current) at 50 Hz (hertz).

Most equipment used in everyday life is designed to operate at 50Hz. Generators in the NEM all make electricity at 50 Hz and depend on system frequency being maintained at close to 50 Hz. Protection systems are installed on all generators to disconnect them, known as "tripping", if the frequency deviates too far from 50 Hz.

Frequency is maintained at the correct level when supply matches the demand. As soon as an imbalance between supply and demand occurs, the frequency starts to change. If demand is higher than supply, frequency will fall; if supply is higher than demand, frequency increases. Falls or increases in frequency can compromise system security.

Power System voltage, frequency and transmission line flows must be controlled in order to maintain secure operation within the technical boundary.

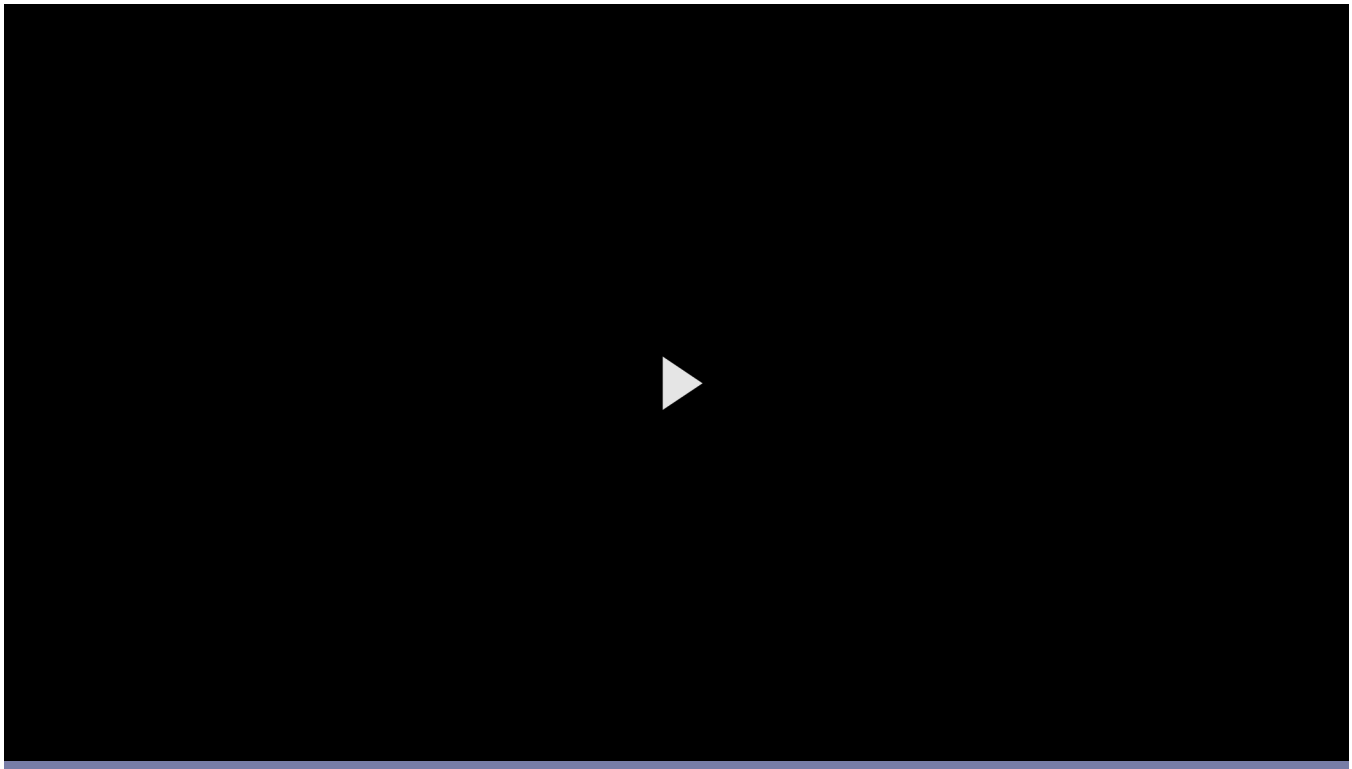
CONTINUE

Credible contingencies

A credible contingency is an event that is considered reasonably possible to occur in the power system based on current conditions. It generally involves the unexpected failure or disconnection of a single element, such as a generator, transmission line, or transformer. These events are anticipated in normal system operation and must be planned for to maintain system security. The power system is therefore operated so it can continue to function safely and reliably if a credible contingency occurs.

Credible contingencies can be caused by events such as lightning strikes, equipment faults, bushfires or storms damaging lines, and sudden mechanical or control failures in generators or transformers.

To avoid damage to elements of the power system or subsequent failures of multiple elements (known as cascading failures), the power system needs to remain secure following a credible contingency. To maintain a secure power system AEMO applies 'constraint equations' which model the power system limitations for one or multiple credible contingencies.



UNFORESEEN ELEMENTS

PLANNED OUTAGES

Natural events including lightning, storms and bushfires can force parts of the transmission network out of service and adversely affect the remaining sections supplying customers. Extreme weather conditions can produce record-high customer demand and may also affect power system operation. For example, cooling systems of generators and transformers may be unable to cope in high temperatures, and transmission lines may ice during low temperatures.

UNFORESEEN ELEMENTS

PLANNED OUTAGES

Most components of the power system are periodically taken out of service for planned maintenance, or may be disconnected from the network by a protection system designed to remove equipment affected by a fault. This includes generators, load blocks and network

elements. These disconnections change the configuration of the power system and its characteristics. The power system may be more vulnerable while some parts are disconnected, so it is important to be prepared to ensure the system remains in a secure operating state. AEMO plays an important role in ensuring that different transmission companies don't undertake maintenance at the same time in a way that would compromise the whole system.

Emergencies and contingency plans

AEMO plans for power system emergencies through contingency plans or involuntary load shedding to maintain power system security and reliability.

Contingency plans

When a transmission plant is taken out of service for maintenance, it usually means that one path through which power flows is not available. The remaining paths become more critical, and should one of these paths fail, a contingency plan is put in place. These plans outline the measures required following such an event, and aim to maintain or restore a secure operating state. Measures may include, but are not limited to, arrangements with generators, network switching and load shedding.

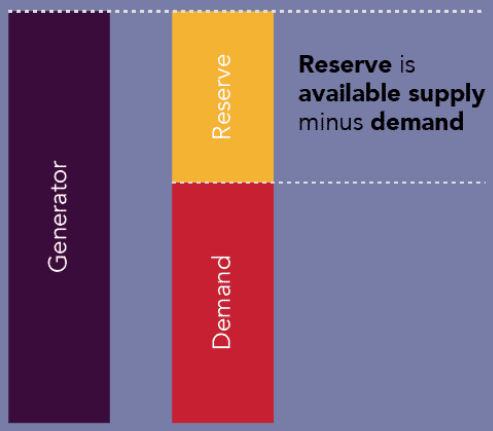
Involuntary load shedding

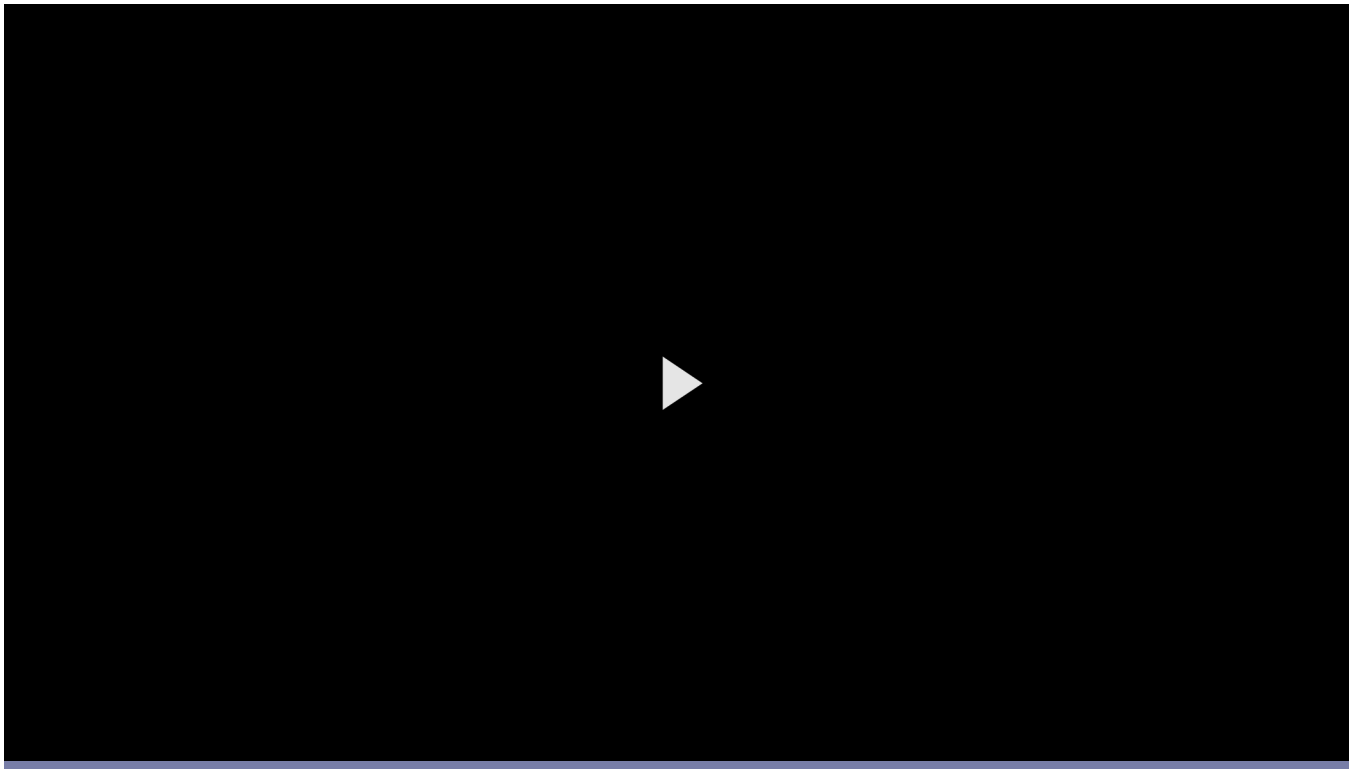
In extreme circumstances it may be necessary to disconnect loads from the power system to restore the supply/demand balance. This is known as involuntary load shedding. For example, when a generating unit or transmission network failure occurs and all reserves have been used, AEMO may need to direct customers to reduce their power consumption so that demand matches the reduced supply. Involuntary load shedding occurs only after all alternatives have been exhausted and it is the only course of action available to maintain operation within standards.

CONTINUE

Maintaining adequate reserves

The power system must maintain a certain level of reserve in order to meet the required standard of supply reliability across the NEM. Reserve calculations take into account supply from the interconnections with other regions. An adequate level of reserve should enable the power system to withstand the loss of the two largest generating units within a region.





Reserve trading

When there is sufficient notice of an upcoming shortfall of supply that threatens to compromise minimum reserve margins, AEMO may tender contracts for electricity supply from sources beyond those factored into the usual forecasting processes. Emergency generators and other generators connected directly to the distribution network who submit tenders may enter contracts to boost supply in the NEM so the widespread supply interruptions that may otherwise have occurred can be avoided. In the same way, some electricity consumers may offer (for a fee) to decrease their demand at times of supply shortfall so that demand and supply are brought into balance.

Lack of Reserve (LOR)

If reserve levels fall below the amount required to maintain a reliable operating state, AEMO will notify market participants of reserve inadequacy according to the following levels:

Lack of Reserve Level 1 (LOR1)

Reserve levels are lower than the two largest supply resources in a region. LOR 1 signals a reduction in pre-determined electricity reserve levels, encouraging generators to offer more supply, or large industrial and commercial consumers to reduce their demand. At this stage, there is no impact to power system security or reliability and AEMO continues to monitor reserve levels to maintain adequate supply.

Lack of Reserve Level 2 (LOR2)

Reserve levels are lower than the single largest supply resource in a region. At this level, there is no impact to the power system, but supply could be disrupted if a large incident occurred. AEMO has the ability to direct generators or activate the Reliability and Emergency Reserve Trader (RERT mechanism to improve the supply/demand balance.

Lack of Reserve Level 3 (LOR3)

A deficit in the supply/demand balance. This condition exists when the available electricity supply is equal to or less than the operational demand. This means there are no reserve supplies available. Controlled load shedding may be required as a last resort to protect system security and prevent long-term damage to system infrastructure.

Low Reserve Condition (LRC)

When AEMO considers that a region's reserves fall short of what is required to meet the reliability standard.

CONTINUE

Ancillary services

Ancillary services can be thought of as an 'insurance policy' for the power system. Certain generators and other participants are paid to generate (or maintain the potential to generate) reserve levels of power in case they are required. This means that if a bushfire or other unexpected event occurs, the power system can draw on this 'backup' service to maintain supply.

AEMO uses ancillary services to manage the power system safely, securely and reliably. Ancillary services are used to maintain key technical characteristics of the system, including frequency, voltage, network loading and system re-start processes.

AEMO operates ten separate markets for the delivery of Frequency Control Ancillary Services (FCAS), and purchases Network Support and Control Ancillary Services (NSCAS) and System Restart Ancillary Services (SRAS) under agreements with service providers.

Payments for ancillary services include payments for availability and for the delivery of services. The market participant/s responsible for causing a situation that requires ancillary services must pay for the services. This occurs whenever FCAS are needed to keep frequency within the normal operating band of 49.85 — 50.15 Hz.

NSCAS are primarily used to control active and reactive power flow into or out of an electricity transmission network, in order to:

- Maintain power system security and reliability of supply of the transmission network in accordance with the power system security standards and the reliability standard (Reliability and Security Ancillary Services - RSAS); or
- Maintain or increase power transfer capability of the transmission network to maximise the present value of net economic benefit to all those who produce, consume or transport electricity in the market (Market Benefit Ancillary Services - MBAS)

SRAS are used in contingency situations in which there has been a major supply disruption or where the power system must be restarted.

Ancillary service costs vary depending on the amount of service required and the time at which the service is required.

CONTINUE

Demand forecasting

AEMO conducts electricity demand forecasts to assist in operating the NEM.

AEMO uses a variety of forecasting processes to determine the level of demand for every dispatch interval (every five minutes) in the NEM. Then, using the submitted bids to generate electricity, AEMO produces a generation schedule (or timetable) to ensure that the forecast demand will be met. This schedule dispatches the least expensive generators into production first, and ensures the power system remains in a secure operating state.

To maintain the supply/demand balance, it is important that AEMO is informed in advance of any limits affecting generator or network capacity. This enables other market participants to respond to potential supply shortfalls by increasing their generation or network capacity. Market participants are able to signal pending supply limitations using a variety of forecasting tools designed to improve the overall efficiency of the market.

Types of forecasts

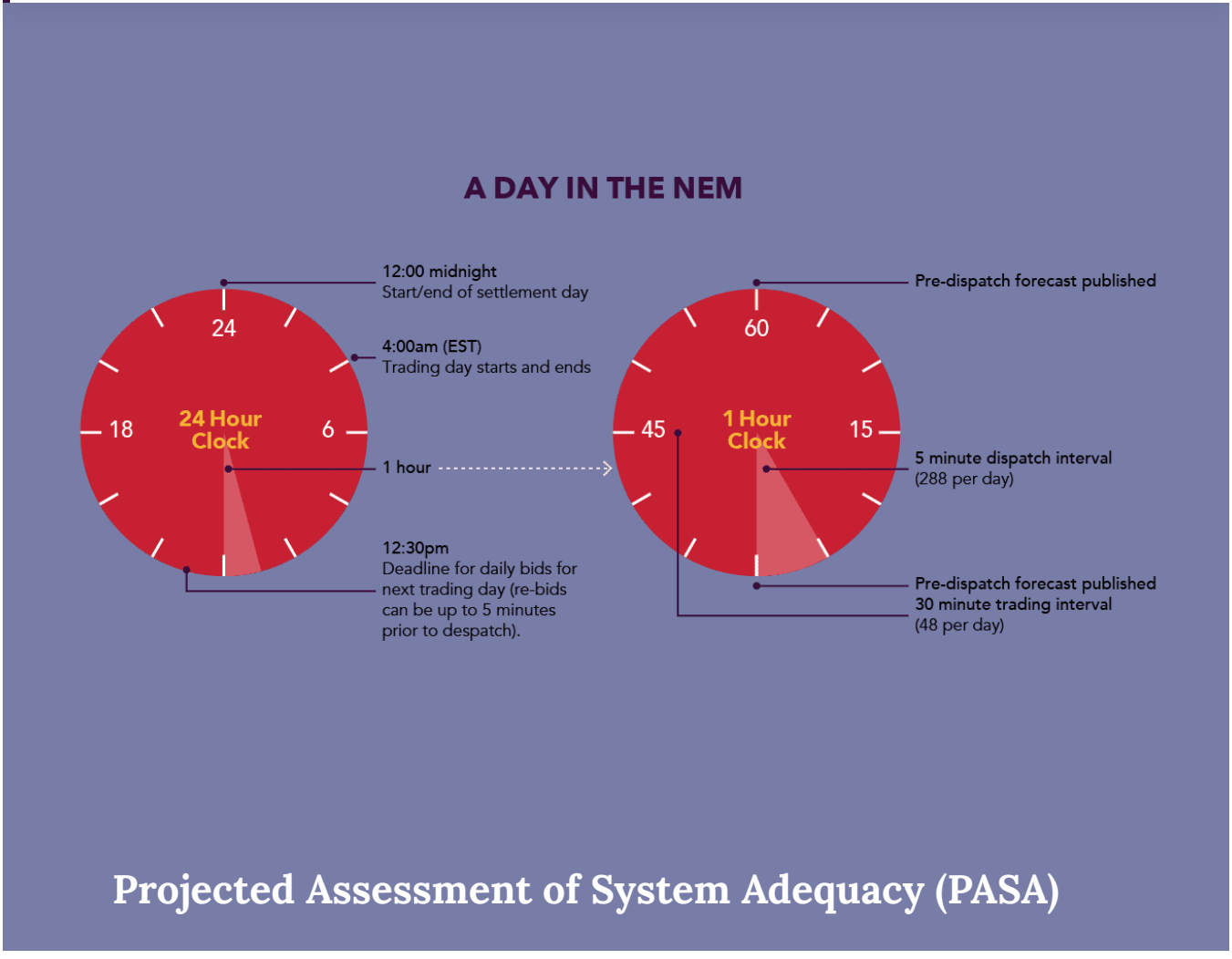
Pre-dispatch forecasting —

Pre-dispatch is a short-term forecast of supply and demand in the market. It is used to estimate prices and demand for the upcoming trading day, and the volume of electricity expected to be supplied through the interconnectors between regions.

Generators and network operators are required to notify AEMO of their maximum supply capacity and availability, and this information is matched against regional demand forecasts. All offers to supply are then collated so that potential supply shortfalls can be identified and published. Market participants use this information as the basis for any re-bids of the capacity they wish to bring to the market.

Five-minute matching of supply and demand

Generators are scheduled and dispatched into production to match supply with prevailing demand every five minutes of every day. This process, in turn, produces dynamic price signals that guide market participants as they bid to supply electricity to the market.



Projected Assessment of System Adequacy (PASA)

AEMO monitors the future adequacy of generating capacity based on the predicted availability of generating units at power plants. AEMO produces both seven-day and two-year forecasts to manage reserves near to real-time and into the longer term. These forecasts are called the short-term and medium-term Projected Assessment of System Adequacy (PASA) respectively. AEMO uses them to ensure that adequate levels of reserve are in the system at all times, and generators and network operators use them to plan augmentation, maintenance and other outages.

Forecast	Forecast period	Updated/published
Short-term PASA	7 days	2-hourly from 4:00am
Medium-term PASA	2 years	2:00pm every Tuesday

PASA forecast timings

CONTINUE

Module summary

Congratulations! You've now completed **Module 4 - The Power System**. This module has examined power system security and reliability, and detailed how AEMO deals with the day-to-day management of the power system from its two national dispatch and security centres.

You may now review any section of this module via the main menu, or exit this module and proceed to **Module 5 - Spot pricing and dispatch**.

Module 5 - NEM Spot Pricing and Dispatch

- Module introduction
- NEM spot pricing and dispatch
- NEM regions and pricing
- Pricing under extreme conditions
- Projected Assessment of System Adequacy (PASA)
- Rights and obligations of NEM participants in pricing and dispatch
- Module summary

Module introduction

This module addresses the following learning objectives:

- To explain the NEM model which links and matches the supply and demand for energy
- To explain how the NEM model maximises the value of trade and delivers an optimal market solution for users
- To explain the dispatch and pricing process
- To identify the relationship between bids, dispatch and the calculation of the spot price
- To explain how network loss factors affect the spot price
- To identify the roles and responsibilities of all parties in dispatch and pricing
- To explain the Spot Market Timetable and market data publishing cycle

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

NEM spot pricing and dispatch

The National Electricity Market (NEM) operates as a Real-time Energy Only Regional Market. These characteristics are defined as below:

Real-Time: The NEM operates as a real-time market, in which prices are determined every 5 minutes. This differs from other markets that may trade on prices set the day ahead or even greater time horizon.

Energy Only: Participants are only paid for energy they provide to the market. Other markets may compensate generators for their capacity in conjunction with the energy they provide.

Regional: The NEM is split into 5 different regions broadly defined by the state boundaries that make up the NEM (QLD, NSW, VIC, SA and TAS) with ACT being included with the NSW region. With each 5 minutes determining a new price for each region.

All wholesale electricity is traded by market participants through a central pool. The dispatch price is set at the value where competitive offers to supply electricity match demand.

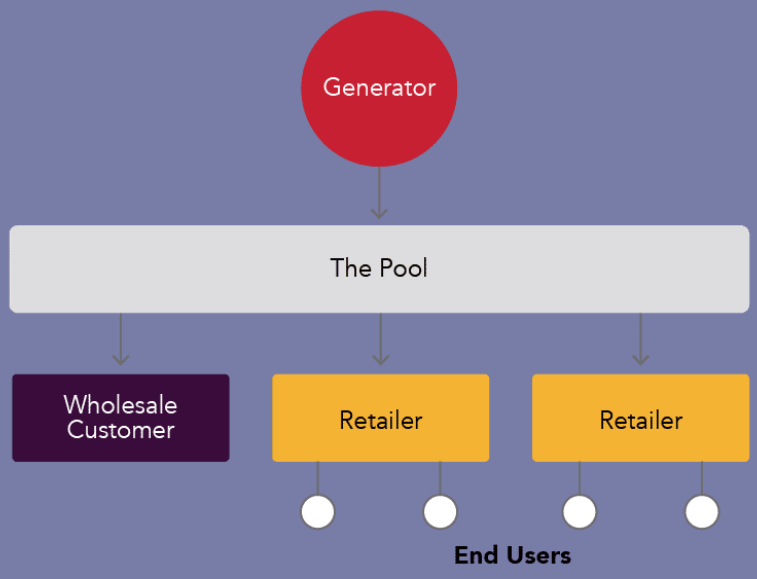
All electricity generated and consumed within a region is paid for at the regional spot price. Trading in the wholesale electricity market can be risky because the spot price can be volatile.

What is the electricity pool?

The electricity pool is made up of a set of rules and procedures managed by AEMO in conjunction with generators, customers and network service providers. This set of rules acts to centrally operate the power system and balance electricity supply and demand.

Electricity is traded using this pool arrangement to allow competition to exist. Electricity can not be stored within the power system for future use, so supply must vary dynamically to match changing demand; and one unit of electricity is indistinguishable from any other unit, so it is impossible to determine which generator produced which electricity.

Electricity supply and demand must be balanced and matched instantaneously to provide a safe supply at acceptable quality standards.

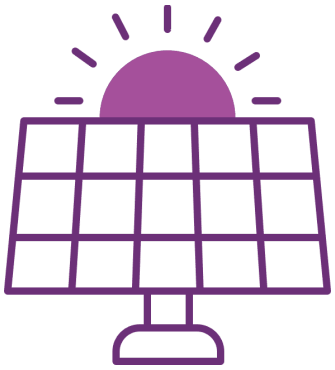


Who trades within the electricity pool?

A range of participant categories for generators and customers are defined in the National Electricity Rules (the Rules). Click through each participant below to see how they each interact within the electricity pool.

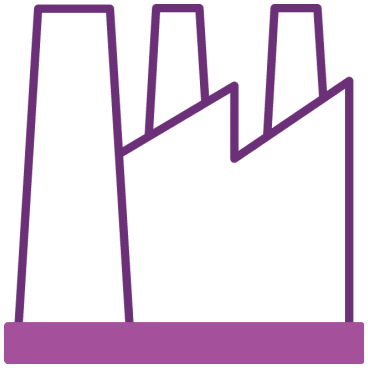
Market scheduled generator

A generator who operates their scheduled generating units in advance with the central dispatch process and who is paid for their generation through the pool at the spot price.



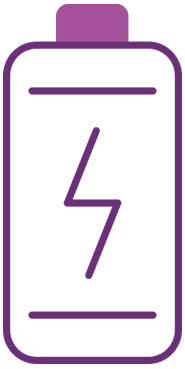
Market semi-scheduled generator

A generator who operates their semi-scheduled generating units in accordance with the central dispatch process and who is paid for their generation through the pool at the spot



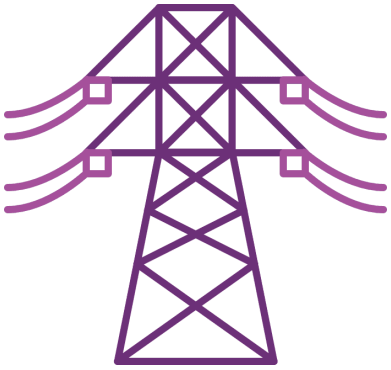
Market non-scheduled generator

A generator who does not participate in the central dispatch process and who is paid for their generation through the pool at the spot price.



Market scheduled load

Scheduled loads are net consumers of electricity that participate in the central dispatch and pricing processes operated by AEMO. They submit price-sensitive bids and pay for the energy used from the

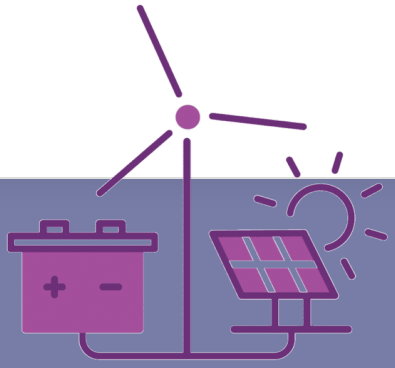


Market scheduled network service provider
An NSP who operates their scheduled network asset in accordance with the central dispatch process. A scheduled NSP must be a

Market customer (retailer)
A market customer (retailer) who buys wholesale electricity from the pool at the spot price and on-sells to end-user customers.

Integrated Resource Provider
Introduced in June 2024, the Integrated Resource Provider (IRP) registration category allows bi

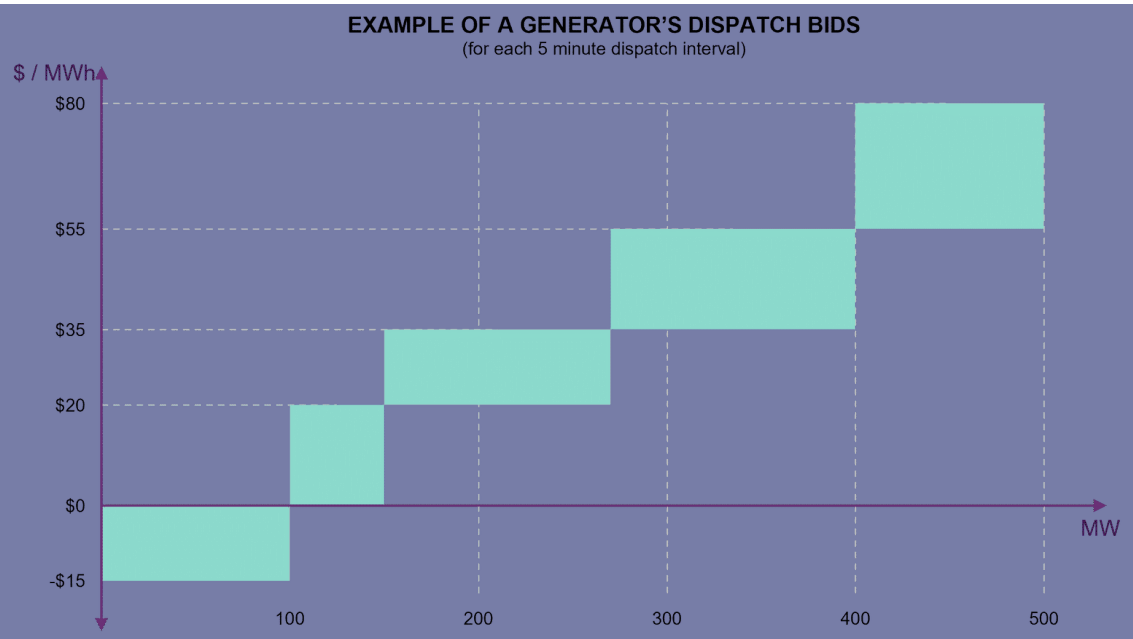
directional resources—such as batteries, hybrid generation and storage



Submitting bids to supply

For AEMO's systems to facilitate supply, scheduled and semi-scheduled generators must submit bids and offers (collectively referred to as bids or bidding) to AEMO specifying the volume of electricity their generating units are prepared to produce for a specified price.

A bid covers a trading day, which is 4:00am to 4:00am AEST. Generators can submit bids, using up to 10 price bands, breaking up the total capacity of the generator into one or more of those price bands. Bids submitted after 12:30pm AEST for the next trading day are called rebids and must be accompanied by a reason. All bids for the current and (after 12:30pm AEST) next trading day are also reflected in pre-dispatch schedules. Pre-dispatch is an indicative forecast of dispatch and pricing for the current trading day (and next trading day, after 12:30pm AEST) to a half-hourly resolution, and updated every 30 minutes. Pre-dispatch shows at a five minute resolution for the next hour.



Submitting bids for demand

Each day, market customers with price-sensitivity or the ability to adjust demand, provide AEMO with a bid for each of their scheduled loads. This bid indicates how much electricity a scheduled load is prepared to buy from the pool at a certain price, for each 5-minute dispatch interval of the relevant trading day.

Central dispatch process

AEMO manages the central dispatch process and pricing cycle which automatically runs every five minutes.

The central dispatch process determines, based on a least-cost optimisation of all bids submitted for supply (and demand), the production required from each generating unit to meet AEMO's forecast of demand at the end of each dispatch interval. This process is called 'scheduling'.

Bids to supply are stacked in order of rising price (along with any bids to reduce demand, in reverse price order), with sufficient generation scheduled to meet the net demand. This results in more expensive generators being scheduled into production only as demand increases.

Dispatch instructions are then issued to scheduled and semi-scheduled generating units and to scheduled loads (collectively called 'scheduled plant'), to produce or consume the instructed amount of energy by the end of the dispatch interval.

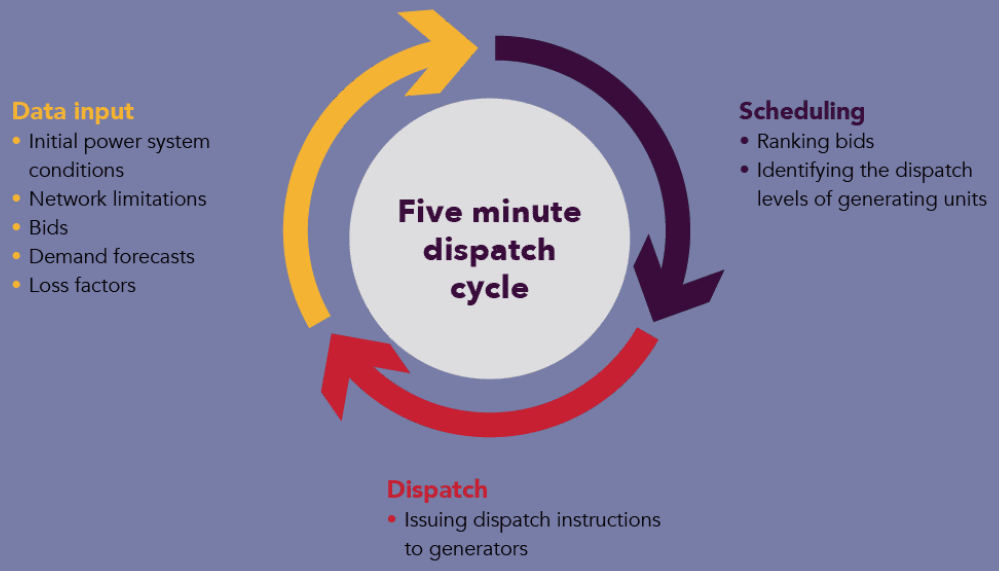
At the same time, the central dispatch process optimally determines, based on bids from ancillary service providers, the capacity reserves required to handle the potential loss of a generating unit, load or transmission network element (refer to Module 6).

The results of the optimisation are used to determine the dispatch price and ancillary service prices for each region.

How does the central dispatch process determine dispatch targets?

In normal circumstances, the central dispatch process starts with the generator who has the lowest priced bid then, in sequence of increasing bid price, schedules enough generation to meet demand, subject to generation capacity and network transfer limits. In other words, scheduled plant is dispatched on an economic or merit order basis.

Electricity demand fluctuates in each five-minute period throughout the day, and the amount of generation scheduled for production also varies. When demand is low, generators with the lowest bid price are in production and when demand is high the more expensive generators are brought into production. This provides an optimal, or least-cost, solution for power supply allocation to meet demand.



The spot market and spot price

The spot market is where market generators are paid for the electricity they sell to the pool and market customers pay for their electricity consumption from the pool. All electricity is traded through the pool at the spot price. The pool is divided into a number of pre-defined regions.

A spot price for each region is determined every five minutes. AEMO uses the spot price to settle all actual energy traded in the NEM.

MARKET PRICE CAP

MARKET FLOOR PRICE

The Rules set a maximum spot price, called the Market Price Cap (MPC). This is also the maximum bid price.

The AEMC's Reliability panel reviews the market price cap and market floor price settings every four years. This process factors in consumer price index information, published by the Australian Bureau of Statistics. In 2025-26 the MPC is $20,300.

MARKET PRICE CAP

MARKET FLOOR PRICE

The Rules set a minimum spot price, called the market floor price. This is also the minimum bid price.

The market floor price is -$1,000 per megawatt hour. The AEMC's Reliability panel reviews the market price cap and market floor price settings every four years.

CONTINUE

Lesson 3 of 7

NEM regions and pricing

The wholesale electricity market is divided into five geographically-based pricing regions. Regions are generally bounded by transmission lines that interconnect major generation or load centres and reach their capacity limits sufficiently often to warrant separate pricing regions.

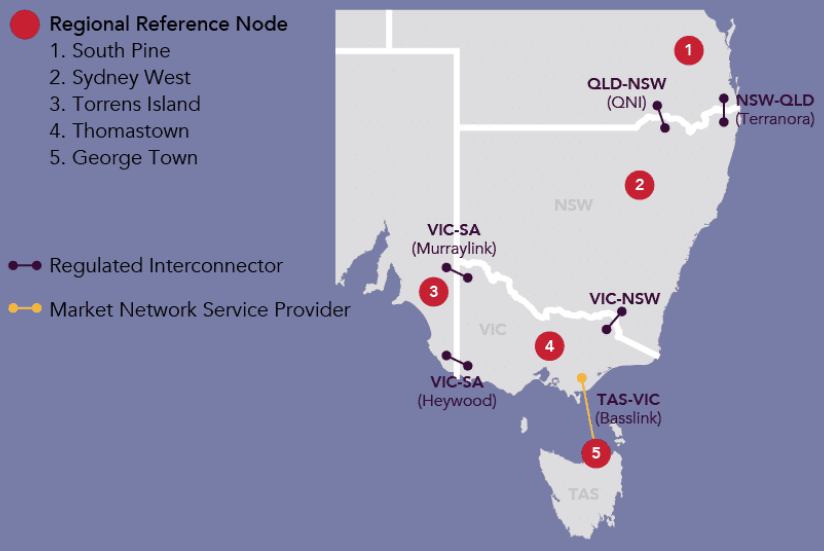
Each pricing region has a unique regional reference node (RRN), which is typically a major load centre (e.g. Melbourne). AEMO calculates a regional reference price (or spot price) for each regional reference node for each trading interval in a trading

day. All prices are published in real time to participants via market systems, and to the general public via the internet.

Market generating units and market loads are allocated to a particular region. Market generators are paid the relevant regional spot price for generation from their market generating units in that region. Market customers pay the relevant regional spot price for consumption by their market loads in that region.

Interconnector capacity

In calculating spot prices, AEMO must account for any limits on the inter-regional flow of electricity. When an interconnector capacity limit is reached, AEMO may need to schedule a more expensive generator to meet electricity demand within the region, even if lower priced generation is available in other regions.



Loss factors

As electricity flows from generators over the transmission and distribution networks to end-users, electrical losses occur due to the electrical resistance of these lines causing some of the electrical energy to dissipate as heat. The flow of electricity over networks is limited to avoid overheating, which can cause asset damage or result in excessive line sag beyond safety clearances.

Losses are incurred as either:

Inter-regional

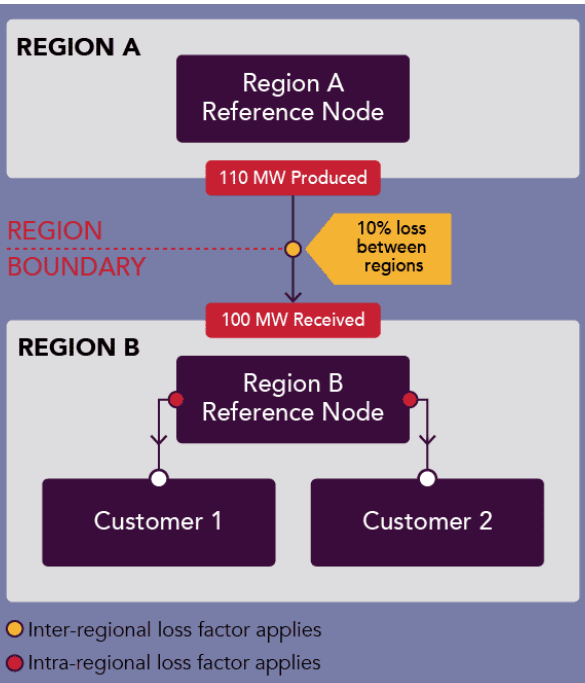
Electrical losses incurred in the transmission network when transporting electricity between regions (i.e. via interconnectors).

Intra-regional

Electrical losses incurred in the transmission network when transporting electricity between generators and their regional reference node or from the regional reference node to customers.

Distribution

Electrical losses incurred when transporting electricity within the distribution network.



Effect of inter-regional loss factors on price

Prices at each regional reference node must be calculated dynamically to reflect the additional generation required to cover electrical losses incurred at different levels of inter-regional flow.

Marginal inter-regional loss factors are used in the central dispatch process to determine inter-regional losses. These losses impose an additional cost on inter-regional flows and increase the effective bid price of generator exports to adjoining regions. If the exporting generator is marginally dispatched to meet demand in the importing region, its effective bid price is used to determine the price for that region.

Intra-regional loss factors

Losses incurred within a region are modelled in the central dispatch process by intra-regional loss factors or marginal loss factors, which represent the marginal losses incurred in transporting electricity from a generating unit to its regional reference node, or from the regional reference node to a customer load.

These are pre-determined annually based on historical analysis of electrical losses. Intra-regional loss factors are used to refer the bid prices submitted for a unit from its local node to the regional reference node, before input to central dispatch. In this way, participants financially cover their own losses in transferring power to or from the regional reference node.

Price difference between regions

Price differences between regions arise when:

- Electrical losses are incurred for flow between regions (also known as inter-regional losses)
- An interconnector flow reaches a limit (also known as a binding constraint)

Where an interconnector flow limit is reached, the inter-regional price difference is typically much greater than price differences arising solely from the electrical losses associated with that flow. This occurs as bid stacks effectively separate, and demand in the downstream region can only be met from local generators at higher prices than could otherwise be obtained inter-regionally.

Spot price as a signal

The spot market is designed to provide both economic and locational price signals, helping market participants make informed decisions about the investment in and operation of electricity generation, demand-side response, and network assets.

CONTINUE

Pricing under extreme conditions

Administered pricing

Spot market prices are naturally volatile, and while the market price cap provides some measure of protection from unbounded prices, there is still potential for periods of sustained high pricing with values at, or close to, the market price cap. This can result from events that the market could not reasonably foresee or insure against - such as a drought or flood.

The Rules provide the spot market with a mechanism to cap the financial exposure of market customers in a region following a period of sustained high pricing. This mechanism is known as administered pricing.

Prices in a region will be limited by an administered price cap for the remainder of the trading day if the sum of spot prices in that region over the previous week exceeds the Cumulative Price Threshold (CPT). The CPT is adjusted for inflation at the start of each financial year.

There is also an administered floor price (the negative of the administered price cap) which floors pricing during the administered pricing period.

Market Suspension in the NEM

AEMO can declare the spot market to be suspended under clause 3.14.3 of the NER if a following situation occurs:

- The power system in the region has collapsed to a black system
- AEMO has been directed by a participating jurisdiction to suspend the market following declaration by that jurisdiction of a state of emergency
- AEMO determines that it has become impossible to operate the spot market in accordance with the provisions of the NER.

The spot market is taken to be suspended at the start of the dispatch interval in which AEMO makes the declaration. This is communicated through NEM Market Notices to registered participants, including the cause of the market suspension, what pricing applies in the suspended region, when those prices are revised, and situation updates.

Pricing, Bidding and Dispatch during Market Suspension

Bidding and dispatch will continue in accordance with NER rule 3.8 where AEMO considers it is practical and reasonable to do so. If it is not reasonable to do so, AEMO may use the most recently published valid pre-dispatch schedule if it is still current.

Spot prices and FCAS prices in a suspended region continue to be set in accordance with rule 3.9 of the NER where reasonable to do so, otherwise they will be based on the relevant Market Suspension Pricing Schedule developed and published in accordance with NER clause 3.14.5(e). Spot prices are capped in regions upstream of a suspended region to minimise the accrual of negative inter-regional settlement residues.

Guide to Market Suspension in the NEM

MARKET SUSPENSION

Here you can find out more about previous market suspension events

NEM EVENTS

CONTINUE

Projected Assessment of System Adequacy (PASA)

AEMO monitors and reports on the future adequacy of generating capacity against forecast electricity demand. This is based on available information supplied by generators and is called Projected Assessment of System Adequacy (PASA).

Generators use this data to make decisions about when to maintain their generating plant.

Click through the below to read about the different types of PASA projections.

Pre-dispatch PASA

Pre-dispatch PASA (PDPASA) is a capacity adequacy forecast covering the same period as pre-dispatch and is updated every 30 minutes. The PDPASA reports, to a half-hourly resolution, total regional generating capacity compared with forecast regional demand.

Short-term PASA

Short-term PASA (STPASA) is a six-day capacity adequacy forecast starting from the end of the PDPASA outlook and is updated every two hours. The STPASA reports, to a half-hourly resolution, total regional generating capacity compared with forecast regional demand.

Medium-term PASA

Medium-term PASA (MTPASA) is a two-year capacity adequacy forecast, starting from Sunday and updated weekly at a minimum. The MTPASA reports, to a daily resolution, total regional generating capacity compared with forecast peak regional demand.

CONTINUE

Rights and obligations of NEM participants in pricing and dispatch

Under the Rules, generators and customers have various rights and obligations concerning the electricity pool managed by AEMO. The following sections outline some of these.

Rights and obligations - scheduled generators and market customers with scheduled loads

- Must participate in the central dispatch, pre-dispatch and PASA processes
- Must provide daily PASA availability and any weekly energy constraints for the next 36 months for use in MTPASA
- Must provide half-hourly PASA and market availability and any daily energy constraints for the next seven days, for use in STPASA
- Can apply to AEMO if they wish to aggregate their scheduled generating units for the purpose of central dispatch
- Must inform AEMO two days ahead of each trading day of available capacity for each trading interval of each trading day
- Must advise AEMO of any expectations that a generating unit will not be able to operate in accordance with dispatch instructions
- Must ensure their ability to dispatch electricity as required under the dispatch schedule
- Must have facilities that can receive dispatch instructions, both electronically and manually, every five minutes
- May submit rebids to change available capacity

- Must respond to dispatch instructions by following energy targets or providing enabled ancillary services if requested

Rights and obligations - semi-scheduled generators

These are the same as for scheduled generators, except:

- Must provide to AEMO an energy conversion model for each semi-scheduled generating unit as part of registration, for use in creating intermittent generation forecasts
- Must provide to AEMO the plant availability of each semi-scheduled generating unit over the next 36 months
- Are not required to separately provide PASA or market availability for use in PASA
- Must respond to dispatch instructions to cap the output of their semi-scheduled generating units to avoid violating power system security or economic dispatch — these instructions are flagged as 'semi-dispatch intervals'
- Cannot participate in reserve contracting

Market information published by AEMO

Real-time

- Actual and forecast prices for energy and market ancillary services for each region
- Actual and forecast energy demand for each region
- Forecast spot price sensitivities to changes in forecast demand

- Generation, actual and forecast, aggregated for all scheduled and semi-scheduled generation unit in each region
- Generation, actual, aggregated for non-scheduled generating units in a region
- Consumption, actual and forecast, aggregated for all scheduled loads in each region
- Ancillary services enabled, forecast aggregated for all service providers, for each region and service
- Inter-regional network flows, forecast, for each modelled network element
- Inter-regional network losses, forecast, for each interconnector
- Generation, actual, for each scheduled, semi-scheduled and non-scheduled generating unit
- Market ancillary services enabled, actual, for each service provider and service
- Market ancillary service offers for each service provider and service
- Consumption, actual, for each scheduled load

Next day

- Ancillary services enabled, forecast, for each service provider
- Generation forecast for each scheduled and semi-scheduled generation unit
- Consumption forecast for scheduled load
- Bids for each scheduled and semi-scheduled generation unit and scheduled load

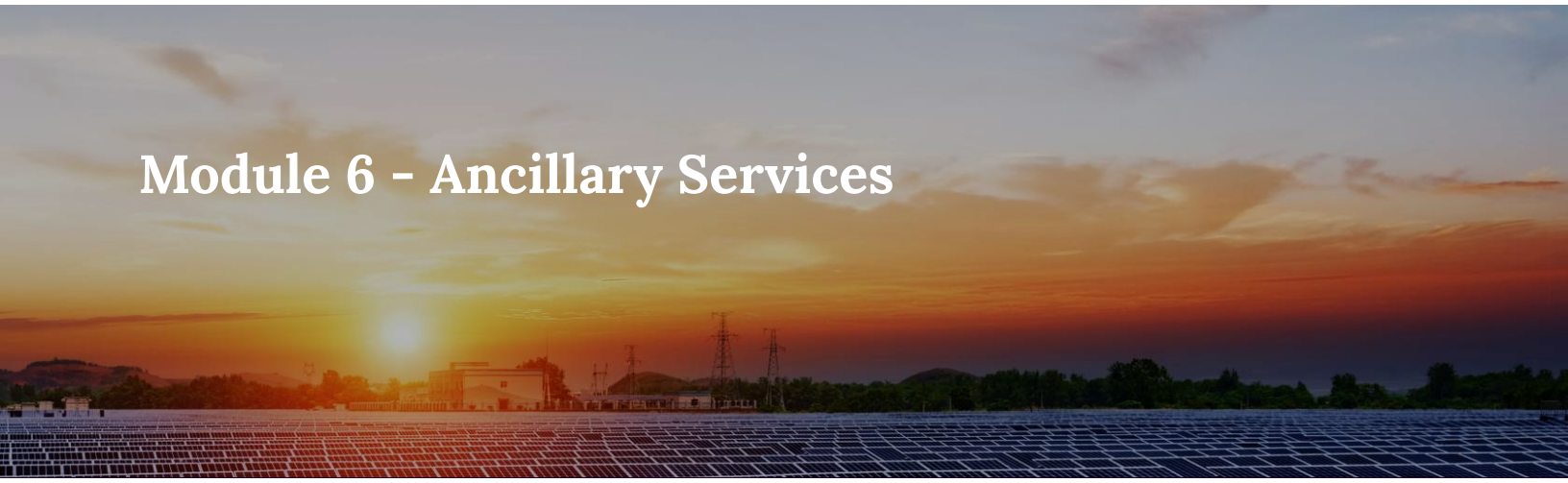
CONTINUE

Module summary

Congratulations! You've now completed **Module 5 - NEM Spot Pricing and Dispatch**

In this module, you were introduced to the concept of the electricity pool, along with the central dispatch process, the spot market, and spot pricing. You also explored NEM regions, loss factors, and interconnector capacity, and how each influences the spot price. Finally, the module covered administered pricing, mandatory restrictions, pre-dispatch, PASA, and the roles and responsibilities of NEM participants.

You can now revisit any section of this module via the main menu, or exit this module and proceed to **Module 6 - Ancillary Services**.



Module 6 - Ancillary Services

- Module introduction
- Ancillary services
- Frequency Control Ancillary Services (FCAS)
- Network Support and Control Ancillary Services (NSCAS)
- System Restart Ancillary Services (SRAS)
- Module summary

Module introduction

This module addresses the following learning objectives:

- To explain the need for ancillary services
- To explain the range of ancillary services
- To explain the pricing and enablement of Frequency Control Ancillary Services (FCAS)
- To explain the concept of co-optimisation between the energy market and ancillary service markets and co-optimisation between regulation and delayed FCAS markets
- To explain the way ancillary service costs are recovered

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

Ancillary services

What are ancillary services?

AEMO is responsible for ensuring that the power system is operated in a safe, secure and reliable manner. To fulfill this obligation, AEMO controls key technical system characteristics, such as frequency and voltage, through ancillary services.

Ancillary services can be thought of as an 'insurance policy' for the power system. For example, certain generators and other participants are paid to provide the capacity to alter their supply in case they are needed. In this way, if an unexpected event occurs, the power system can draw on this 'backup' service to maintain supply.

AEMO purchases the required services from service providers under Ancillary Service Agreements and Market Ancillary Service Arrangements.

Ancillary services

All NEM ancillary services can be grouped into three major categories:

Frequency Control

FCAS are used by AEMO to maintain system frequency close to 50 Hz at all times, in accordance with the NEM frequency standards.

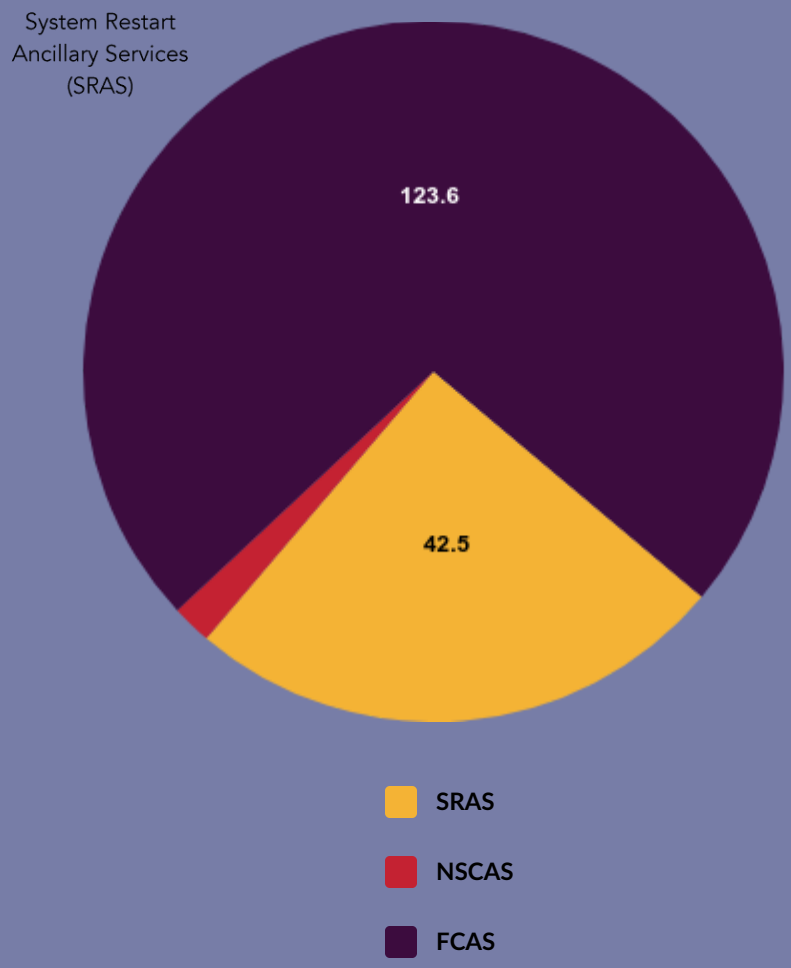
Network Support and Control Ancillary Services (NSCAS)

NSCAS are primarily used to:

- Maintain power system security and reliability of supply of the transmission network in accordance with the power system security standards and the

SRAS are used when a major supply disruption or a whole or partial system blackout occurs and the power system must be restarted.

202 lary Service Payments ($ Millions)



Ancillary service history

Ancillary services have always been a physical requirement of any electrical system, but have only taken on a commercial focus since the introduction of energy markets.

When the NEM commenced, all ancillary services were provided through long-term agreements between AEMO (who purchases on behalf of the market) and ancillary service providers.

However, the National Electricity Code at the time required AEMO to investigate more competitive means of ancillary service arrangements. This led to the introduction of spot markets for Frequency Control Ancillary Services (FCAS), which commenced in September 2001 following a long development phase.

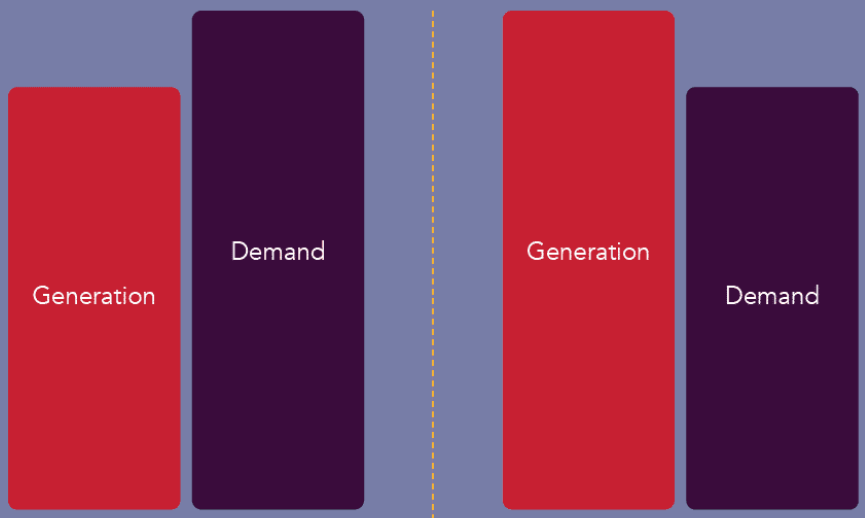
AEMO operates ten separate markets for the delivery of FCAS.
Network Support and Control Ancillary Services (NSCAS) and System Restart Ancillary Services (SRAS) are non-market services, sourced via agreements with service providers.

CONTINUE

Frequency Control Ancillary Services (FCAS)

To understand frequency control, think of a power system like a car engine. If a car traveling at a constant velocity experiences a change in load but no corresponding change to the power input to the engine, it will speed up (if the load decreases, such as encountering a downhill slope) or slow down (if the load increases, such as encountering an uphill slope).

Similarly, if the load changes on a power system without a corresponding variation in the generation feeding that power system, the frequency (speed) will deviate. This was demonstrated in the frequency activity in Module 4.



Frequency control

To maintain the frequency within the NEM standards, FCAS is used to alter either generation or demand to maintain the supply/demand balance.

Frequency control can be divided into two reasonably distinct subsets:

REGULATION FREQUENCY CONTROL

CONTINGENCY FREQUENCY CONTROL

The correction of the supply/demand balance in response to minor deviations in either load or generation.

Regulation services are continually used to correct minor changes in the supply/demand balance. Regulation services are controlled centrally from one of AEMO's two control centres.

REGULATION FREQUENCY CONTROL

CONTINGENCY FREQUENCY CONTROL

The correction of the supply/demand balance following major contingency events such as the loss of a generating unit or a large transmission element.

Providers supplying contingency control services may also be used to assist with regulation frequency control. Contingency services are controlled locally and are triggered by the frequency deviation that follows a contingency

Frequency control services

Regulation frequency control services are provided by generators on Automatic Generation Control (AGC). The AGC system allows AEMO to continually monitor system frequency and send control signals out to regulation service providers to maintain system frequency within the normal operating band of 49.85 to 50.15 Hz.

Contingency frequency control services

Contingency services are provided by technologies that can locally detect any frequency deviation and respond in a manner that corrects the frequency. Some examples of these technologies include:

- Generator Governor Response: where the generator's governor reacts to the frequency deviation by opening or closing the turbine steam valve and altering the power (MW) output of the generator accordingly
- Load shedding: where a load can be quickly disconnected from the power system (this can only correct a low frequency)

- Rapid Generation: where a frequency relay detects a low frequency and correspondingly starts a fast generator (this can only correct a low frequency)
- Rapid Unit Unloading: where a frequency relay detects a high frequency and correspondingly reduces a generator output (this can only correct a high frequency)

FCAS requirements

The NEM's frequency control standards have led to the classification of ten FCAS requirements. These are:

Regulation

- Regulation raise.
- Regulation lower.

Contingency

- **Very Fast raise** and **Very Fast lower** - one second response to slow the immediate frequency deviation. These services were introduced in October 2023.
- **Fast raise** and **fast lower** - six second response to arrest the immediate frequency deviation.
- **Slow raise** and **slow lower** - sixty second response to keep the frequency within the single contingency band.

- **Delayed raise** and **delayed lower** - five minute response to return the frequency to the normal frequency operating band of 49.85 to 50.15 Hz.

FCAS markets

The definition of eight separate and exclusive FCAS requirements has led to the development of eight distinct FCAS markets:

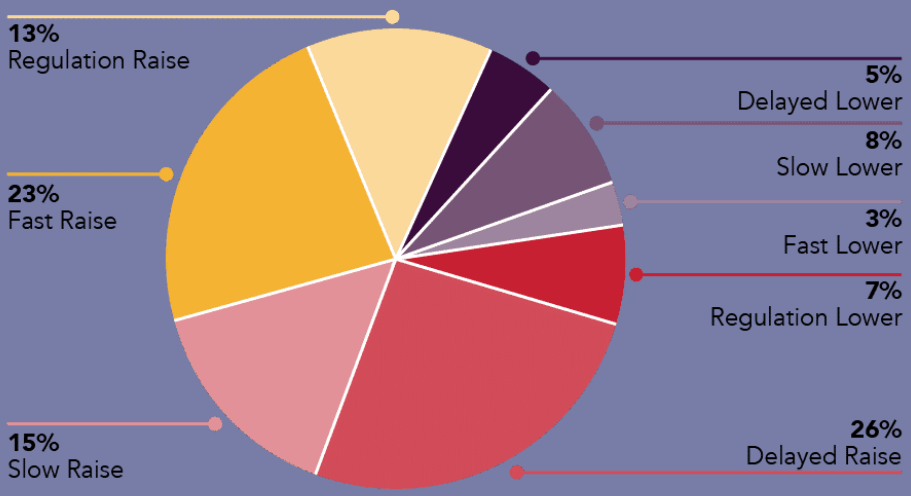
Regulation

- Regulation raise
- Regulation lower

Contingency

- Very fast raise (1-second raise)
- Very fast lower (1-second lower)
- Fast raise (6-second raise)
- Fast lower (6-second lower)
- Slow raise (60-second raise)
- Slow lower (60-second lower)
- Delayed raise (5-minute raise)
- Delayed lower (5-minute lower)

Participants must register with AEMO for each distinct FCAS market within which they wish to participate. Once registered, a service provider can participate in an FCAS market by submitting an appropriate FCAS offer or bid for that service via AEMO's market management systems.

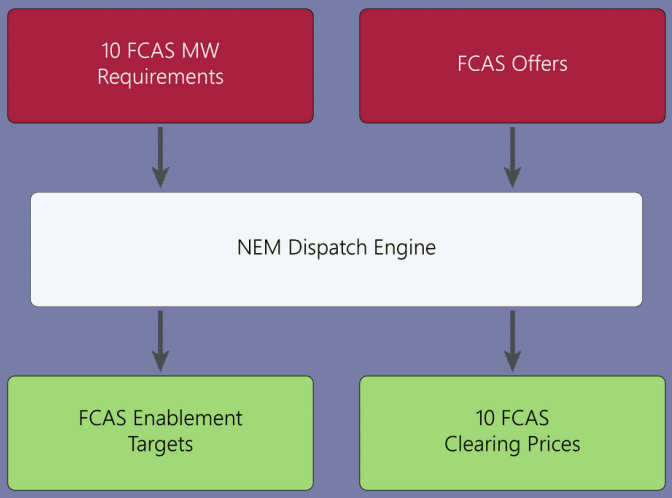


AEMO's NEM Dispatch Engine

An FCAS offer or bid submitted for a raise service represents the amount of megawatts that a participant can add to the system in order to raise the frequency in the given timeframe.

An FCAS offer or bid submitted for a lower service represents the amount of megawatts that a participant can take from the system in order to lower the frequency in the given timeframe.

During every market dispatch interval, AEMOs NEM Dispatch Engine (NEMDE) must ensure there is a sufficient amount of all eight FCAS services to meet the FCAS megawatt requirement.



NEMDE and co-optimisation

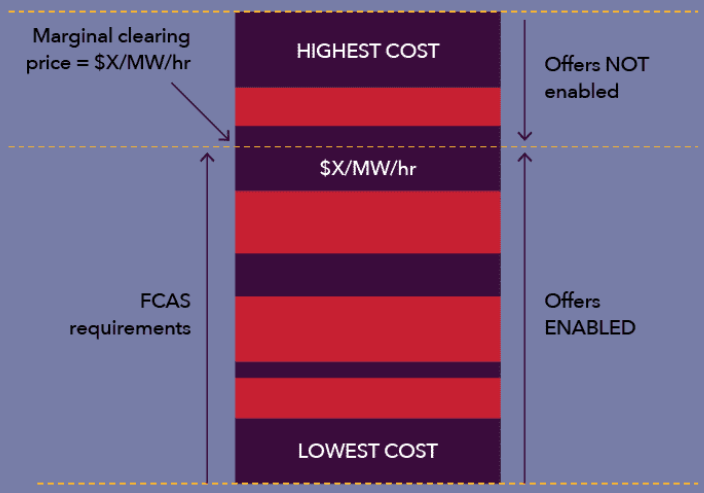
When determining the quantity of Regulation and/or Contingency FCAS, AEMO must calculate the required quantity that:

- can be sourced from any NEM region (global FCAS requirement).
- can only be sourced from nominated regions (local FCAS requirement).

Local FCAS providers are required in abnormal circumstances where only local market participants are practically able to provide FCAS. This most often occurs when a region becomes isolated — or 'islanded' — due to planned or forced transmission element outages.

When the services are required, NEMDE will enable FCAS offers in merit order of cost. The highest cost offer to be enabled will set the price for the FCAS service type.

During periods of high or low demand, it may be necessary for NEMDE to move a scheduled generator or load's energy target to satisfy FCAS requirements and thereby reduce the total cost of energy and FCAS to the market. This process is called 'co-optimisation' and is built into the dispatch algorithm.



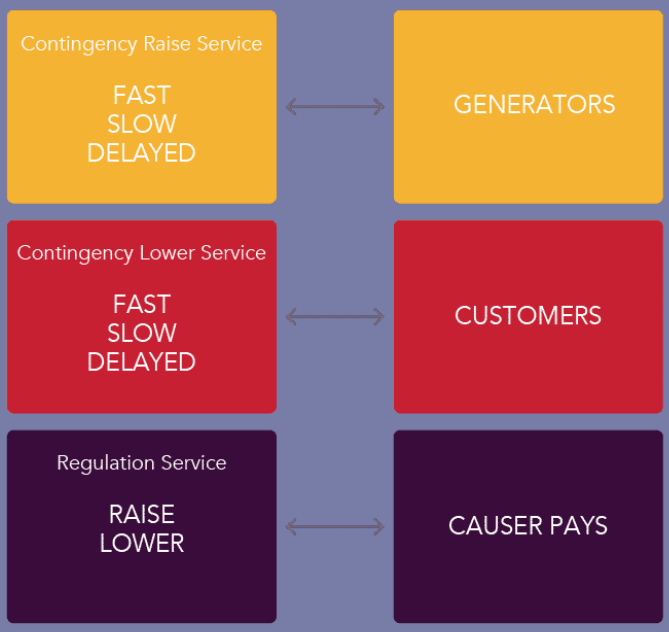
FCAS markets settlements

For each five-minute market dispatch interval, NEMDE determines a clearing price for each of the ten FCAS markets. This price is then used by settlements to determine payments to each FCAS provider.

All payments to FCAS providers are recovered from market participants according to the recovery rules. As contingency raise requirements are set to manage the loss of the largest generator on the system, all payments for the four contingency raise services are recovered from market participants who have sent energy into the grid. On the other hand, as contingency lower requirements are set to manage the loss of the largest load/transmission element on the system, all payments for the four

contingency lower services are recovered from those market customers who have consumed energy.

The recovery of payment for regulation service is based on a "causer pays" factor methodology (Contingency is also a form of causer pays principle). Under this methodology the response of measured generators and loads to frequency deviations is monitored and used to determine a series of factors.



Thumbnail

CONTINUE

Network Support and Control Ancillary Services (NSCAS)

Network Support and Control Ancillary Services (NSCAS) are primarily used to control the voltage at different points of the power network to within the prescribed standards, or keep the power flow on network elements within the physical limitations of those elements.

These services are provided either by NSPs or by AEMO through a non-market tender process. They are not part of the energy market.

Currently, this is divided into two categories:

Reliability and Security Ancillary Service (RSAS)

The purpose of RSAS is to maintain the power system within acceptable technical parameters, or to increase access to supply such that the NEM can maintain power system security and reliability of supply of the transmission network in accordance with the power system security standards and the reliability standard.

RSAS provides AEMO with the tools it needs to operate the NEM consistent with its system security and reliability obligations:

- To maintain the system in a secure operating state during normal operation, consistent with the power system security standards
- To return the system to a secure operating state within 30 minutes following a credible contingency event or protected event, consistent with the power system security

standards

- To ensure each NEM region has sufficient local generation and transmission capacity such that demand can be supplied consistent with the reliability standard

Market Benefit Ancillary Service (MBAS)

The purpose of MBAS is to maximise the present value of net economic benefit to all those who produce, consume or transport electricity in the electricity market. One way that MBAS can be delivered is by increasing power transfer limits in order to reduce the impact of constraint equations on NEM dispatch. AEMO uses constraint equations to model power system limits in the National Electricity Market dispatch engine (NEMDE), with each constraint equation providing a mathematical representation of a physical limit of the transmission network.
Examples of power system limits addressed through MBAS include, but are not limited to:

- Transmission thermal limitations
- Voltage upper and lower limitations
- Voltage stability
- Transient stability
- Oscillatory stability
- System strength limitations

CONTINUE

System Restart Ancillary Services (SRAS)

System Restart Ancillary Services (SRAS) enable the power system to be restarted following a complete or partial system blackout. There are a total of twelve system restart ancillary services and these align with designated electrical 'sub-networks'.

NSCAS and SRAS payments

Both NSCAS and SRAS are provided to the market under long-term ancillary service contracts negotiated between AEMO and the participant providing the service. These services are paid for through a mix of:

- Enabling payments: made only when the service is specifically enabled
- Availability payments: for every trading interval that the service is available
- Testing payments: made when a test is successfully conducted
- Usage payments: made when the service is successfully delivered in response to an instruction from AEMO

NSCAS payments are recovered from market participants who have consumed energy only, whereas SRAS payments are recovered equally (50:50) from those who have consumed and those who have generated.

Drag the terms on the left to match their definitions on the right.

System Restart Ancillary Service (SRAS)	Used to restart the system following a complete or partial system failure
Market Benefit Ancillary Service (MBAS)	Provides small corrections to keep frequency within the normal operating band
Contingency Frequency Control Ancillary Service (FCAS)	Can be used maximise the economic benefit to producers, consumers or transporters
Regulation Frequency Control Ancillary Service (FCAS)	Used to bring the frequency back within the operating band after a contingency event
Reliability and Security Ancillary Service (RSAS)	Extra services to support the security and reliability of the transmission system

SUBMIT

CONTINUE

Module summary

Congratulations! You've now completed **Module 6 - Ancillary Services**.

You should now have a reasonable understanding of the three types of ancillary services within the NEM: Frequency Control Ancillary Services (FCAS); Network Support and Control Ancillary Services (NSCAS) and System Restart Ancillary Services (SRAS). The module examined the markets that support FCAS, as well as the separate NSCAS and SRAS payment systems.

You may now review any section of this module via the main menu, or exit this module and proceed to **Module 7 - Metering**.

Module 7 - Metering

- Module introduction
- Metering within the NEM
- Metering Types
- Wholesale metering
- Retail contestability
- Net System Load Profile (NSLP)
- Metering operations
- Data accuracy
- Module summary

Module introduction

This module explores:

- Why we need metering
- Meter types
- Retail contestability
- Issues in the reconciliation of metering information
- The need for timely, accurate metering data

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

Metering within the NEM

Why do we have metering?

Metering can be thought of as the NEM's "cash register," capturing and reconciling the transactions of market participants. It enables the measurement, recording, and storage of electrical energy produced and consumed. Metering data is essential for accurately determining the volume of energy generated and used across the NEM.

AEMO uses metering data to perform the settlements process, enabling:

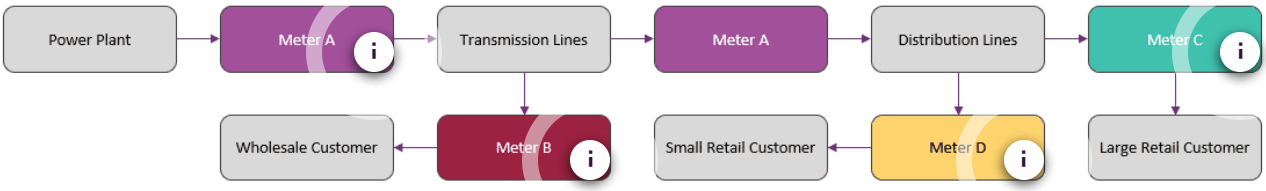
- Generators to be paid for the energy they supply to the NEM
- Retailers to be accurately billed for the energy they purchase on behalf of their end-use consumers
- Wholesale customers to be billed for the energy they purchase directly from the pool

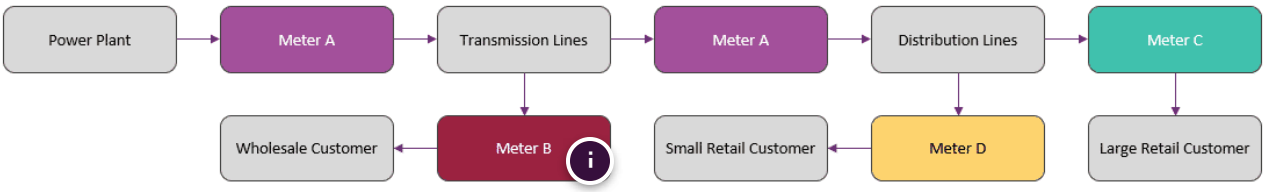
Transmission and distribution network service providers use metering data to determine their Use-of-System charges for transporting electrical energy across their networks. Retailers use metering data to bill their end-use consumers.

Metering within the NEM

Here is a simplified model of the NEM, showing energy flowing from generators through to wholesale and retail customers. As you can see, there are specific metering points designed to measure all energy moving through the system.

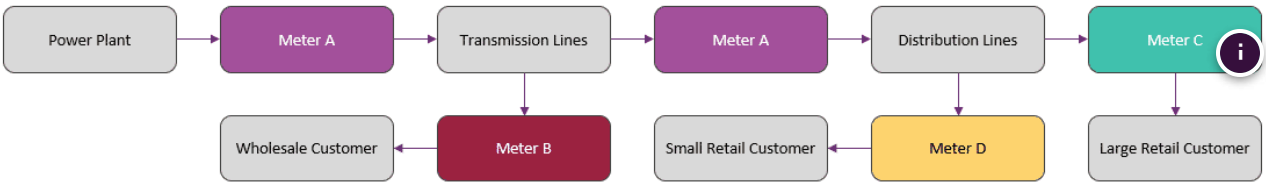
Click on the each of the markers below to learn more about each of the metering points.





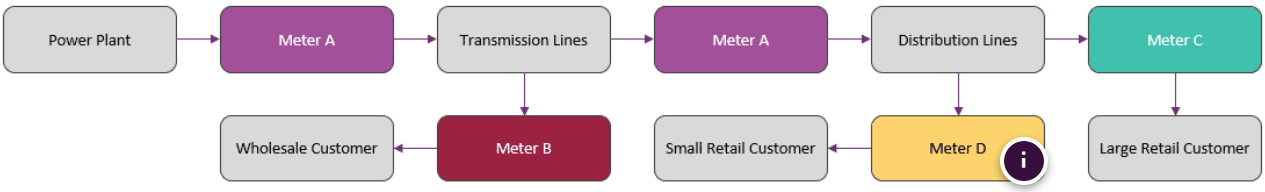
Wholesale customer

Aluminum smelters, mines and other large industrial users require large volumes of energy, so it's usually in their interest to purchase energy 'wholesale' directly from the Pool. Metering points are set up between the high-voltage transmission lines and these wholesale customers. These meters record energy consumption in 5-minute increments that are aligned to the price setting arrangements from the Pool.



Large consumer

Consumers such as factories, supermarkets, large office buildings and hospitals require a large volume of energy and will typically be contracted via a retail offering linked to the spot market price. These highly accurate meters record energy consumption in 5-minnute increments that are aligned to the price setting arrangements of the Pool. The metering information also relates to demand, enabling large customers to effectively manage their electrical load.

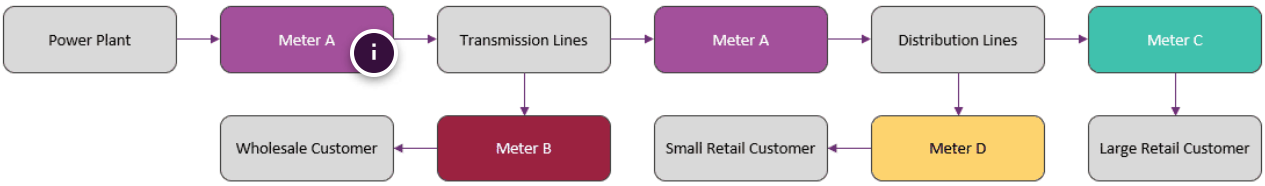


Small consumer

Most small consumers (such as houses, school buildings and small commercial operations) purchase their energy from a retailer offering a range of fixed-tariff options based on a combination of the following:

- the configuration of the consumer's electrical installation (this includes how their premises or property is set up, what electrical equipment they typically use, how and by whom the electrical equipment is controlled, and for how long)
- how much energy they require
- The tariff options and conditions made available to the retailer by the distributor for transporting energy on their distribution network

As small consumers rarely participate directly in the spot market, the complexity of metering installations can vary based on the consumer's agreement with their retailer and the jurisdictional requirements (which vary across the NEM). By the end of 2030, most small customers will have a smart meter (type 45) installed at their homes. These meters are read remotely and provide metering data in 5-minute intervals.



Generation and transmission metering

Measuring energy flow into and out of the transmission network, these meters record very high volumes of electrical energy throughput, so they must be extremely accurate. The readings taken from these meters drive the wholesale settlement of energy in the NEM.

National Metering Identifier (NMI)

In order to uniquely identify each connection point (including generation, transmission, distribution, wholesale and domestic customers) a National Metering Identifier (NMI) is assigned. The NMI is used throughout the market to identify a specific connection point. Over time that NMI will always stay with that connection point, irrespective of who occupies the premises, or whether metering equipment is replaced. NMI's have 10 characters and are issued by network service providers.

CONTINUE

Metering Types

To meet the demands of the different metering requirements across the NEM, and to deliver those services as cost effectively as possible, the National Electricity Rules (the Rules) specify the different categories of metering installations within the NEM.

The metering types all differ in their physical equipment specifications as well as the systems and processes that support their maintenance and information management.

Let's examine these metering types more closely.

Metering types 1, 2, 3 and 4

The first four types of meters measure heavy-traffic areas of the NEM. Typically this includes the energy flowing across the transmission network from generator to wholesale customers, to distributors and to consumers who have requirements for an interval metering arrangement at their premises.

The typical characteristics of metering types 1, 2, 3 and 4 are:

- Metering information is extracted from the meter remotely on a daily basis. This is typically done via a connection using Global System for Mobile communication (GSM), General Packet Radio System (GPRS) or other similar data transfer type.
- Energy is measured and recorded in 5-minute intervals for all metering types 1 to 3 and for most type 4's, enabling reconciliation with the Pool price setting

mechanisms.

TYPE 1 METERS

TYPE 2, 3 AND 4 METERS

Measurement points with an annual throughput of energy over 1,000 GWh have the highest metering standards applied. These are classified as type 1. Type 1 meters must be accurate to within 0.5%, and require a 'check', or duplicate metering installation, to verify the accuracy of the metering information.

TYPE 1 METERS

TYPE 2, 3 AND 4 METERS

Type 2, 3 and 4 metering installation requirements decrease in complexity and are applied in graduated steps based on annual energy throughput, typically down to 160 MWh, 150 MWh or 100 MWh, depending on Jurisdictional requirements.

Type 2 meters must be accurate to within 1.0% and require check metering to be installed. Types 3 and 4 meters must be accurate to within 1.5%.

Metering types 4S, 5 and 6

Metering types 5 and 6 measure energy usage in homes, small businesses, and other market sectors with relatively small to moderate annual energy throughput. Type 6 metering is typically used for domestic connections and type 5 for small commercial connections. Meter readers visit type 5 and 6 meters to obtain readings monthly or quarterly.

Type 5 meters record energy in 30-minute time segments and a meter reader must sight and download data from the meter.

Type 5 and 6 meters are being gradually replaced by type 4S meters under the Power of Choice program. Most Type 4S meters record energy in 5-minute time segments. Any Type 4S meter that was installed prior to 1 December 2018 can record in 30-minute time segments and did not need to be updated to recording in 5-minute intervals. The metering data from these meters is collected remotely on a daily basis.

Type 6 meters record energy consumption on a continuous basis and the cumulative value of energy consumption is displayed on the meter in kilowatt hours. A meter reader sights and directly reads the meter. The energy consumption for the period between readings is obtained by subtracting the previous reading from the current reading.

These less complex metering requirements minimise costs while maintaining high standards of accuracy.

By the end of 2030 most type 5 and type 6 meters will be replaced by Type 4S meters.

Metering installation for type 7

There are loads in the NEM where it is not practical or economic to meter on a connection by connection basis, but whose energy consumption can be calculated to a reasonable level of accuracy using an algorithm. Currently, this method is restricted to street lighting and, in some jurisdictions, traffic signals.

Using street lighting as an example, consumption is determined using industry-agreed load values (measured in Watts), on and off times (related to sunset and sunrise times) and the number and type of street lights.

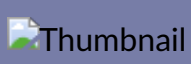
This way, although there is no physical meter for a type 7 "metering installation", energy consumption is calculated for billing and settlement purposes.

Metering type 9

Advances in metering technology have enabled the introduction of a new Type 9 metering installation, which came into effect on 31 May 2026. Type 9 metering supports the measurement of energy consumption for unmetered loads, including public lighting and certain categories of street furniture, such as electric vehicle charging infrastructure.

Under the revised arrangements, existing Type 7 metering installations may be replaced or upgraded to Type 9 metering installations. This reduces reliance on calculated consumption profiles and allows for the use of actual metered data for settlement purposes.

Certain unmetered loads, such as public infrastructure (for example public barbeques, and bus stop shelters) are non-contestable and are allocated to Local Retailers in settlements, with their consumption incorporated into AEMO's Global Settlements framework.



CONTINUE

Wholesale metering

Having explored the range of needs for metering data across the NEM, and the arrangements in place to support them, we will now examine how this data is integrated into the operation of the wholesale energy market.

Besides AEMO, there are five parties interested in metering for the wholesale market:

Generators

Generate electricity and sell it to the pool.

Transmission Network Service Providers (TNSPs)

Transmit electricity from generators to the Transmission Node Identifiers, known as TNIs. TNIs are the connection point between the transmission network and a distribution network - there may be multiple TNIs for a distribution network. TNSPs recover revenue based on the measured volumes of energy.

Wholesale customers

Purchase electricity from the spot market.

Local retailer

The retailer associated with a TNI — purchasing energy at that connection point from the Pool.

Distribution Network Service Providers (DNSPs)

Transport energy from the TNI to each connection point within their distribution network and recover revenue based on energy flows measured at each TNI and each connection point.

Key wholesale metering points

There are three key points across the wholesale market where measuring energy flow is critical:

- **Generation** - accurately measures energy flow from generators so they can be paid
- **Wholesale customers** - allows customers to trade directly in the spot market purchasing energy at pool prices
- **Transmission Node Identifier (TNI)** - sometimes referred to as boundary metering, measures consumption from the pool so that the total amount of energy consumed through the TNI can be measured and billed to the appropriate retailers (there is more information about this later in this module).

After applying the Marginal Loss Factor to metering data to account for transmission losses to and from the Regional Reference Node, we can see that once the metering

arrangements in the wholesale market are in place, selling and purchasing energy is reasonably straightforward.

CONTINUE

Retail contestability

Prior to the electricity retail market opening to competition in December 1998, all end-use consumers purchased energy from their Local Retailer.

With the introduction of full retail contestability across jurisdictions within the NEM in January 2002, energy consumed downstream from a TNI is no longer exclusively tied to the Local Retailer's customers. Most consumers can now select the retailer from which they purchase energy, and many have transferred away from their Local Retailer.

Where this occurs, the new retailer is known as a tier 2 retailer. Tier 1 retailers are the Local Retailer.

From a retail billing perspective, the NEM metering arrangements facilitate this process well, but to operate the settlements process we need to look beyond the wholesale market and into the retail market to determine how to facilitate retail contestability.

Settling the market

The 'old' way: Settlement by differencing

To enable retail contestability, whilst ensuring the wholesale market remained uncomplicated and transparent, the NEM originally operated a process called

Settlement by differencing to calculate the weekly energy bills for retailers.

Under Settlement by differencing, the Local Retailer was required to pay for all energy used within its distribution area, apart from the metered energy use of customers who had transferred to another retailer - i.e. the *difference* between the consumption for all tier-2 retailers and the total energy measured through the TNI.

This model was suitable at the commencement of the NEM because of the high proportion of customers contracted to a Local Retailer, and the large quantity of type 5 and 6 meters that needed to be read manually.

Global Settlement

Over time, the number of consumers still using the Local Retailer diminished. When coupled with the gradual replacement of older type 5 and 6 meters for remotely-read meters and technological advances that increased data processing power, it was now feasible to calculate the individual usage for Local Retailers.

On 1 May, 2022, the NEM switched from using Settlement by differencing to **Global Settlement**. Under Global Settlement, the weekly statements of Local Retailers are now calculated in the same way as for tier 2 retailers, using the usage information of the NMI's of their customers.

In this process, each TNI has a Local Area. Usage data for NMI's within that area is aggregated by retailer, to be settled in their weekly statement.

Once all of those individual NMI's have been aggregated by Retailer, all that 'known' energy is then subtracted away from the total aggregated energy for the Local Area, and any balance is called Unaccounted for Energy (UFE). This is then apportioned across all of the Retailers in the Local Area. UFE then becomes a line item on the Retailers settlement account.

Distribution Loss Factors

End-use customers' metering data must be multiplied by an appropriate Distribution Loss Factor (DLF) to account for any losses across the distribution system and ensure that the TNI settlement calculation is accurate.

Load Profiling

Because Settlement processes occur weekly while energy is traded in 5-minute time segments, correctly calculating retailers' statements needs to account for the relevant 5-minute window within which consumer uses energy. This is complicated by the reality that many users still have meters that either record energy use in 30-minute blocks, or don't record time of use at all (i.e. type 6 meters).

To convert metering data for each connection into 5-minute intervals, AEMO uses two complex profiling methods known as **Net System Load Profile** (NSLP) and **Controlled Load Profile** (CLP).

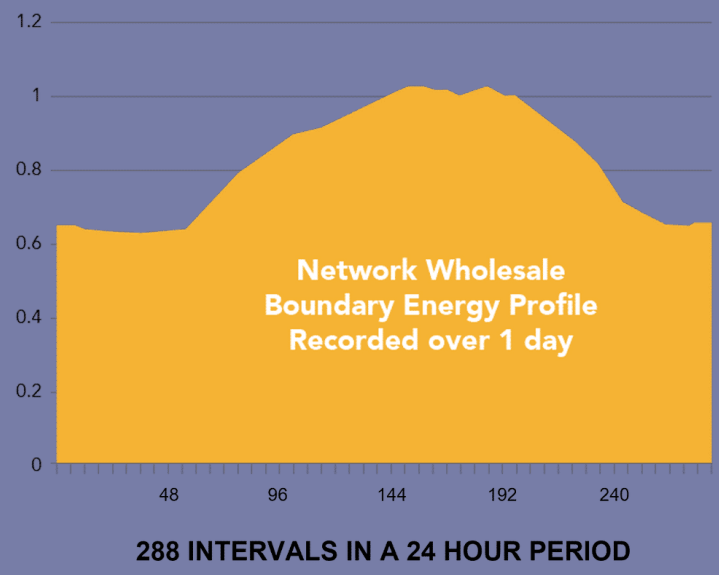
CONTINUE

Net System Load Profile (NSLP)

The graph below is a typical representation of the energy throughput profile at a TNI across a day, measured and recorded in 5-minute intervals. There are 288 5-minute intervals in a 24-hour day.

The NSLP represents the aggregated load consumed by non-interval metered connection points. "Known" 5-minute metering data from type 1-4, 4S and 5 meters and calculated metering data from type 7 metering installations is subtracted from the TNI profile, which is recorded and stored in 5-minute intervals, and the residual is the NSLP. The NSLP is used to convert non-interval metering data into 5-minute intervals to enable settlements to be calculated using the 5-minute spot price.

Let's examine the calculation of NSLP.

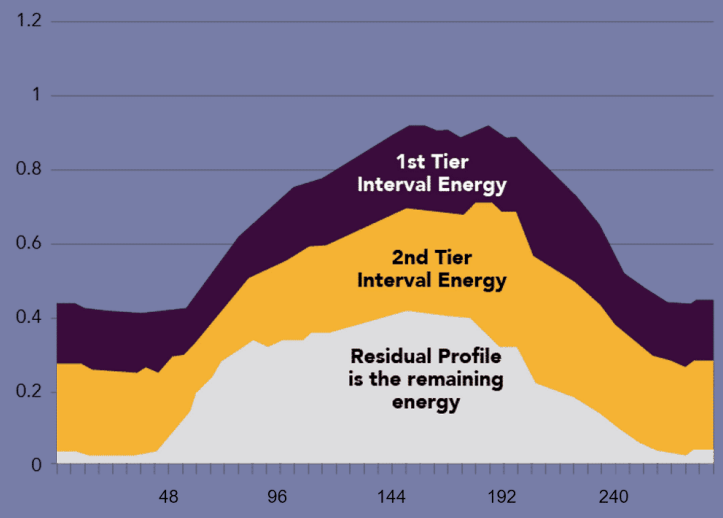


The market is settled on 5-minute interval data.

The market needs to convert non-interval (accumulated / basic) data into 5-minute intervals for settlements.

NSLP - 'known' consumption

Energy consumption for type 1, 2, 3, 4, 4S, 5 and 9 meters downstream from the TNI can be subtracted directly from the profile at the TNI. This "known" consumption information is already recorded in 5-minute intervals. The profile shape of the remaining energy throughput, including the energy consumption NOT recorded in 5-minute intervals, changes significantly.

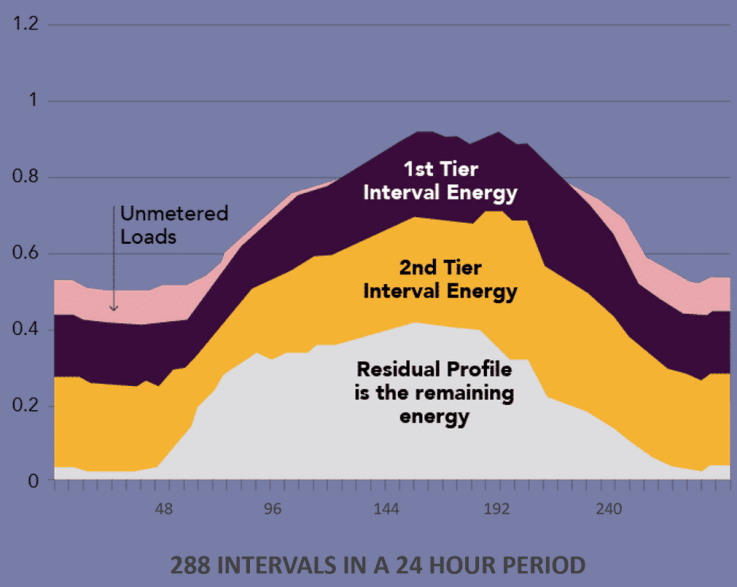


288 INTERVALS IN A 24 HOUR PERIOD

Known internal data is identified.
The light grey section (the residual profile) is decreasing in size.
The overall profile shape remains the same.

Type 7 loads

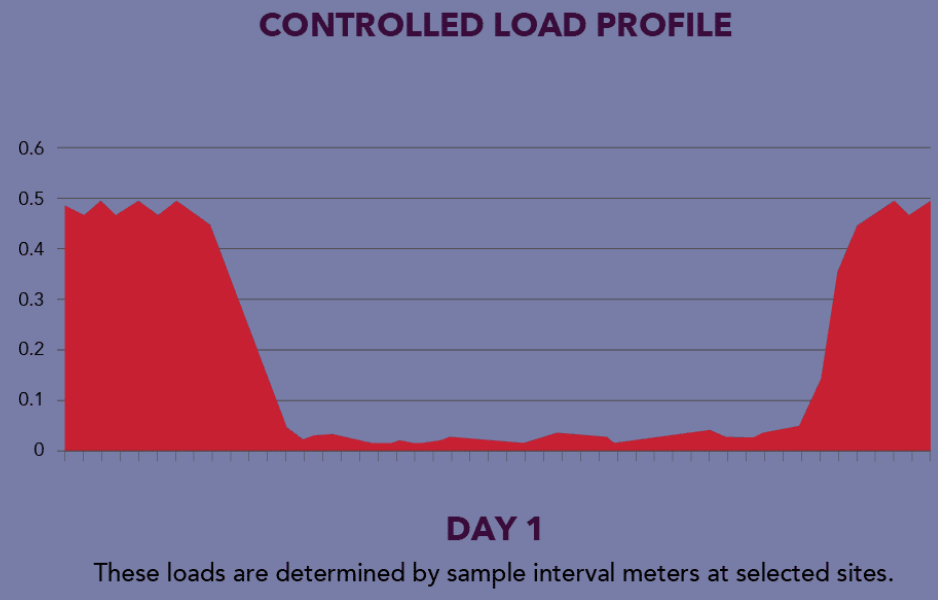
AEMO can then identify and subtract the type 7 metering loads at the TNI, as this "unmetered" energy consumption can be calculated in 5-minute intervals based on an established algorithm.



Unmetered loads primarily consist of public street lighting.
Unmetered loads are the next known data to be subtracted.

Controlled load profile

Where required by a jurisdiction, the next step is to identify the Controlled Load Profile (CLP) that relates to designated domestic or commercial loads such as water heating and storage heating, on and off times being controlled by the DNSP and the profile determined by sample meters that record energy consumption in 5-minute intervals. The CLP is then subtracted from the remaining energy profile at the TNI.



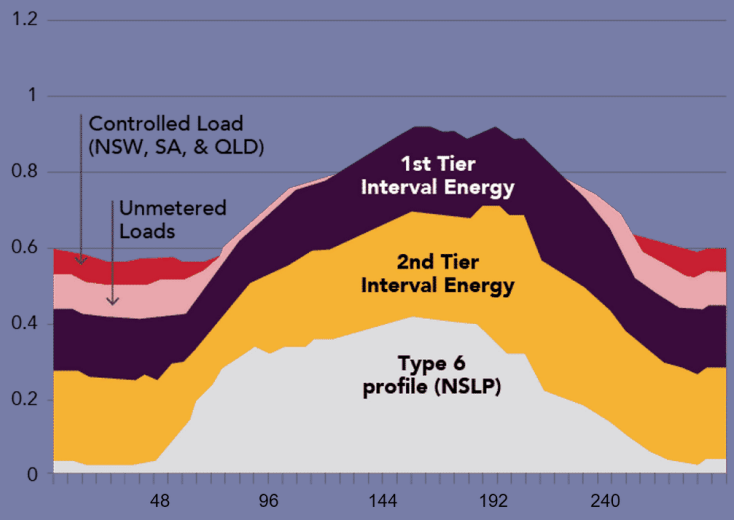
These loads are determined by sample interval meters at selected sites.

Controlled loads are only calculated in QLD.

Determining the NSLP

The residual profile is now the NSLP and is applied to all metering points where 5-minute interval metering data is not available. This determines the energy purchase price for all retailers to enable accurate and transparent settlement at the TNI.

The NSLP for type 6 metering installations will diminish over time as type 6 meters are replaced with type 4S meters.



288 INTERVALS IN A 24 HOUR PERIOD

Overall shape remains the same for the day. NSLP is now defined.

Controlled load profile intervals identified (QLD only).

Unmetered loads with a pre-determined interval profile identified.

Residual is the interval profile for the type 6 installations.

CONTINUE

Metering operations

How does AEMO ensure that metering rules are clearly understood, correctly applied, and operate effectively and efficiently?

To support this, the NEM defines specific roles and responsibilities for the provision and management of metering services.

Financially Responsible Market Participant

The Financially Responsible Market Participant (FRMP) is a registered participant (usually a generator or retailer) and must ensure that a connection point has a metering installation.

Retailers bill customers and pay AEMO for energy used, while generators produce energy and get paid by AEMO for the energy they have supplied.

The FRMP is responsible for engaging the Metering Coordinator.

Local Network Service Provider

The Local Network Service Provider (LNSP) is responsible for establishing connection points, allocating the National Metering Identifier (NMI), providing the distribution loss factors and determining Transmission Node Identifiers (TNI) for connection points.

Metering Coordinator

The Metering Coordinator is formally responsible for the provision of metering installations and ensuring that they meet all obligations under the Rules. It is the Metering Coordinator's role to engage Metering Providers and, except when the Metering Coordinator is also the TNSP, the Metering Coordinator also engages the Metering Data Provider. This is to ensure that the metering installation and the parties providing metering services and metering data services are adhering to Rules requirements.

The role of the Metering Coordinator is determined firstly by metering type and then by formal agreement:

- The Metering Coordinator for type 1-4, 4S and 9 metering installations will be engaged by the retailer for the connection point
- The Metering Coordinator for type 5, 6 and 7 metering installations is always the Local Network Service Provider

Metering Providers (provide, install and maintain metering)

Metering Providers are registered and accredited by AEMO. They are engaged by the Metering Coordinator to provide Rules-compliant metering services for installation and maintenance of physical metering equipment.

Metering Data Providers (collect, process and transfer data)

Metering Data Providers are registered and accredited by AEMO and engaged by Metering Coordinators to collect, process and transfer metering data. This includes manual meter reading for types 5 and 6 metering installations, and remote collection for types 1, 2, 3, 4, 4S and 9 metering installations.

Metering Data Providers also produce calculated metering data for type 7 metering installations. They provide metering data to AEMO and participants.

Local Retailer

Also known as the Host Retailer or 1st Tier Retailer (all other retailers are considered 2nd Tier). They are defined geographically by Distribution Areas. There are only six Local Retailers in the NEM.

Retailer of Last Resort

The Retailer of Last Resort (ROLR) is the default retailer that is assigned to a customer if their retailer of choice is suspended from trading in the NEM. Historically this was the Local Retailer, but is now determined by the Australian Energy Regulator (AER) in all jurisdictions.

CONTINUE

Data accuracy

Quality metering data delivered to the market on time is the foundation for all metering and settlements processes in the NEM. Accordingly, Metering Data Providers must undertake the following:

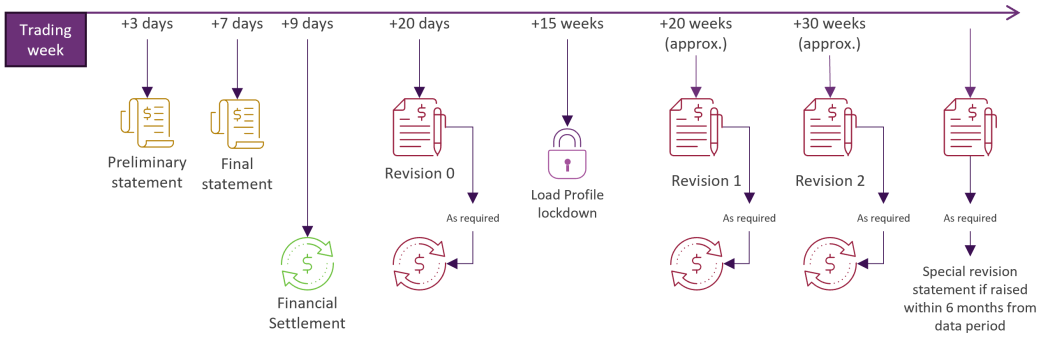
- Deliver a minimum of 99% of meter data to AEMO for settlements
- No more than 1% of this data can be substitute data

Metering Data Providers will substitute data if they are unable to obtain actual data from the meters. AEMO sets the substitution rules to ensure the best possible quality data available.

Settlement calendar

The chart shows how AEMO deals with different data delivery deadlines and how these relate to the settlement cycles. AEMO initiates five settlement runs per trading week - preliminary, final, revision 0, revision 1 and revision 2. Let's examine each of these more closely.

METERING DELIVERY AND SETTLEMENT CYCLES



An overview of the approx. 6-month settlement cycle carried out for each trading week

+3 business days

Preliminary statement

Preliminary settlement statements are issued to participants three business days after the end of the trading week. Given that type 5 and 6 meters are only read monthly or quarterly, metering data at this stage includes a large number of forward-estimated values.

+7 and +9 business days

Final statement and Financial settlement

AEMO issues final settlement statements seven business days after the end of the trading week, with financial settlement by the ninth business day. Settlement bills are issued and

money changes hands: retailers pay AEMO by 10:30am (Sydney time), and generators are paid by AEMO by 2pm. This allows an initial exchange of money between NEM participants.

+20 business days

Revision 0 settlement

Revision 0 settlement occurs twenty business days after the trading week. Credits and debits are issued to retailers between the final and revision 0 settlement dates.

+15 weeks

NSLP lockdown

Given the dynamic nature of Net System Load Profiles (NSLP) - they change with every settlement run - they are locked in on a weekly basis, approximately 15 weeks after the settlement week.

+20 weeks

Revision 1 settlement

Revision 1 settlement occurs approximately twenty weeks after the end of the trading week. The credits and debits cover variances cover variances for data received between Revision

0 and Revision 1.
Because all manually read meters should have been read within this time frame (except those with access problems), the meter data is more accurate.

+30 weeks

Revision 2 settlement

Revision 2 settlement is the last settlement, run approximately thirty weeks after the initial trading week. This settlement acts as the final 'clean up' correcting any anomalies. Special revisions are run if there is more than 5% difference between revision 1 and revision 2.

Revision 0 was introduced on August 9, 2026 when the Settlement Cycle was shortened by an AEMC rule change.

Prior to this, the settlement cycle consisted of a Preliminary Statement at 5 business days after the end of the billing week, the Final Statement at 18 business days after the billing week, and Financial Settlement at 20 business days. The timing of Revisions 1 and 2 was not changed.

Because there are still meters that are only read monthly or quarterly, Revision 0 was introduced (coinciding with the original

financial settlement at 20 business days) to help manage the financial implications for Participants until there is a greater deployment of smart meters across the NEM.

AEMO data validation and issue resolution

Prior to submitting data for settlements, AEMO performs a set of validations to ensure quality data. These validations include identifying any errors or anomalies and resolving them with participants and Metering Data Providers.

AEMO quality management

In addition to the data validation processes, AEMO undertakes a number of quality assurance measures for the NEM:

- All Metering Service Providers must be accredited by AEMO
- Accreditation of Metering Service Providers is based on their ability to comply with the Rules and AEMO Service Level Procedures, and to maintain ISO 9000 series accreditation
- AEMO performs regular audits of Metering Service Providers to ensure ongoing compliance
- AEMO monitors performance and reports on all parties' activities and obligations related to metering

AEMO manages various industry reference and working groups, ensuring regular, proactive engagement with participants, and subject matter expertise is pooled wherever possible for the operation, development and overall benefit of the NEM.

To ensure AEMO meets its obligations, an independent auditor audits AEMO on a regular basis.

CONTINUE

Module summary

Congratulations! You've now completed **Module 7 - Metering**.

You should now have a reasonable understanding of the relationship between metering information and the settlement process, as well as the roles, responsibilities and obligations of all parties in metering. The module also examined the wholesale market, retail market, retail contestability and issues in the reconciliation of metering information. Finally, we examined how timely, accurate metering data is important for the operation the NEM.

You may now review any section of this module via the main menu, or exit this module and proceed to **Module 8 - Settlements and Prudentials**.

Module 8 – NEM Settlements and Prudentials

- Module introduction
- Introduction to settlements
- Settlements in the physical energy market
- Settlement statements
- Energy sales and purchases
- Compensation
- Application of GST to NEM transactions
- Adjustments
- The billing cycle
- Payments

- Disputes
- Activity - settlements terminology
- Prudential supervision
- Daily prudential supervision
- Settlements residue
- Settlements in the financial electricity market
- Activity - prudentials terminology
- Module summary

Module introduction

This module addresses the following learning objectives:

- To explain the role of settlements in the NEM
- To explain AEMO electricity settlement statements
- To describe the main transactions that occur in the NEM
- To identify calculations required for key settlement transactions
- To explain the quality control of settlement statements and the process for adjustment to settlement statements
- To explain the billing and payments cycle, settlement calendar, timetable and reporting
- To explain prudential supervision in the NEM
- To explain how settlement residue arises

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

Introduction to settlements

Settlement is the process of determining financial payments, billing and settling for the amounts of electricity produced or used. AEMO calculates the financial liability or receivable for all market participants on a daily basis and settles transactions for all trade in the NEM weekly.

This involves AEMO collecting all money due for electricity purchased from the pool from market customers, and paying generators for the electricity they have produced. The spot price is the basis for all these financial transactions.

Settlement transactions administered by AEMO include but are not limited to:

- Spot market energy transactions
- Reallocation transactions
- Ancillary service transactions
- Participant fees

The role of metering in settlements

In the previous module, we learnt about the importance of metering in the NEM. In particular, metering is important for settlements, as energy settlement calculations are based on metering information.

Market participants are required to have appropriate meter and communication methods that register and store electricity readings and upload this data to AEMO. Market participants must contract an AEMO-registered meter data provider (MDP) to provide the data forwarding service to AEMO.

The MDP is responsible for the following:

- Acquiring metering data for each meter for a market participant
- Validating metering data and, if necessary, applying substitutions to gaps in the data
- Setting the next scheduled read date for basic (non-interval) meters
- Applying the relevant connection point distribution loss factors to meter readings provided to participants
- Forwarding metering data to the market participant and AEMO settlements using the ownership rules contained in MSATS (Market Settlement and Transfer Solution)

MDPs also provide metering data directly to the market participant financially responsible for the connection point. Market participants can use this information to verify aggregation and settlement calculations.

CONTINUE

Settlements in the physical energy market

The physical market involves the real time dispatch and consumption of electricity to meet demand while maintaining system security. The settlement systems and function then uses information from the physical market, such as energy generation, consumption, and pricing to calculate the financial liabilities and credits of market participants. Based on this information, AEMO issues statements and facilitates financial settlement through its clearing house processes, in accordance with the Rules.

The settlement process ensures that registered participants are appropriately paid for energy supplied to the grid and charged for energy consumed, in accordance with the Rules.

Settlement receipts and payments by AEMO are made through Austraclear (operated by the Australian Securities Exchange). AEMO is not liable for any shortfall in funds received from participants in order to pay out to others.

CONTINUE

Settlement statements

Settlement statements are issued electronically to Market participants through the markets portal in a detailed settlement report. An accompanying high-level settlement summary shows the aggregated amounts for each category of transaction.

The settlement statement that AEMO issues to participants could include, but is not limited to:

- Payment to and from Market participants for Sent Out and Consumed Energy
- Payment and recovery of ancillary services
- Any compensations
- Any billing adjustments
- Payment of NEM participant fees

Who pays participant fees?

Participant fees in the National Electricity Market (NEM) are paid by all Market participants, including retailers, generators, Integrated Resource Providers (IRPs), and Transmission Network Service Providers (TNSPs).

AEMO determines how these fees are structured, including how they are calculated and how costs are shared across different participant groups. This ensures that each participant class contributes an appropriate proportion of the overall fees.

The fee structure is reviewed in consultation with industry every five years to ensure it remains fair and appropriate.

How are participant fees calculated?

Participant fees are calculated using a structure developed by AEMO in consultation with industry. This structure varies depending on the type of participant.

For Market Customers and non-registered generator sites, fees are typically based on energy usage. This means that the more energy a participant consumes or generates, the higher the fees they will pay. In addition, some fees are influenced by the number of connection points a participant has. In recent years, the rate has been approximately $0.50 per megawatt hour (MWh) of energy consumed.

For wholesale participants, fees are calculated differently. They are based on a combination of:

- The amount of energy produced
- The maximum capacity of registered generators and Bi-directional Units (BDUs)

These two components are weighted equally (50/50) in the calculation.

Once the total fee amount is determined, AEMO notifies each wholesale participant, and the amount is then paid on a daily basis.

CONTINUE

Energy sales and purchases

Energy sales and purchases are determined for each five-minute dispatch interval by region. This is represented by:

Trading amount = adjusted gross energy* x transmission loss factor x regional reference price

* The adjusted gross energy is the raw metered energy at the connection point plus any adjustments for allocated losses in the distribution network.

The trading amount can be positive or negative. Energy imported into the NEM (in other words, sales from generators) is positive. Energy taken from the NEM (in other words, being purchased by retailers) is negative. Positive trading amounts represent a payment by AEMO to a market participant. Negative trading amounts represent a payment to AEMO by a market participant.

CONTINUE

Compensation

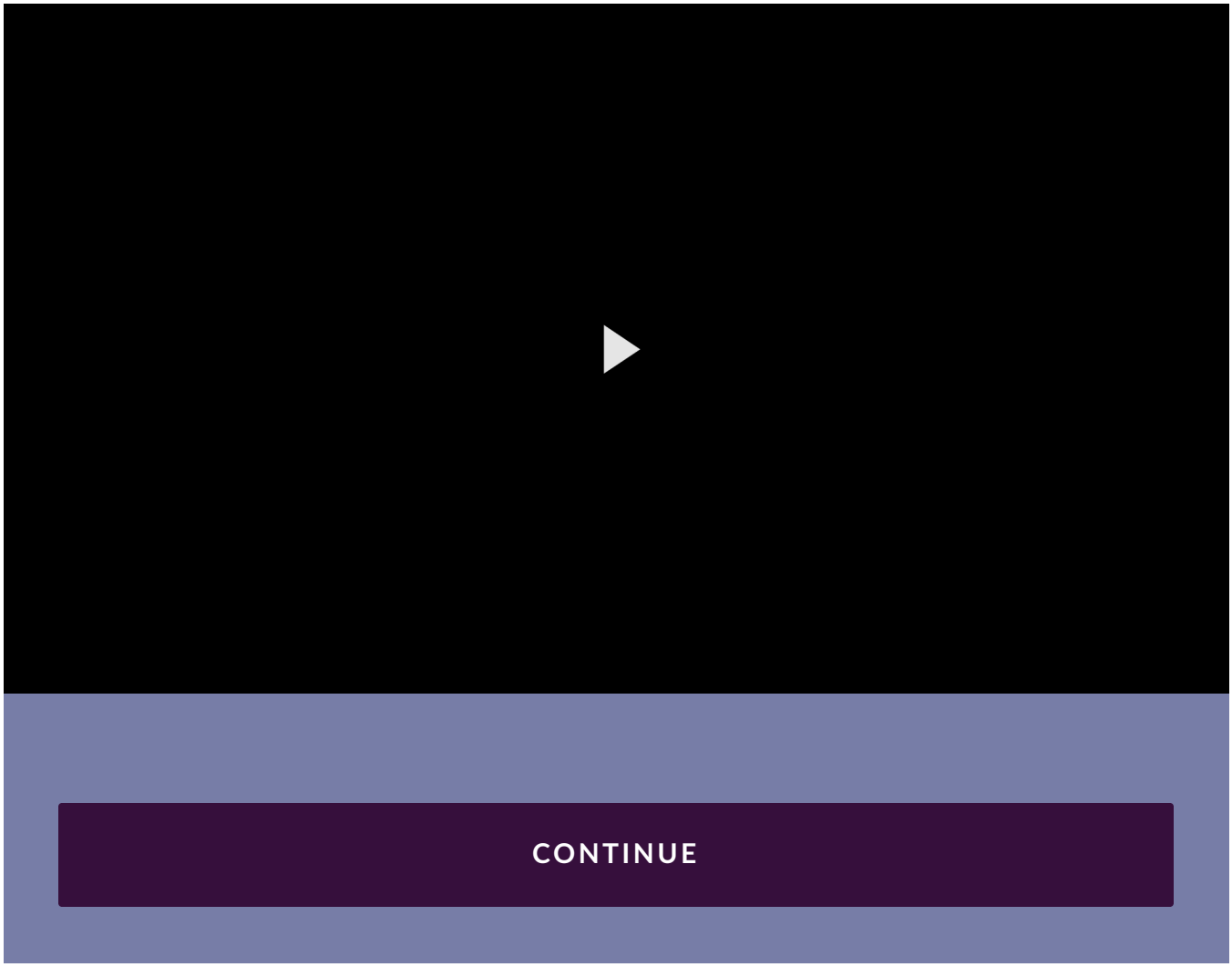
Conditions which may attract compensation for a participant include, but are not limited to, intervention compensation, the administered price cap and ancillary services.

Intervention compensation

To maintain adequate reserve in the power system, AEMO could issue a direction to a generator regarding one of its scheduled generating units. This may cause the generator to be in a different financial position to the one it would have been in had the intervention not occurred. Compensation may also be payable when AEMO makes a system security direction or an ancillary services direction.

Administered price cap

The Rules require AEMO to cap the regional reference price during administered price periods to a level below the spot price calculated by the market dispatch and pricing system. Under these conditions, compensation may become payable to generators that were dispatched even though their bid prices were higher than the capped value. These compensation payments are recovered from market participants (mainly retailers) in proportion to their energy purchases during the affected periods.



CONTINUE

Application of GST to NEM transactions

The Goods and Services Tax (GST) is applied to most NEM transactions, including participant fees.

AEMO publishes GST-exclusive regional reference prices. This approach has been supported unanimously by Market participants and also external interested parties - mainly financial institutions. With GST-exclusive pricing, calculating and publishing the regional reference prices continues as it did prior to the introduction of GST. Taxable transaction amounts are increased by GST (currently 10%) at the time of the statement issue.

NEM transactions range across all three GST categories:

- Taxable supply
- Supplies that are input taxed
- Supplies that are GST free

AEMO issues tax invoices and recipient-created tax invoices with each final statement. This enables market participants to claim input tax credits and/or recognise their GST liability. Adjustment notes and recipient-created adjustment notes are issued with each revised settlement statement.

CONTINUE

Adjustments

AEMO processes two types of adjustments:

- For amounts not calculated automatically by the system in the current billing period
- For amounts being differences between a revised settlement statement and the previous statement for the same billing period. These amounts attract an interest payment in recognition of the delay in payment and the time value of money

CONTINUE

The billing cycle

AEMO settles the wholesale electricity market over a seven-day billing period. The period is from Sunday to Saturday of each week and consists of 2,016 trading intervals, each lasting five minutes.

Each market participant receives a preliminary statement on the third business day after the end of each billing period. This statement is accompanied by supporting data relating to the transactions in that billing period and the prices at which electricity was bought or sold.

A final statement is issued on the seventh business day after the end of each billing period. This statement contains the amount payable or receivable by the market participants for that billing period.

The billing cycle

In the National Electricity Market (NEM), billing refers to the process of aggregating settlement amounts for each billing week. Settlement statements are issued according to a billing cycle defined in the National Electricity Rules (NER).

AEMO first issues a preliminary statement on the third business day after the billing week. This statement is for information purposes only. It is followed by the final statement, issued on the seventh business day. The amount on the final statement must be paid and is financially settled two business days later.

To ensure accuracy, revision statements are also issued after the final statement. These occur at 20 business days, 20 business weeks and 30 business weeks after the billing week. The billing week is only considered fully complete once these revision processes are finalised.

CONTINUE

Payments

Market participants must pay AEMO in cleared funds, any net amount payable by 10:30am (Sydney time) on the ninth business day after each billing period. This obligation exists even if the Market participant disputes the amount. All payments are made using the Austraclear electronic payment system.

AEMO pays Market participants any net amount receivable, in cleared funds, by 2:00pm (Sydney time) on the same day payment is due from other Market participants.

AEMO's prudential dashboard

Market Participants in the NEM have access to multiple sources of information to understand and manage their financial obligations. When AEMO issues a final statement, Market participants rely on this document as the primary record of the amount owed for a given billing period. These statements provide a formal and authoritative summary of liabilities.

For periods where a statement has not yet been issued, AEMO provides a range of electronic information services to ensure Market participants still have timely visibility of their financial

position. These tools enable Market participants to monitor their obligations on an ongoing basis, rather than waiting for formal statements.

One key resource is the Prudential Dashboard, an online platform that consolidates critical financial and risk-related information. Through this dashboard, Market participants can access:

- A summary of all outstanding settlement periods
- The amount owing for each individual period
- Their total outstanding balance
- Their current maximum credit limit (MCL)
- Details of the credit support (such as bank guarantees or cash security) they have in place

This comprehensive view allows Market participants to actively manage their financial exposure and ensure compliance with prudential requirements. By providing both detailed and real-time information, AEMO supports transparency and helps maintain the financial stability of the market.

CONTINUE

Disputes

Market participants must use reasonable endeavours to resolve any dispute regarding the preliminary statement within 15 business days of the end of the relevant billing period.

They must raise disputes of the final statement or routine revised statement within six months of the relevant billing period.

Market participants will receive revised settlement statements approximately 20 business days, 20 business weeks and 30 business weeks after each billing period based on resubmitted metering data. These statements include any adjustments from disputes that have been resolved.

Disputes

The preliminary statement is issued three business days after the end of the billing week. While it is provided for information purposes only, it is highly valuable to participants for two key reasons:

- It gives an early indication of the amounts likely to be payable or receivable in the final statement, which is issued on the seventh business day after the billing week.
- It includes supporting data that participants can use to reconcile and validate the amounts.

Participants can review this information and raise any questions or disputes with AEMO during the period between the preliminary and final statements. This step is important because the amounts in the final statement must be financially settled, even if they are under dispute. Early identification of issues allows AEMO to investigate and, if necessary, correct them before the final statement is issued.

If a dispute is not resolved before the final statement, adjustments can still be made through revision statements. Standard revision statements are issued at 20 business days, 20 business weeks and 30 business weeks after the end of the billing week. These revisions adjust the final statement amounts that have already been financially settled. In some cases, AEMO may also issue a special revision statement to address significant issues identified after the final statement.

The revision process is important for several reasons. It allows time for all actual meter data to be included before a billing week is closed. For example, many residential meters are read quarterly, so the revision window ensures this data is incorporated into settlements.

Additionally, the extended revision timeline means that Market participants remain financially responsible for their activity in the NEM for more than six months. This applies even if they have become inactive or have been removed from the market.

CONTINUE

Activity - settlements terminology

Drag each of the terms on the left to match it's definition on the right.

Preliminary Statement	Issued to market participants on the 3rd business day after the end of a billing period
Revision Statements	Responsible for acquiring metering data from each meter for a market participant
Final Statement	Issued on the 7th business day after the end of each billing period
Billing Period	The period from Sunday to Saturday each week, comprising 2,016 dispatch intervals

Meter Data Provider	Adjustments to the amount payable made at approx. 20 business dasys, 20 and 30 business weeks after the billing period

SUBMIT

CONTINUE

Prudential supervision

Settlement transactions are typically around $150-$350 million per week and generators expect reasonable assurance that these amounts will be paid on the due date. To assure these payments, the Rules require AEMO to monitor the liability that Market participants have to the market and compare it to the credit support amounts provided by each Market participant.

The NEM is a 'blind market', which means suppliers (generators) are unable to identify the customers (retailers) who purchase their output. Unlike other markets, it is therefore not possible for the supplier to undertake the normal customer credit risk assessments and set prices and terms of payment accordingly. For this reason, there are minimum acceptable credit criteria provided under the Rules. These are designed to ensure the credit risk of Market participants is not factored into the determination of the market spot price for electricity transactions.

Prudential arragements in the NEM

In any marketplace, it is essential to maintain market integrity and ensure sellers have confidence they will be paid for the goods and services they provide. For electricity generators in the NEM, timely payment is vital and without it, their financial viability and continued operation would be at risk.

However, the NEM presents unique challenges that complicate these requirements. The NEM operates as a blind market, meaning generators do not know the identity of buyers purchasing their electricity. Instead, generators sell their energy to AEMO,

which then aggregates and on-sells this electricity to retailers, resulting in generators having no direct visibility or control over who ultimately purchases their energy.

A further complication is the timing of payments. Generators are typically paid up to two weeks after supplying electricity to the market. This delay increases financial risk, as generators must rely on the assurance that payments will be made in the future, despite not knowing the specific counterparties involved.

Without mechanisms to manage these risks, generators may respond by incorporating additional risk premiums into their pricing. For example, beyond accounting for operational costs such as fuel and maintenance, generators might increase prices to hedge against the possibility of delayed or non-payment. This would lead to price distortion, where electricity prices reflect not only production costs but also perceived financial risks within the market.

To address this issue and promote price efficiency, the NEM employs a system of prudential arrangements. These are mandatory financial safeguards imposed on electricity buyers (primarily retailers) to ensure they can meet their payment obligations. By requiring retailers to demonstrate financial capacity and provide appropriate guarantees, these arrangements provide generators with confidence that they will be paid on time.

Prudential arrangements play a crucial role in stabilising the market by reducing financial uncertainty, preventing unnecessary price inflation, and supporting the reliable operation of the electricity system by ensuring that all Market Participants can transact with confidence.

Credit Requirements and Maximum Credit Limit

All Market participants with net liability to AEMO must satisfy the acceptable credit criteria, or must provide AEMO with an unconditional bank guarantee or cash security (up to $20m, effective from November 2026). A bank guarantee must be in the form specified by AEMO, from an acceptable credit support provider, for an amount which must not be less than the market participant's calculated maximum credit limit (MCL).

The MCL is set to ensure that a 2% prudential standard is met. This standard relates to the probability of a Market participant breaching its trading limit on a given day and subsequently its outstandings exceeding its prudential margin in the next 7 days if the trading limit breach is not remedied.

The credit period is 28 days (21 day outstandings period and 7-day reaction period). MCL Regional parameters (Volatility Factors and Average Price) are published in advance for all regions on AEMO's website.

The review periods are seasonal:

- Summer (1 Dec - 31 Mar)
- Winter (1 Apr - 31 Aug)
- Shoulder (1 Sep - 3 Nov)

Maximum Credit Limit Review

AEMO conducts regular prudential reviews of each Market participant to ensure that their financial obligations are appropriately managed. These reviews typically occur every three to four months, following the seasonal calendar (Winter, Summer, Shoulder). Although AEMO may undertake them more frequently if there are notable changes in a Market participant’s market activity.

As part of this process, AEMO calculates a key metric known as the Maximum Credit Limit (MCL). The MCL represents the minimum amount of credit support that a Market participant must provide to cover the risk of non-payment within the market, such as cash security or a bank guarantee.

In determining the MCL, AEMO considers several important factors:

- **Load and generation levels:** AEMO assesses the participant's historical electricity consumption (for retailers) and generation output (for generators). This historical data provides a baseline for estimating future exposure.
- **Trends in activity:** AEMO also evaluates any observable changes in a participant's operations. For example, a retailer acquiring new customers would likely experience increased demand (load), which in turn raises its financial exposure in the market.
- **Spot price levels and volatility:** In addition to examining average historical prices, AEMO places significant emphasis on **price volatility** in the NEM. Where prices have shown greater variability, AEMO may increase its price assumptions to reflect a higher level of financial risk.

By incorporating these factors, the MCL is designed to ensure that Market participants maintain sufficient financial backing to meet their expected obligations under typical market conditions.

However, it is important to note that the MCL is not intended to cover extreme or rare pricing events, such as those that may trigger administered price caps in the market. Instead, it is calibrated to provide adequate coverage for reasonably expected market conditions, while recognising that extraordinary events fall outside standard prudential settings.

Overall, the MCL framework plays a critical role in maintaining financial stability in the NEM, ensuring that participants are appropriately resourced to manage their risks while supporting efficient market operation.

CONTINUE

Daily prudential supervision

In order to minimise the risk of shortfall in the market the prudential framework establishes a trading limit below which a Market participant must maintain its liabilities. These liabilities are referred to as outstandings and are calculated each business day.

The trading limit is determined as the total amount of credit support held by AEMO, less the Market participant's prudential margin. The prudential margin is designed to provide a seven-day buffer in credit support to cover outstandings that may accrue whilst a defaulting Market participant is being removed from the NEM. The provision of a prudential margin reduces the risk of shortfall in payments to generators.

AEMO's Prudential Dashboard

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For periods where a statement has not yet been issued, AEMO provides a range of electronic information services to ensure Market participants still have timely visibility of their financial position. These tools enable Market Participants to monitor their obligations on an ongoing basis, rather than waiting for formal statements.

One key resource is the Prudential Dashboard, an online platform that consolidates critical financial and risk-related information. Through this dashboard, Market participants can access:

- A summary of all outstanding settlement periods
- The amount owing for each individual period
- Their total outstanding balance
- Their current maximum credit limit
- Details of the credit support (such as bank guarantees or cash security) they have in place

This comprehensive view allows Market participants to actively manage their financial exposure and ensure compliance with prudential requirements. By providing both detailed and real-time information, AEMO supports transparency and helps maintain the financial stability of the market.

Monitoring Outstandings and Trading Limits

AEMO conducts daily monitoring of participant outstandings against their trading limit. Participant outstandings mainly comprise of the following:

- Actual settlement amounts payable to AEMO in respect of prior billing periods which remain unpaid
- AEMO's estimate of the aggregate amount payable regarding trading intervals having already occurred for which actual settlement amounts are not yet available

When a Market participant's current outstandings exceed its trading limit, AEMO may issue the Market participant with an interim statement and a call notice. AEMO allows

a Market participant to bring its outstandings below its trading limit by 11:30am each business day before issuing a call notice.

Interim statements cover any transactions for trading intervals not already included in issued statements. The interim statement amount is AEMO's estimate of the dollar value of transactions entered into by the market participant since the date covered by the latest preliminary statement.

If a call notice has been issued the Market participant must do one of the following, before 11:00am on the next business day:

- Provide an increase in credit support, using either a bank guarantee or cash as credit support (up to $20m)
- Pay AEMO the call amount using a security deposit (cash)
- Submit and authorise an ex-post reallocation to the value of the call amount
- Any combination of the above to the value of the call amount

If the Market participant elects to use a security deposit or cash credit support in response to the call notice, they will receive interest earned on the monies paid.

Consequences of payment default in the NEM

Under the National Electricity Rules (NER), AEMO is only able to distribute payments to credit participants, typically electricity generators, using funds it receives from debit participants, primarily retailers. This structure means that the financial integrity of the market depends heavily on all participants meeting their payment obligations.

A failure by a retailer to pay its debts is therefore considered a highly serious event within the NEM, although such occurrences are rare. If a payment default were to occur, AEMO would have limited options and may be required to short pay generators, effectively passing the financial shortfall onto them. Such an outcome

could have significant consequences, potentially undermining the financial stability of the market and the confidence of participants.

To mitigate this risk, the NEM relies on prudential arrangements, which are designed to proactively prevent payment failures. These arrangements require retailers to provide financial security, often in the form of a bank guarantee, to ensure that sufficient funds are available to meet their obligations. In the event of non-payment, AEMO can draw upon this guarantee to recover outstanding amounts and minimise any deficit.

A retailer default carries serious consequences. Upon failure to meet payment obligations, AEMO will take immediate action to recover the debt and enforce compliance. If the retailer continues to default, further regulatory measures may be triggered, including suspension from the market.

In such cases, the situation may escalate the removal of customers from the defaulting retailer. This is typically managed through the Retailer of Last Resort (RoLR) process, whereby customers are transferred to alternative retailers to ensure continued energy supply. Ultimately, these mechanisms are in place to protect both market stability and consumer interests, ensuring that disruptions caused by retailer failures are effectively contained.

Retailer of Last Resort (RoLR)

When an energy retailer fails, its customers are automatically reassigned to new retailers through a regulated process designed to ensure continuity of supply, similar to customer transfers between retailers. However, customers are not involved in selecting their new provider.

This mechanism is called Retailer of Last Resort (RoLR), in which AEMO is responsible for appointing one or more substitute retailers to take over the affected customer. Once the failed retailer is formally suspended from the market, the transfer of

customers occurs automatically through systems operated by AEMO. This ensures a seamless transition, preventing any interruption to electricity or gas supply.

Following the transfer, the RoLR contacts each customer directly to establish new contractual arrangements, including tariffs, payment terms, and other conditions. Although customers are initially assigned to a RoLR, they retain full choice and flexibility and may switch to another energy retailer at any time.

This framework ensures both stability in the energy market and protection for consumers during retailer failure events.

CONTINUE

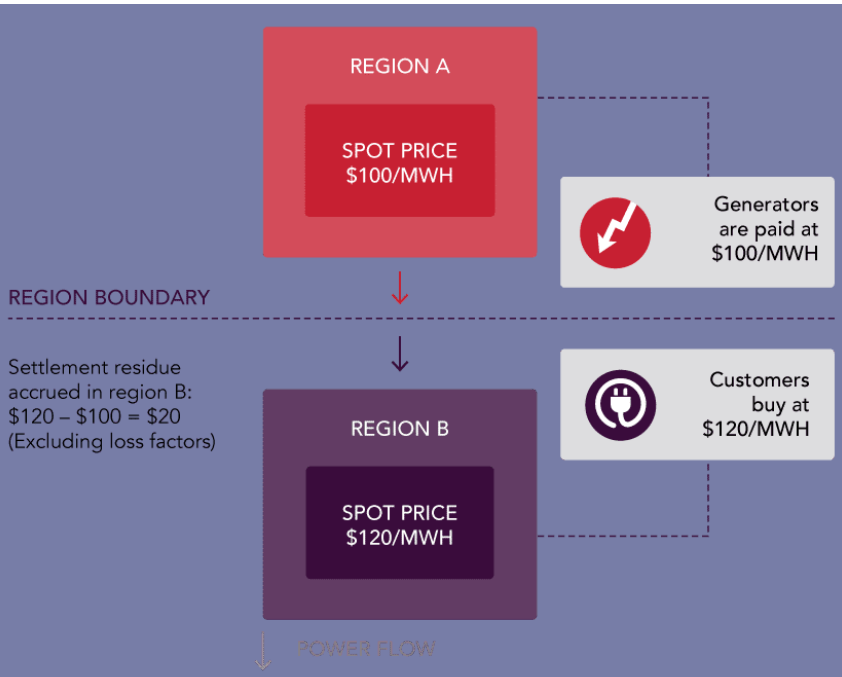
Settlements residue

The spot price for electricity in each region of the NEM is determined by a number of factors, including supply and demand, the physical limitations of interconnectors, and the loss factors for both the transmission and distribution networks. This can result in significant spot price differences for any trading interval across NEM regions.

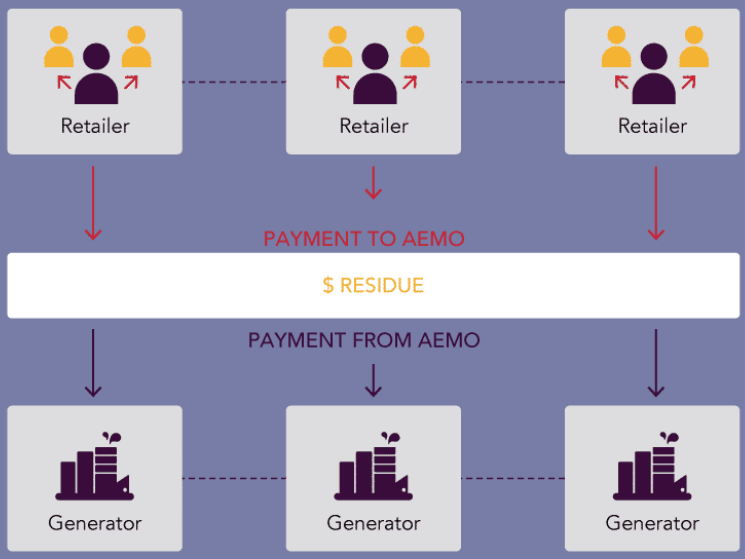
The difference between the value of electricity in the region where it is generated and its value if sold in another region is called the inter-regional settlements residue.

At the intra-regional level (within a single region) settlements residue are caused by the following:

- The adjustment of prices using marginal loss factors between the regional reference node and transmission network connection points tends to recover more from retailers than would be the case if average loss factors were used
- Variations in metering results within the accuracy limits



Inter-regional settlements residue

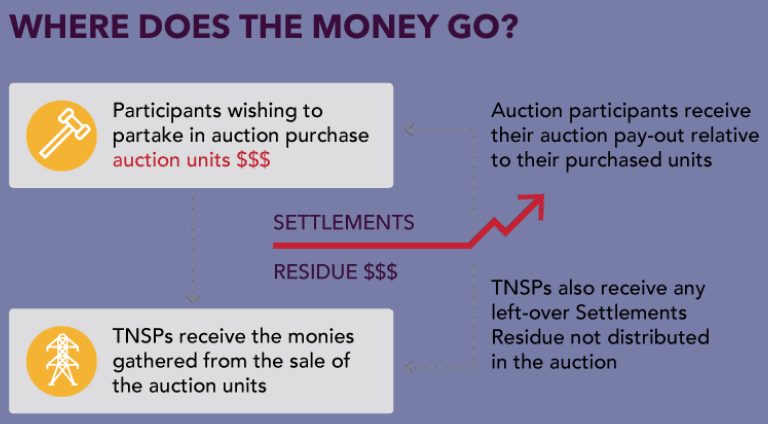


Settlements residue

The settlements residue auction

The inter-regional settlements residue that accumulates is made available to the market by auction, known as the Settlements Residue Auction (SRA). In each auction a number of units can be bid on, with each unit representing a share of the settlements residue over a given calendar quarter. The auction process establishes the market value of the residue, and contributes to interregional trade by providing market participants with a mechanism to manage the risk associated with different price outcomes between trading regions.

Market participants who purchase auction units obtain access to a share of the residue. If a settlement residue occurs in a given interval it is paid according to the ownership of units through the standard billing processes. In this way, the premium paid for the auction units provide protection against high price differences between regions in the wholesale market.



CONTINUE

Settlements in the financial electricity market

In addition to the settlements for the physical production and consumption of energy in the wholesale market (managed by AEMO), there is a financial market which offers participants a range of financial instruments to manage the financial risks inherent in providing electricity services.

The electricity spot price can be volatile due to a range of supply and demand factors. As such, buyers and sellers of electricity may seek to reduce this uncertainty by entering into long-term or short-term contracts that set agreed prices for electricity.

One basic form of contract is a bilateral hedge contract where two parties agree to exchange cash against the spot price. Two typical types of hedge contracts are the two-way hedge (swap) and one-way contract (option). Subject to general competition laws, these contracts may be kept confidential and may take any form the parties wish to negotiate.

Financial market contracts are financial instruments only and do not affect the way AEMO manages the power system. AEMO has no involvement or knowledge of these contracts between buyers and sellers.

Reallocation agreements

Participants that have entered into financial contrats are able to register a reallocation agreement with AEMO that reflects their bilateral, swap or option

contract and allows them to minimise cash flow timing risk and prudential obligations to AEMO. The reallocation between the two participants applies matching credit / debit amounts to their prudential position.

Risk management and hedging in the NEM

Retailers participating in the NEM are exposed to the spot price of electricity, which can be highly volatile. To manage this risk, it is standard practice for most retailers to adopt hedging strategies that reduce their exposure to unpredictable price fluctuations.

Hedging can take several forms. Retailers may enter into long-term power purchase agreements (PPAs) with generators, securing electricity at agreed prices over extended periods. Alternatively, they may use over-the-counter (OTC) financial contracts negotiated directly with counterparties, or trade standardised hedge products through exchanges such as the Australian Securities Exchange (ASX). Each of these instruments allows retailers to stabilise costs by locking in or smoothing prices for a portion of their electricity demand.

When hedging arrangements are established directly between Market Participants, typically between a retailer and a generator. AEMO provides a mechanism known as reallocations, which allows Market Participants to formally register their OTC hedge contracts with AEMO.

By registering these hedges, AEMO is able to recognise that a retailer's exposure to the spot market is partially reduced. In practical terms, this means that for the portion of electricity covered by a hedge, the retailer is no longer fully subject to spot price volatility and instead benefits from a more predictable, fixed-price arrangement.

As a result, AEMO can adjust the retailer's prudential requirements, specifically by reducing the amount of credit support the retailer is required to provide. This reflects the lower financial risk associated with a hedged position.

Overall, hedging and reallocations play an important role in the NEM by enabling retailers to manage price risk more effectively while also supporting efficient and proportionate prudential arrangements.

CONTINUE

Activity - prudentials terminology

Drag each of the terms on the left to match its definition on the right.

Prudential Margin	Issued when a market participant's outstandings exceed their trading limit
Settlement Residue Auction	The amount a market participant can owe to the market at any one time
Maximum Credit Limit	The amount calculated to cover a participant's expected outstandings + a reaction period
Call Notice	The process of selling the rights to residue created by inter-regional price differences

Trading Limit	The difference between the Maximum Credit Limit and the Trading Limit

SUBMIT

CONTINUE

Module summary

Congratulations! You've now completed **Module 8 - NEM Settlements and Prudentials.**

This module has introduced and explained the settlements process in the NEM, including the billing cycle, payment processes and dispute resolution. The module then examined prudential supervision within the NEM, providing an overview of the prudential framework used to control credit risk. Inter-regional settlements residue was introduced, and the settlements residue auction process explained.

You may now review any section of this module via the main menu, or exit this module and proceed to **Module 9 - Financial Markets**.

Module 9 - Financial Markets

- Module introduction
- Financial markets
- Derivative products
- Over-the-Counter contracts (OTCs) vs futures contracts
- Module summary

Module introduction

This module addresses the following learning objectives:

- The inter-relationship between the NEM (spot market) and financial markets
- How contract positions might affect bidding behaviours and spot market outcomes
- To identify the different types of hedging instruments used in the financial markets
- To identify and explain some of the risks associated with hedges

There is no formal assessment for this module, however you will be required to complete some simple activities at key stages to demonstrate your understanding.

CONTINUE

Financial markets

Managing risk through financial contracts

National Electricity Market (NEM) participants need to manage the financial risks associated with the significant degree of spot price volatility that occurs during trading periods.

They typically achieve this by using financial contracts that lock in a firm price for electricity that will be produced or consumed at a given time in the future. These contracts can substantially reduce market participants' financial exposure and contribute to revenue stability.

They are known as derivatives as they derive their underlying value from the spot market, and consist of either swaps or options. These can be traded between two parties directly, through an intermediary or on the Australian Securities Exchange.

AEMO has no involvement or knowledge of these contracts and they do not affect the way the physical power system is managed by AEMO. However, they do affect generator bidding behaviour.

Why do we have financial markets?

There is a significant amount at stake in the NEM. An electricity retailer that is not adequately hedged may be exposed to costs of up to a million dollars per minute

during periods of extreme market prices. For a retailer, this level of exposure can erode an entire year's profit within a matter of hours.

Financial markets operate in tandem with the electricity market, providing both retailers and generators with mechanisms to manage price risk. By enabling greater price certainty and stability, these markets support the investment signals necessary to maintain a reliable and sustainable energy system.

Movements in spot price

All electricity traded in the NEM is settled at the regional spot price, which is determined by matching total demand across the interconnected network with generators' price bids.

Because electricity supply and demand must be balanced in real time, generation must occur precisely when it is needed by consumers. Surplus supply at one point in time cannot be used to meet demand at another. This dynamic creates the potential for significant fluctuations in the spot price between dispatch intervals.

Financial contracts for electricity

Financial contracts are the most common method used by market participants to manage financial and physical risks caused by spot price volatility. These generally take the form of electricity derivatives which lock in firm prices for electricity that will be produced or consumed at a given time in the future.

As future supply or demand of electricity is difficult to predict ahead of time there is typically a residual exposure to spot risk. Electricity derivatives can reduce participants' financial exposure and lead to more stable spot price outcomes.

Medium Term Projected Assessment of System Adequacy (MT PASA) gives a two year look ahead for outages (supply forecast), so the challenge is typically

unplanned outages and the increase in intermittent generation. Demand can be forecast, but a retailer will have a changing profile based on customers coming and going, and changes in behaviour over time.

The Australian Securities and Investments Commission (ASIC) is responsible for regulating these financial instruments.

Structure of the financial markets

Similar to markets for other commodities such as gold, wheat and oil, the financial markets for electricity comprise a range of players including brokers, financial market intermediaries, speculators, electricity generators, retailers and distribution companies. The bulk of the business transacted in the NEM is between generators and electricity retailers who principally use derivatives to reduce their exposure to spot price volatility.

Over-the-counter (OTC) financial contracts can be entered into as either short- or long-term agreements, and are based on an agreed contract or strike price for specific volumes of electricity. They operate independently of both the NEM and AEMO, and are separate to the process of balancing supply and demand when generators are scheduled and dispatched into production.

Retailers

A retailer that purchases electricity in the NEM on behalf of its customers is exposed to the risk that spot prices may rise above the level it can recover from those customers. To manage this risk, retailers may use financial derivatives such as futures contracts, electricity swaps, or options to lock in a fixed price for the volume of energy they expect to purchase.

By adopting these hedging strategies, retailers can reduce their exposure to increases in the spot price while securing a price level that supports a sustainable and predictable profit margin.

Generators

Similarly, electricity generators may enter into financial contracts to manage the risk of sustained low spot prices. Generators need to ensure that the price they receive for electricity remains sufficiently above their operating cost, such as fuel expenses, to recover their significant capital investments.
To achieve this, generators can use financial contracts to secure a fixed price for the electricity they produce. In doing so, they may take the opposite position to retailers in the financial markets.
While hedging activities between retailers and generators make up the majority of electricity financial market transactions, a range of other participant, including brokers and financial institution, also play a role in managing market risk. These participants may offer products such as outage insurance and weather derivatives.

Brokers and banks

Brokers play a key role in facilitating transactions in electricity financial markets. Using services provided by organisations such as Reuters and Bloomberg, they disseminate buy and sell prices for a range of financial contracts on behalf of their clients. Through these platforms and their networks of end-use customers, brokers help connect counterparties and support the negotiation and execution of trades.

Brokers also publish indicative prices, providing Market participants with a reference point for current market conditions. They do not take physical or financial positions themselves; instead, their revenue is typically generated through transaction fees, often calculated as a fixed cost per megawatt hour of electricity traded.

Financial intermediaries such as banks are also essential to the functioning of these markets. Banks may act as market makers by quoting both buy and sell prices, or they may take trading positions directly. Given their scale, banks can also perform a warehousing role - temporarily assuming market risk to facilitate client transactions and holding these positions until they can be offset in the market.

CONTINUE

Derivative products

The financial instruments previously described (electricity swaps, futures and options) are commonly referred to as electricity derivatives. This is because their value is derived from, or based on, an underlying commodity or physical asset. Derivatives are sometimes known as "synthetic" products, i.e Contract for Difference (CFD), because the derivative holder gains access to cash flows (positive or negative). These gains arise from differences between the derivative's contract price and the relevant commodity or security spot price on which the derivative is based.

A key reason for the widespread popularity of derivatives is that they are highly flexible and can be tailored to match an individual's risk management needs. In the case of an options contract, for example, the strike price, method of conversion, and maturity date can all be altered to suit a user's particular risk or reward objectives. This ability to adjust certain components depending on the particular risk being hedged holds true for other derivatives too.

Another advantage of electricity derivatives is that they are often leveraged instruments. This means that for a comparatively small financial outlay, a user can obtain a hedge instrument with a notional value that is many times higher than the outlay. Most derivatives do not require payment when the contract is established. Instead, the parties simply agree to exchange cash regularly (typically weekly) based on the difference between electricity spot price movements and the fixed, strike price agreed to in the swap.

Contract examples

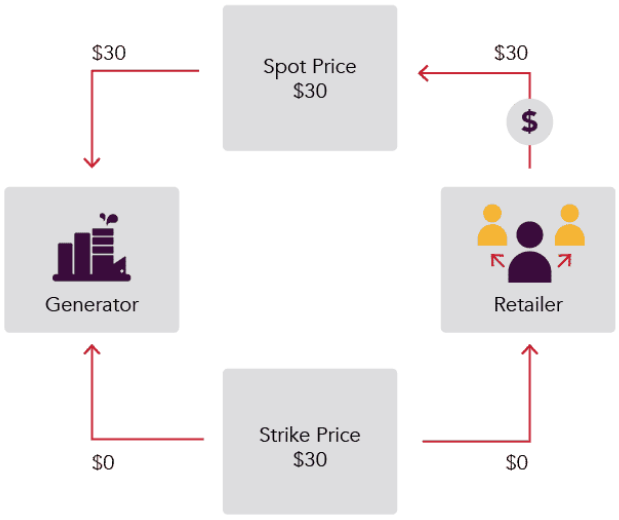
The following example shows how a generator or retailer might use an electricity derivative contract for protection against an adverse spot price movement.

AGREEING TERMS | HIGHER SPOT PRICE | LOWER SPOT PRICE

Retailer ABC needs to purchase 100 megawatts of electricity over the next six months to cover electricity purchases made by its Customers.

To guard against the risk of electricity spot prices in their NEM region rising sharply over that period, retailer ABC purchases a flat profile electricity swap for the full 100 megawatts of load at a strike price of $30 per megawatt hour. By purchasing the swap, the retailer locks in an average price of $30 per megawatt hour for the electricity they intend to purchase through the spot market.

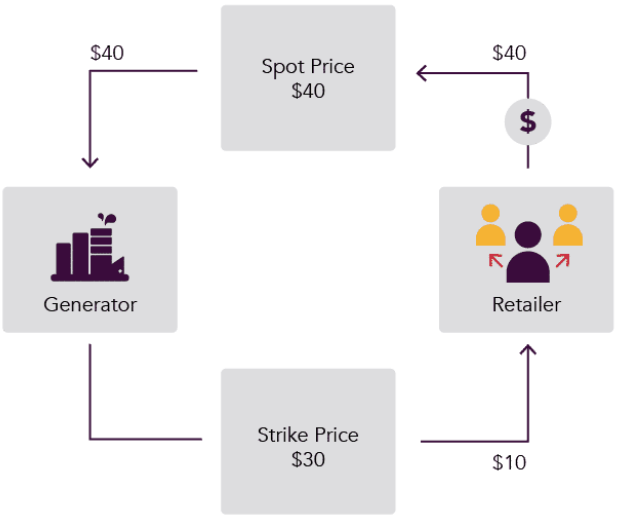
The counterparty to this transaction might well be a generator in the same NEM region who is similarly seeking to set a firm price for the electricity they intend to produce over the same six-month period. By entering into this swap agreement, the two parties are essentially committing to exchange cash based on any rise or fall in the spot price (from the swap price position of $30 per megawatt hour).



AGREEING TERMS | HIGHER SPOT PRICE | LOWER SPOT PRICE

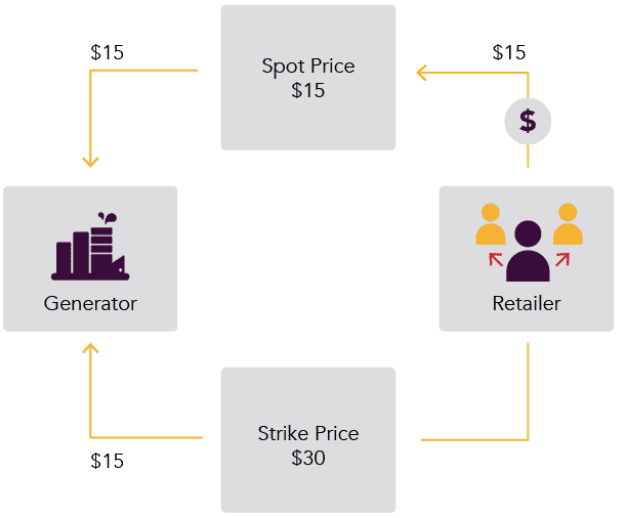
On the first settlement day of this swap contract, the average spot price for electricity in that region has risen to $40 per megawatt hour. In this instance, the generator will compensate the retailer $10 for each megawatt hour of electricity it purchases at the spot price (up to the 100 megawatts of the swap agreement).

The net cost of energy purchased by the retailer will therefore be equivalent to the fixed price of $30 per megawatt hour agreed under the swap. AEMO will pay the generator the spot price of $40 per megawatt hour, and the generator will pay the retailer $10 per megawatt hour to satisfy the terms of the swap agreement.



AGREEING TERMS | HIGHER SPOT PRICE | LOWER SPOT PRICE

On the second settlement day of this swap contract, the average spot price for electricity falls to $15 per megawatt hour. This time, the retailer will compensate the generator $15 for each megawatt hour of electricity it purchases at the spot price (up to the 100 megawatts of the swap agreement). The net cost of energy sold by the generator will therefore be equivalent to the fixed price of $30 per megawatt hour agreed under the swap. AEMO will pay the generator the spot price of $15 per megawatt hour, and the retailer will pay the generator $15 per megawatt hour to satisfy the terms of the swap agreement.



Thumbnail

Determining cash flows

Market participants need to specify how the cash flows for the swap will be determined as part of the swap agreement. The participants are free to specify the periods which suit them, however they tend to use standardised time definitions. For example, if a swap contract specifies that settlement be based on "peak" prices, this means that cash flows will be calculated using spot price outcomes during peak period hours. The peak period hours are usually defined as being from 7:00 am to 10:00 pm Monday to Friday, excluding public holidays.

If the basis of the swap is "off-peak" prices, it will use spot prices that are determined outside of defined peak periods. "Flat"' price contracts represent price outcomes at any time of the day or week, and cover both the peak and off-peak periods.

Callable contracts

While electricity swaps are the most widely traded financial instruments and provide users with price certainty against spot prices, they do not provide risk mitigation against movement in the forward prices (the price at which the swaps are traded). Participants seeking to protect against adverse movements in the forward price often use other derivative instruments such as callable options.

Callable options, calls and puts, give buyers the right - but not the obligation - to pay or receive an agreed price for energy up to and including a specified future date. For this right, the option buyer pays the option seller a premium. The cost of the premium depends on factors such as time to maturity, the strike price, and the amount of spot price volatility in the underlying market.

A major benefit of options is that they give buyers the flexibility to abandon or discard their undertaking to buy electricity under the option agreement should the spot price move in a way that makes exercise of the option uneconomical. For example, a retailer who buys an option to gain protection against a price rise in the forward market can subsequently abandon that option (for the loss of the premium paid) should the forward price fall rather than rise. In this circumstance, the retailer would be able to take full advantage of the lower-than-anticipated forward price outcome, and their loss would be limited to the premium paid. The option seller keeps the premium regardless of whether the buyer eventually takes up the option or not.

Caps, floors and collars

The options most commonly traded in the electricity market are known as caps, floors and collars. A cap option allows the holder to set an upper limit on the price

that they will pay for electricity. For example, a cap with a strike price of $300 per megawatt hour ensures that no matter how high the electricity spot price may rise, the maximum price the option holder will pay for electricity covered under the option will be $300 per megawatt hour. Similarly, a floor contract struck at $40 per megawatt hour means that no matter how low spot prices fall, the minimum price for the quantity of electricity specified in the option will be $40 per megawatt hour.

Collars essentially combine both a cap and a floor option. Using a collar, a retailer or generator can set a price band for which they agree to produce or consume electricity in the future. Price bands selected are fully negotiable between both parties at the time of the collar option purchase. Frequently, collar options are struck so that a zero premium payment is required and the premium, or cost of buying the cap option, is offset by the premium earned through the sale of the floor, or vice versa. A zero-cost collar struck at the same cap and floor price is the same as a swap.

Options are widely used in the electricity market because contracting parties can set firm prices while still being able to abandon these prices should more favourable forward prices eventuate. Retailers typically buy caps to secure firm maximum prices for future electricity purchases, while generators use floors to lock in a minimum price to cover future generation output. Intermediaries, such as banks and traders, are also active option users, with option sellers looking to profit from the premium earned when options are abandoned by buyers.

Exchange-traded futures

As well as swaps and options traded in the Over-the-Counter (OTC) market, electricity futures contracts are also traded on the Australian Securities Exchange (ASX). Like other risk management products, futures allow generators and retailers to lock in firm future prices for the electricity they produce or consume, protecting against future spot price uncertainty.

Futures contracts differ from OTC instruments in that they are highly standardised in terms of contract size, minimum allowable price fluctuation and maturity dates. All these parameters are specified by the exchange when the futures contract is purchased.

Although they are not as flexible as OTC instruments, an advantage of futures is that they are cleared and settled through a centralised exchange clearing mechanism. In

this process, the clearing house becomes a central counterparty to the transaction, which means that there is no credit risk between the two contracting parties. By trading through the exchange the identities of the parties are also kept confidential from each other.

'Exchange traded' and 'Over-the-Counter' markets

When it comes to the different types of financial markets there is no better or worse – you can't compare the two. Both mechanisms have their advantages and disadvantages for different types of participants. The exchange traded market is typically very structured; they offer a range of different products, however the participants are constrained in those types of products. The advantage is there is no counterparty risk as they are traded with a central counterparty. The other advantage is that the payments go through a process of margining, which can assist a participant in managing their cash flow when there are fluctuations in the spot price.

For the OTC market, one of the key advantages is flexibility. The OTC contracts are generally completed within a legal framework that sets standard terms and conditions, but the price and volume contracted can be structured in various ways which gives participants a lot of flexibility. However, one of the disadvantages of OTC contracts is that the participant is exposed to counter-party risk.

CONTINUE

Over-the-Counter contracts (OTCs) vs futures contracts

The advantage of Over-the-Counter contracts (OTCs)

OTCs generally allow unlimited flexibility for users to structure a deal that is tailored to each party's load characteristics and risk objectives.

THE ADVANTAGE OF FUTURES

CASH FLOW DIFFERENCES

Exchange clearing houses are highly creditworthy, with their mutual guarantee structure being typically sourced from the combined creditworthiness of several large, highly-rated financial institutions. As a result, electricity market participants may use exchange clearing houses as credit intermediaries in hedge transactions instead of using lesser-rated counter-parties or participant counter-parties with whom existing credit limits are already reached. By using futures contracts—and the services of an exchange clearing house—market participants subject to the liquidity of the futures market can broaden the number of counter-parties available to deal in hedge transactions. In trading Future contracts, mark-to-market (MTM) is the daily process where the exchange adjusts the value of your open contracts to match the current market price. At the end of each trading day, gains are credited and losses are debited directly from a margin account to ensure all obligations are settled.

THE ADVANTAGE OF FUTURES

CASH FLOW DIFFERENCES

OTC trades do not typically start settling until the start of the spot price period. This means that there is no exchange of cash flow until the underlying period has started. This means that the product cannot be finally settled until after the end of the spot price settlement period. Futures products have daily margins where the movement in the forward curve is settled every day through margins. This means that changes in the forward curve are settled immediately. This also means that if a company buys and sells the same product it effectively has no deals on its books to settle at a later stage. This is particularly attractive to speculative traders.

Counterparty Risk

Counterparty risk is the risk associated with the other party to a financial contract not meeting its obligations. For instance, if that counterparty suffers any type of financial stress and worst case, fails, then effectively that risk is passed onto the participant. There are various ways that participants in the financial market may manage counterparty risk; generally they will:

- seek to understand their counterparty and the counterparty's credit rating.
- establish limits above which they don't trade with any single counterparty.
- require the counterparty to provide security to them that could be used in the event the counterparty fails.

Settlement residue auction

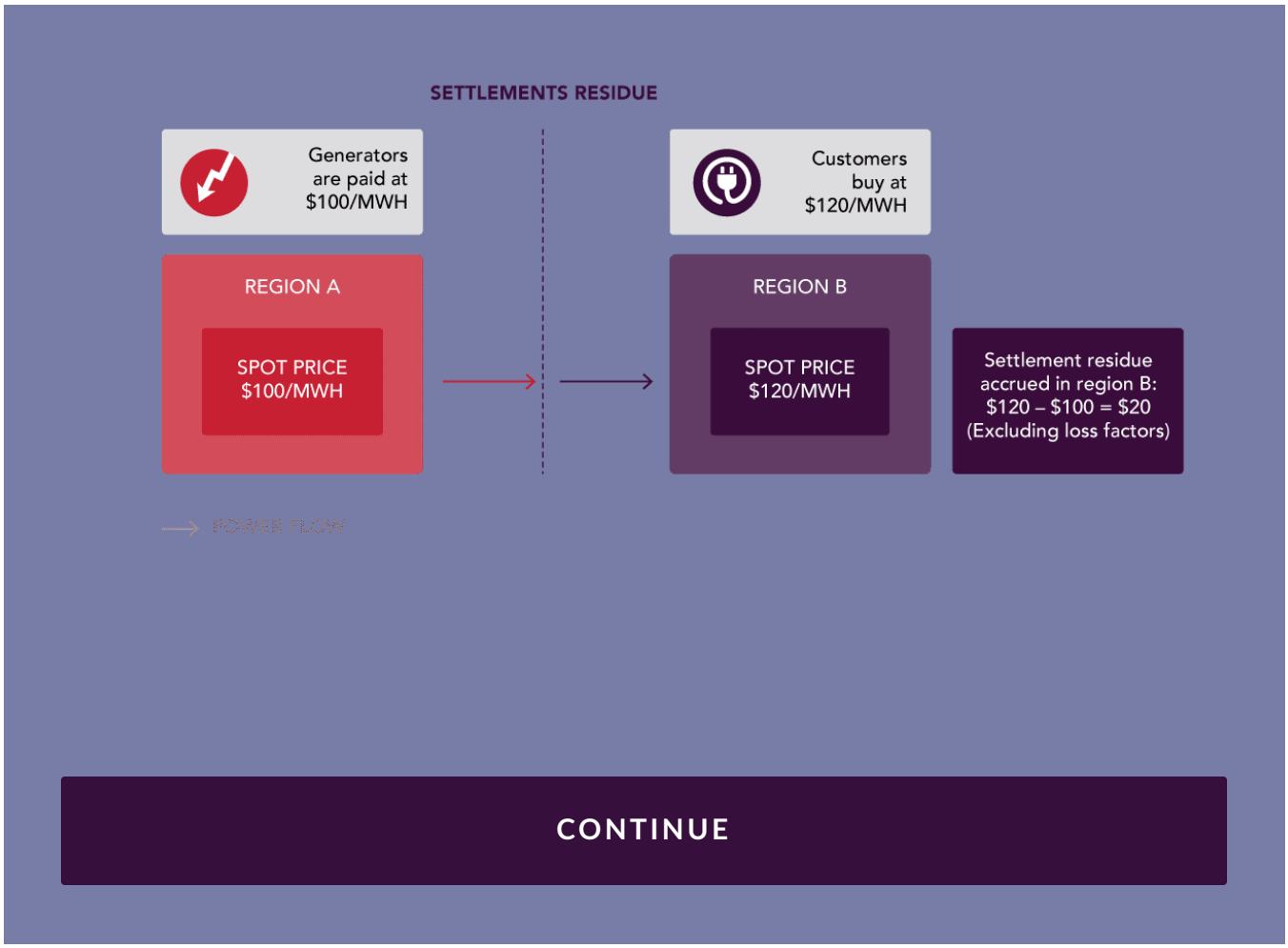
While most financial market activity engaged in by generators and retailers occurs through the OTC and futures markets, participants can also use AEMO's settlement residue auction process to manage the risk involved in inter-regional trade.

Because the spot price for electricity in each region of the NEM is determined by supply and demand, the physical limitations of interconnectors, and transmission network loss factors, there may be significant differences in the spot price across NEM regions.

The difference between the value of electricity in the region where it is generated and its value when sold in a different NEM region is known as the inter-regional settlement residue (IRSR). The IRSR that accumulates is made available to the market by conducting a quarterly auction which provides participants with a mechanism to manage the risk associated with different price outcomes between trading regions.

In these auctions, market participants lodge bids for an entitlement to any IRSR that may accumulate across a designated NEM interconnector. They purchase auction units to obtain access to a share of the residue by a premium as protection against high spot prices.

The volume of IRSR available through the auction depends on the actual level of interconnector flows which may, on occasion, be highly variable. IRSR is a financial surplus or deficit generated in the NEM when electricity is transferred between regions with different spot prices. "Firmness" refers to the reliability and financial certainty of IRSR units (like SRA units) as hedging instruments against wholesale price volatility. SRA units are defined as “Non-Firm” as they require both energy flow (in a particular direction) and associated price separation between regions . Consequently IRSR auction units are considered less reliable tools for risk management than other types of financial instruments available to market participants.


SETTLEMENTS RESIDUE
Generators are paid at $100/MWH
Customers buy at $120/MWH
REGION A
SPOT PRICE $100/MWH
REGION B
SPOT PRICE $120/MWH
Settlement residue accrued in region B: $120 – $100 = $20 (Excluding loss factors)
POWER FLOW
CONTINUE

Module summary

Congratulations! You've now completed **Module 9 - Financial Markets**.

This module has explained the role of the financial markets in relationship to the NEM, outlining the structure of the financial market and describing the role of retailers, generators, brokers and banks. The module also examined some example contract scenarios, before reviewing exchange-traded futures and the settlement residue auction.

You may now review any section of this module via the main menu, or exit this module and proceed to **Module 10 - How do I find out more?**.

Module 10 - How do I find out more?

Well done - you've nearly completed the National Electricity Market Foundations eLearning course. As with any complex system, there is a vast amount of data and information available - far more than can be covered in an introductory course.

With this in mind, the aim of this module is to provide you with a selection of resources to help you navigate and further explore the evolving Australian energy landscape.

- **Module introduction**
- **AEMO planning reports**
- **Data**
- **AEMO Website - other resources**
- **Additional sources of information**
- **Energy Education courses**
- **Module summary**

Module introduction

This module addresses the following learning objectives:

- To identify information sources you can utilise to find out more detail about the NEM
- To introduce some of the major reports and planning documents that AEMO publishes for the development of the NEM
- To explore ways you can continue to build your understanding of the NEM and the wider Australian energy sector.

There is no formal assessment for this module.

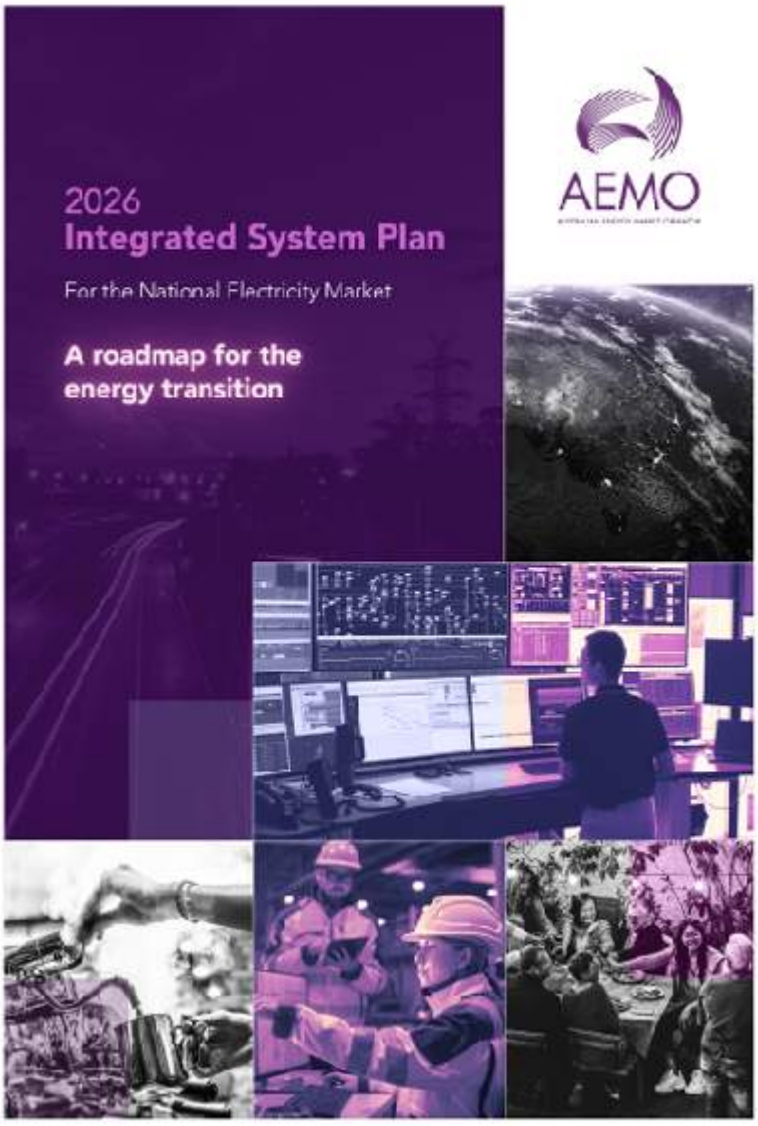
Handy tip: Click here to access AEMO's dedicated glossary of terms for a helpful list of acronyms, abbreviations, measurements and other terminology commonly used across the Australian energy sector.

CONTINUE

AEMO planning reports

In addition to its role as the operator of various energy systems and markets - including the NEM - one of AEMO's other major functions is the National Transmission Planner. To enable AEMO to function independently and transparently, it does not own any generation or transmission assets. In order to run markets and systems effectively, and to transport energy from generators to consumers, AEMO is, therefore, reliant on the market's natural development.

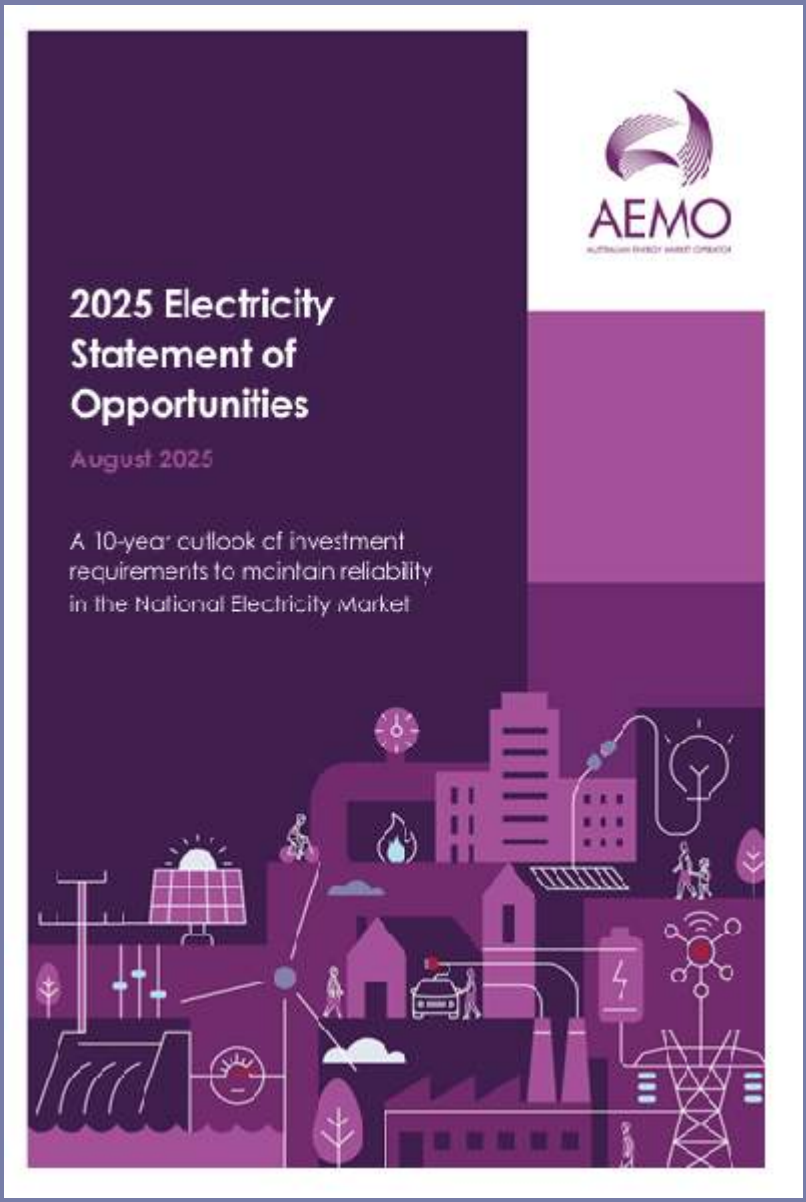
AEMO publishes a number of reports designed to provide information to industry and stakeholders to make informed short- and long-term operational, investment, and policy decisions.



Integrated System Plan (ISP)

The Integrated System Plan (ISP) is a whole-of-system plan that provides an integrated roadmap for the efficient development of the National Electricity Market (NEM) over the next 20 years and beyond. The ISP is produced every two years.

Its primary objective is to optimise value to end consumers by designing the lowest cost, secure and reliable energy system capable of meeting any emissions trajectory determined by policy makers at an acceptable level of risk. It fully utilises the opportunities provided from existing technologies and anticipated innovations in Distributed Energy Resources (DER), large-scale generation, networks and coupled sectors such as gas and transport.

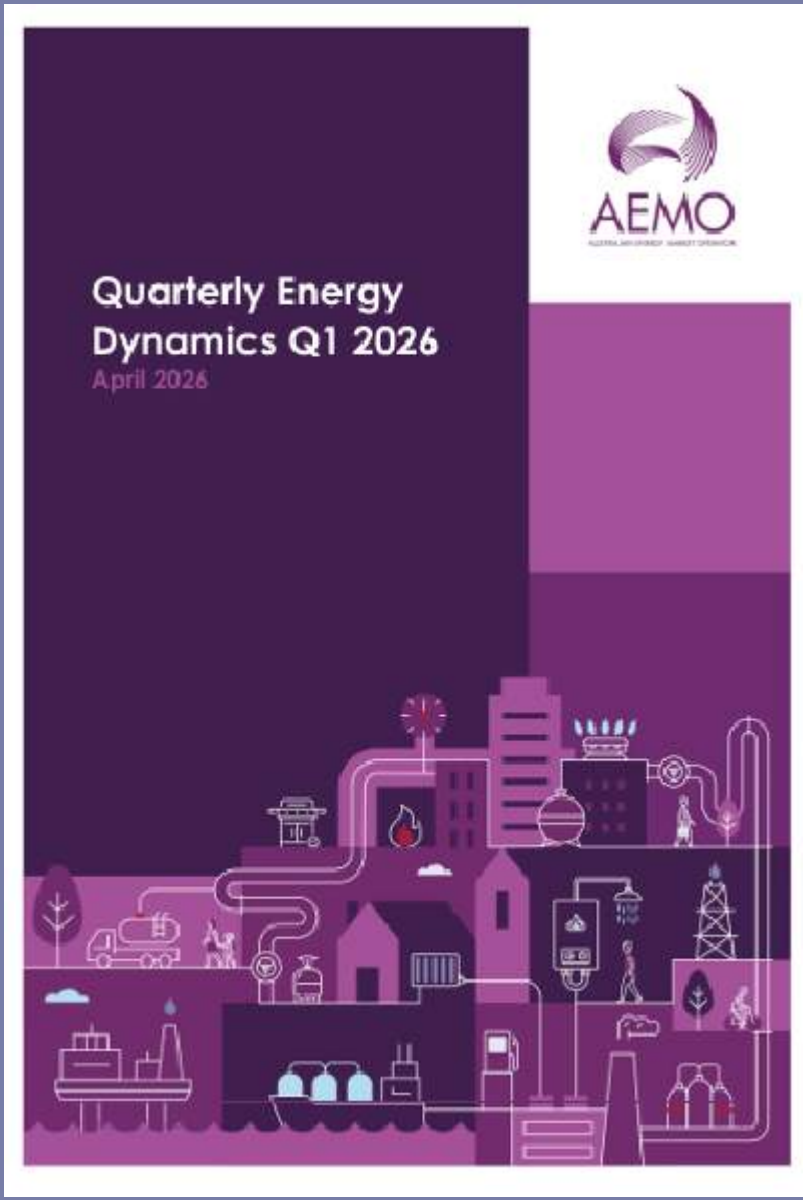


Electricity Statement of Opportunities (ESOO)

The Electricity Statement of Opportunities (ESOO) provides technical and market data for the National Electricity Market (NEM) over a 10-year period to inform the planning and decision-making of market participants, new investors, and jurisdictional bodies.

Whereas the ISP is a roadmap for the future design and development of the NEM, the ESOO can be considered more a forecast of the ability of the NEM to provide sufficient energy to meet the needs of consumers. This concept of sufficient supply is called Reliability, and is

related to the Projected Assessment of System Adequacy (PASA) reports that we learned about earlier in this course.



Quarterly Energy Dynamics (QED)

AEMO's Quarterly Energy Dynamics reports detailed market dynamics, trends and outcomes in Australia's electricity and gas markets. It is produced quarterly, and most commonly

compares results for the quarter against other recent quarters, either consecutive (e.g. Q's 1-4 2025) or year-on-year (e.g. Q1 2023, Q1 2024, Q1 2025).

Inputs, Assumptions and Scenarios Report (IASR)

AEMO delivers a range of planning and forecasting publications to inform decision making including the Electricity Statement of Opportunities (ESOO), the Gas Statement of Opportunities (GSOO) and the Integrated System Plan (ISP).

AEMO strives to ensure its planning and forecasting publications are of the highest quality, with data that is fit for purpose and industry-reviewed, and processes that are transparent. To build confidence in the forecasts, and help decision makers understand the outcomes, AEMO consults on and publishes its scenarios, inputs, assumptions and methodologies. Confidential information is withheld from public release, but is incorporated in the modelling conducted, if necessary, to deliver assessments that are as accurate as possible.

The current IASR can be accessed through AEMO's website from the individual sections dealing with the ISP, ESOO and GSOO.

Other AEMO reports

As you've learned, the NEM is not the only energy system operated by AEMO. AEMO is also responsible for Western Australia's Wholesale Electricity Market (WEM) and South West Interconnected System (SWIS), the gas Short Term Trading Markets (STTM) in Sydney, Brisbane, and Adelaide, the Victorian Declared Wholesale Gas Market (DWGM) and Declared Transmission System, as well as Gas Supply Hubs and Pipeline Capacity Trading. Together, these inter-related markets play a vital role in meeting Australia's energy needs.

In addition to the ESOO produced for the NEM, AEMO publishes a similar Electricity Statement of Opportunities for the WEM (WEM ESOO). For the gas sector, AEMO

produces annual Statements of Opportunities for both the east coast (GSOO) and Western Australia (WA GSOO). As system operator of the Victorian Declared Transmission System, AEMO also publishes the Victorian Gas Planning Report (VGPR).

Integrated System Plan (ISP)
View current and past ISP's, as well as the status of the next scheduled ISP, including details of consultation with industry and stakeholders

ISP

Electricity Statement of Opportunities (ESOO)
View current and past ESOO documents

NEM ESOO

Quarterly Energy Dynamics (QED)
View the most recent, and past, QED reports

QED

Western Australia Electricity Statement of Opportunities (WEM ESOO)
View current and past WEM ESOO documents for the Wholesale Electricity Market (WEM) and South-West Interconnected System (SWIS)

WEM ESOO

East-coast Gas Statement of Opportunities (GSOO)
View current and past GSOO documents

GSOO

Western Australia Gas Statement of Opportunities (WA GSOO)

View current and past WA GSOO documents

WA GSOO

Victorian Gas Planning Report (VGPR)
View current and past VGPR documents

VGPR

Summary

In this lesson we have explored some of the publications and reports that AEMO produces.

Next, we'll look at some of the data sources available for various aspects of the NEM.

CONTINUE

Data

Accurate, up-to-date data is essential for the effective day-to-day and long-term management of the NEM. To support transparency and informed decision-making, AEMO publishes a wide range of information, including short and long-term demand and supply forecasts, network connection data, real-time (five-minute) market updates, and detailed daily reports on generator bids, offers, and dispatch targets.

Broad access to this data promotes consistency across the market and supports the delivery of affordable, secure, and reliable energy for all Australians.

Market Data

NEM Data Dashboard

A simple NEM data dashboard is available on AEMO's website. This dashboard offers views of the current state of the NEM, including an overview of the dispatch in each region, Price and Demand outcomes and forecasts for each region, the current mix of generation sources, and more.

NEM Data Dashboard

Go to AEMO's NEM Data Dashboard to see live wholesale prices, interconnector flows, forecast prices and demand, fuel technology mix, etc...

DATA DASHBOARD

NEMWeb

More detailed market data is available via AEMO's NEMWeb page. This information is publicly available for download as CSV files. The vast amount of information available through NEMWeb means it can be daunting to find a single piece of information and analyse it, so AEMO has created user guides and other resources to help.

NEMWeb
Access to market data for the NEM

NEMWEB

NEMWeb Help
User guides, descriptions and definitions of the various elements of information available in NEMWeb

NEMWEB HELP

Data (NEM)
NEMWeb and the Help pages are located with the Data (NEM) section of AEMO's website. Other information is available from elsewhere within this section.

DATA (NEM)

In-house and third-party analysis tools

Manually finding, downloading and analysing data can be quite time-consuming, especially for those who require frequent updates and/or large volumes of data.

Many market participants have developed in-house analysis and visualisation tools, and are able to extract data from AEMO systems via an automated feed.

Alternatively, some organisations may use third-party, commercially available tools for visualisation of NEM data on a subscription basis due to the potential complexity of developing an in-house tool.

There are also a number of additional free visualisation tools that display various aspects of NEM information in powerful, easy to understand ways.

AEMO does not endorse any individual analysis or visualisation tool. Organisations and individuals seeking to utilise these should research to find the solution best suited to their individual needs.

Forecasting data

NEM Generation

NEM Generation data is collected from generation industry participants, and published every three months, or as required. The data includes capacity of existing, withdrawn, committed, and proposed generation projects in the NEM.

AEMO also publishes NEM generation maps that illustrate the volumes of connection interest within a particular NEM region and sub regions. These maps are helpful to

identify potential network gaps and help inform future investment in network expansion.

NEM Generation Information

NEM GENERATION

NEM Generation Maps

GENERATION MAPS

Electricity Demand Forecasts

Prior to 2012 electricity demand forecasts for the NEM were produced by the Transmission Network Service Providers for each NEM region and published annually by AEMO. AEMO then took on the role of producing these forecasts for the NEM, and since 2015 has published them on the AEMO Forecasting Data Portal. These forecasts include operational consumption, maximum demand, and since 2016, minimum demand and have been published in a number of forecasting documents which can be found on AEMO's website.

Electricity Demand Forecasts

Links to previous forecasting documents

DEMAND FORECASTS

AEMO Forecast Data Portal

DATA PORTAL

Summary

In this lesson we have explored some of the sources of data that AEMO produces.

Next, we'll look at some of the other information resources available on the AEMO website.

CONTINUE

AEMO Website - other resources

To support transparency and ensure information is accessible to a broad range of industry participants and stakeholders, AEMO publishes a wide range of data and insights on its website.

Initiatives

AEMO carries out a number of pilot programs and research to enhance Australia's current and future energy system.

An example of this is the NEM Distributed Energy Resources (DER) Program. AEMO's DER program aims to understand and integrate high levels of DER into Australia's energy system and market through trialing Virtual Power Plants offering Frequency Control Ancillary Services, assessing technical settings, and assessing DER behaviour in response to power system disturbances.

AEMO has also partnered with AusNet Services and Mondo to launch Project Edge, a DER marketplace demonstration.

View current AEMO initiatives

INITIATIVES

Consultations and Industry Forums and Working Groups

AEMO regularly runs consultations with stakeholders to discuss issues and share information on proposed changes to the energy systems and markets.

AEMO also runs a number of regular industry forums and working groups to ensure stakeholders are involved in the planning and implementation of changes to the energy system and markets. These groups allow stakeholders an opportunity to discuss issues and challenges, and knowledge share with AEMO.

AEMO's website lists all open and closes consultations conducted by AEMO, as well as current industry forums and working groups.

Consultations and Industry forums and working groups

CONSULTATIONS

Operating Procedures

As you have learned in the previous modules, the National Electricity Market is a complex and vast power system. AEMO's system operating procedures are developed to ensure AEMO can provide information related to the management of the NEM, and secure and reliable supply.

Operating procedures include Power System Security Guidelines, Predispatch, Dispatch, Non-market Ancillary Services, and more.

Power System Operating Procedures

PROCEDURES

Subscribe

To keep industry stakeholders informed on initiatives, consultations, changes to operating procedures and much more, AEMO offers a number of different subscription options.

These subscriptions include NEM updates on topics such as market event reports, network connections, and incident reports. They also include market notices on Market Interventions, Market Suspensions, and Administered Price Caps.

It's quick and easy to sign up to a number of subscriptions on the AEMO website.

Subscribe

SUBSCRIBE

AEMO On Air podcast

AEMO On Air explores a wide range of topics across Australia's energy sector, including renewable energy growth, battery storage, rooftop solar, and grid stability.

The podcast is available on all major podcast platforms, or can be accessed via AEMO's website.

Summary

In this lesson we have outlined some of the various information points available on the AEMO website.

Next we'll look at additional sources of information.

CONTINUE

Additional sources of information

Recognising that AEMO is just one part of Australia's broader energy landscape, there are many other sources of information available for different purposes.

The list below highlights a selection of useful resources, though it is not exhaustive. You may recognise some of these from previous modules.

Market Bodies and other regulatory bodies

The Australian Energy Market Commission (AEMC) is responsible for making and amending the National Electricity Rules that underpin the NEM.

The Australian Energy Market Regulator (AER) monitors and enforces the compliance of market participants, as well as AEMO with the National Energy Rules.

The Energy and Climate Change Ministerial Council (ECMC) was established by the National Cabinet, to replace the former Energy National Cabinet Reform Committee, which in turn was preceded by the COAG Energy Council. The ECMC is a forum for the Commonwealth, Australian states and territories, and New Zealand to work together on priority issues of national significance and key reforms in the energy and climate change sectors.

Australian Energy Market Commission

AEMC

Australian Energy Regulator

AER

Energy and Climate Change Ministerial Council

ECMC

Transmission and Distribution networks

Each of Australia's Transmission and Distribution networks provides a range of information dealing with their local network and areas of responsibility.

Below is a table of each transmission and distribution network providers for each NEM region.

Jurisdiction	Transmission	Distribution
Australian Capital Territory	Transgrid	Evoenergy
New South Wales		Ausgrid Endeavour Energy Essential Energy

Queensland	Powerlink	Energex Ergon Energy
South Australia	ElectraNet	SA Power Networks
Tasmania	TasNetworks Basslink (regulated TNSP from 1 July 2026)	
Victoria	AusNet (assets) VicGrid (planning)	AusNet CitiPower & Powercor Jemena United Energy

Government & Policy

Each of the levels of government plays an important role in the Australian energy system, including, but not limited to, developing and delivering policy and legislation to manage the sector, regulation and enforcement, and engaging with and representing consumers.

At the federal level sits the Department of Climate Change, Energy, the Environment and Water (DCCEEW). This department was established in July 2022 to help address climate change and build a more innovative, future ready energy system, among other things.

Each state and territory within the NEM also has its own department(s) dealing with energy related matters. In some cases the different elements of energy can be split across multiple departments rather than residing within a single entity.

The list below is to the major energy related department in each jurisdiction, but doesn't include other departments or regulatory bodies that may be involved in

aspects of the energy system.

Jurisdiction	Government Department
Federal	Department of Climate Change, Energy, the Environment and Water (DCCEEW)
Australian Capital Territory	Environment, Planning and Sustainable Development
New South Wales	NSW Department of Climate Change, Energy, the Environment and Water (DCCEEW) Energy Corporation of NSW (EnergyCo)
Queensland	Department of Energy and Public Works
South Australia	Department for Energy and Mining
Tasmania	Department of Renewables, Climate and Future Industries
Victoria	Department of Energy, Environment and Climate Action (DEECA)

Summary

In this lesson we have explored additional sources of information including Government and regulatory bodies and transmission and distribution network owners.

Next we'll look at what other energy education courses that AEMO has on offer.

CONTINUE

Energy Education courses

Now that you're close to completing this eLearning course, you might want to consider some of the other courses that AEMO offers. There are a mix of courses designed for beginners, those broadening their knowledge a bit further, and some for those wanting to delve into some quite technical concepts.

Foundation and Overview courses

NEM Basics: short video course - this course gives a high-level summary of the NEM, takes approximately 1.5 hours to complete, and is self-paced so that participants can complete it their own time.

NEM Overview - one of AEMO's most popular courses, the NEM Overview covers how the NEM operates as a physical power system and a wholesale market for electricity. The course is delivered either in-person on a single-day, or virtually via Microsoft Teams over two half-day sessions.

NEM Operational Forecasting Overview - launched in June 2023 this on-demand video course provides an overview of how AEMO produces short-term operational forecasts in the NEM. It takes approximately 2 hours to complete and is self-paced so that participants can complete it in their own time.

NEM Technical Courses

Network and Frequency Control Ancillary Services (FCAS) Constraints in the NEM - participants will gain a working-level understanding of why constraints are used to manage the power system, and the effect they may have on the market. The course is delivered in a hybrid mode over two full-days (with options for in-person or online attendance), or fully online across four half-days.

Understanding Market Settlement and Transfer Solution (MSATS) - this course explains what the MSATS system is used for and associated processes and concepts. It is delivered in a hybrid mode over two days, with options to attend either in-person or virtually via Microsoft Teams, and includes an eLearning module that should be completed prior to the instructor-led portion of the course.

Connecting and Registering Generators in the NEM - this course is designed to support individuals and organisations involved in connecting and registering generators and integrated resource systems in the National Electricity Market (NEM). Delivered as a half-day course by AEMO subject matter experts, the program provides a practical, end-to-end overview of the connections and registration process. It equips participants with the knowledge, context, and tools needed to navigate complex technical and regulatory environments.

Metrology for the National Electricity Market (NEM) is designed for metering provider categories A and B (MPAs and MPBs), and provides a thorough understanding of how metrology operates in the NEM. This course is typically only delivered in a face-face setting, but the virtual mode may be considered in some circumstances.

Other Energy systems

As well as the NEM, AEMO runs the energy systems for Gas in various regions, and power systems in Western Australia. Courses are available on the following topics:

Wholesale Electricity Market:

- Introduction to the Wholesale Electricity Market (WEM)
- WEM Reserve Capacity Mechanism
- WEM SCED & Constraints

Gas:

- Gas Markets Foundations - eLearning course
- Overview of the Short Term Trading Market (STTM)
- Overview of the Declared Wholesale Gas Market (DWGM)

How to enroll

If you would like to enroll in any of AEMO's education courses above, please follow the link below to the AEMO website where you can find more information on the individual course including dates (if applicable) and price. From the website you will be prompted to enroll through the AEMO Learning Academy.

For those whose organisation requires an invoice produced prior to payment or a Purchase Order number to be used, this can be arranged by emailing energyeducation@aemo.com.au

AEMO Industry courses

COURSES

AEMO Learning Academy

LEARNING ACADEMY

Summary

In this lesson we have outlined the AEMO energy industry courses as well as how to enroll.

Next we will summarise the module, and then you will have completed the course!

CONTINUE

Module summary

Congratulations! You've now completed **Module 10 - How do I find out more?**

This module has introduced you to some of the additional resources and information sources available to NEM market participants, stakeholders and the public.

You may now review any section of this module via the main menu on your left, or exit this module.

Feedback

The following link with allow you to provide feedback on this NEM Overview eLearning course

FEEDBACK